Open-FDD Documentation

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1 Home
2 Open-FDD
  2.1 What it does
  2.2 Quick start
    2.2.1 Protecting the entire API with Caddy
  2.3 Documentation
  2.4 Stack
  2.5 Behind the firewall; cloud export is vendor-led
3 System Overview
4 System Overview
  4.1 Architecture
  4.2 Services
  4.3 Campus-based architecture
  4.4 Data flow
  4.5 Ways to deploy
  4.6 Key concepts
5 Getting Started
6 Getting Started
  6.1 Do this to bootstrap
  6.2 Prerequisites
  6.3 What the bootstrap script does
  6.4 After bootstrap
  6.5 Next steps
7 Using the React dashboard
8 Using the React dashboard
  8.1 Navigation
  8.2 Common workflows
9 BACnet Overview
10 BACnet Integration
  10.1 Components
  10.2 Ports

10.3 Discovery and getting points into the data model
10.4 Data acquisition
10.5 Configuration
11 BACnet, RDF, and BRICK
12 BACnet, RDF, and BRICK
12.1 Roles
12.2 Flow
12.3 Try it (Open-FDD Swagger)
12.4 Summary
13 BACnet
14 BACnet
14.1 Setup (do this before the platform scrapes data)
15 Overview
16 Data model flow
16.1 Flow
16.2 TTL structure
16.3 Data modeling process (discover → tag → import → validate)
16.4 Data-model API
16.5 Validation
17 Sites
18 Sites
18.1 Structure
18.2 Scoping
18.3 API
19 Equipment
20 Equipment
20.1 Structure
20.2 Hierarchy
20.3 API
21 AI-assisted data modeling
22 AI-assisted Brick tagging
22.1 Workflow (export → tag → import)
22.2 API contract (import)
22.3 LLM prompt and agent guidelines
22.4 Rules for this project (what to send the LLM)
22.5 See also
22.6 In-house AI agent (OpenAI API Assist)
22.6.1 How the agent works: retry and prompt chaining
22.7 Automated tagging via the API (summary)
22.8 Possible extension: AI assist on Data Model Testing
23 Points
24 Points
24.1 Structure
24.2 Time-series data

24.3 Layers and mapping
24.3.1 BACnet addressing
24.4 API
24.5 Queries
25 LLM workflow (export + rules + validate → import)
26 LLM workflow: one prompt, export JSON, rules, and validated import
26.1 What you upload to the LLM
26.2 Where the rules live
26.3 Feeds / fed_by (HVAC topology)
26.4 Validate before import (so backend CRUD accepts it)
26.5 Mechanical engineer flow (short)
26.6 See also
27 SPARQL cookbook
28 SPARQL cookbook
28.1 Data Model Setup vs Data Model Testing
28.2 How to run SPARQL
28.3 Prefixes and graph contents
28.4 Recipe 1: Platform config
28.5 Recipe 2: Sites and counts
28.6 Recipe 3: Equipment and points by site
28.7 Recipe 4: BACnet devices and objects
28.8 Recipe 5: FDD — points and rule mapping
28.9 Recipe 6: FDD — time-series references for rules
28.10 Recipe 7: Integrity and completeness
28.11 Predefined buttons (Data Model Testing)
28.12 Saved queries
28.13 Related docs
29 Data modeling
30 Data modeling
31 Overview
32 Fault rules
32.1 Where rules live: config path and how to manage them
32.2 Hot reload (edit → run → view)
32.3 Rule types
32.4 YAML structure
32.5 Input resolution (100% Brick)
32.6 Running rules
32.7 Cookbook
33 Expression Rule Cookbook
34 Expression Rule Cookbook
34.1 100% Brick-model driven (no column in rules)
34.2 How to define expressions

34.3 AHU rules (reference-style)
34.3.1 Rule A — Duct static below setpoint at full fan speed
34.3.2 Rule B — Blended air temp below expected band
34.3.3 Rule C — Blended air temp above expected band
34.3.4 Rule D — Discharge air cold when heating commanded
34.3.5 Rule E — SAT too low with full heating
34.3.6 Rule F — SAT/MAT mismatch in economizer mode
34.3.7 Rule G — Ambient too warm for free cooling
34.3.8 Rule H — Ambient vs blended mismatch (econ + mech cooling)
34.3.9 Rule I — Ambient vs blended mismatch (econ-only)
34.3.10 Rule J — Discharge above blended in cooling
34.3.11 Rule K — Discharge above setpoint in full cooling
34.3.12 Rule L — Cooling coil delta-T when inactive
34.3.13 Rule M — Heating coil delta-T when inactive
34.4 Central plant
34.4.1 Differential pressure at max pump speed
34.4.2 Plant flow high at max pump
34.4.3 Plant supply temp outside deadband
34.4.4 Chiller runtime over daily limit
34.5 Heat pumps
34.5.1 Discharge cold when heating
34.6 VAV zones
34.6.1 Excessive heating during warm weather
34.6.2 Damper or valve at full open
34.6.3 Zone and IAQ bounds
34.7 Opportunistic rules (economizer & ventilation)
34.7.1 Economizing when outdoor conditions are unfavorable
34.7.2 Mechanical cooling when econ could suffice
34.7.3 Low ventilation (estimated OA fraction)
34.7.4 Preheat over-conditioning
34.7.5 Blended air damper deviation
34.8 Weather station
34.8.1 Unrealistic temperature spike
34.8.2 RH bounds
34.8.3 Wind gust vs sustained
34.9 Sensor validation (bounds & flatline)
35 Fault rules for HVAC
36 Fault rules for HVAC
37 Cloud export example
38 Cloud export example
38.1 What the example does
38.2 Run the example
38.3 Use it as your starting point

39 Concepts
40 Concepts
41 How-to Guides
42 How-to Guides
43 Grafana SQL cookbook
44 Grafana SQL cookbook —
44.1 BACnet
44.1.1 1. Site Variable
44.1.2 2. Device Variable
44.1.3 3. Point Variable
44.1.4 BACnet only data
44.2 BACnet Plus Fault Data
44.3 Host stats (stack-host-stats)
44.3.1 Dashboard variable (Host)
44.3.2 Host — Memory (used / available / total)
44.3.3 Host — Load (1 / 5 / 15 min)
44.3.4 Host — Swap (used / total)
44.3.5 Host — Disk space (free / used / total)
44.3.6 Host — Disk available (gauge, latest)
44.3.7 Containers — CPU %
44.3.8 Containers — Memory %
44.3.9 Containers — Table (latest)
44.4 Weather
44.4.1 Temp RH Dewpoint
44.4.2 Wind Speed Gusts Direction
44.4.3 Wind Speed Gust Direction
44.4.4 Solar Radiation
44.4.5 Cloud Cover
44.5 Faults
44.5.1 Fault Definitions
44.5.2 Fault Definition Count
45 Grafana dashboards
46 Grafana dashboards (optional)
46.1 React frontend parity
47 Configuration
48 Configuration
48.1 Platform config (RDF + CRUD)
48.2 Platform keys (RDF / GET-PUT /config)
48.3 Services that read config from the API (BACnet scraper)
48.4 BACnet: single gateway, remote gateways, central aggregator
48.5 Edge / resource limits
48.6 Environment
49 Contributing

50 Contributing to Open-FDD
50.1 Table of Contents
50.2 Code of Conduct
50.3 I Have a Question
50.4 I Want To Contribute
50.4.1 Legal notice
50.5 Reporting Bugs
50.5.1 Before submitting a bug report
50.5.2 How to submit a good bug report
50.6 Suggesting Enhancements
50.6.1 Before submitting
50.6.2 How to submit a good enhancement suggestion
50.7 Contributing Rules (Expression Rule Cookbook)
50.8 Your First Code Contribution
50.9 Improving the Documentation
50.10 Styleguides
50.11 Git workflow (branch → PR → sync)
50.12 Commit Messages
50.13 Join the Project Team
50.14 Attribution
51 API Reference
52 API Reference
52.1 High-level overview
52.2 Overview AI context and behavior
52.2.1 What the agent receives on each request
52.2.2 What the agent does *not* use
52.2.3 Models and keys
53 Appendix
54 Appendix
55 Frontend Evaluation
56 React frontend evaluation
56.1 Quality and stack
56.2 Feature coverage vs OpenAPI CRUD
56.3 Grafana cookbook parity
56.4 WebSockets: keep or remove?
56.5 Legacy static UI and BACnet tree
56.6 Caddy and security
56.7 Summary
57 Plan Rules Crud And Hot Reload
58 Plan: Default Rules, Fix Phantom Definitions, Rules CRUD + Frontend (Hot Reload)
58.1 Goals
58.2 1. Fix phantom definitions (backend)
58.3 2. Default rules (no code change)

58.4 3. Backend: Rules CRUD (upload / delete)
58.5 4. Frontend: Upload, Download, Delete
58.6 5. Hot reload (no change)
58.7 6. Implementation order
58.8 7. Security / validation
58.9 8. Summary
59 Security
60 Security and Caddy Bootstrap
60.1 Architecture: frontend, API, and Caddy
60.2 Frontend API key (Bearer)
60.3 Quick bootstrap with Caddy and basic auth
60.4 Caddyfile for protecting the entire API
60.5 Default password and how to change it
60.6 Services: encryption and exposure
60.7 Hardening best practices
60.8 Throttling and rate limiting
60.8.1 1. No API rate limiting by default
60.8.2 2. Outbound: OT/building network is paced
60.8.3 3. Inbound: rate limiting at the reverse proxy (e.g. Caddy)
60.9 TLS (optional): self-signed certificates
60.10 Troubleshooting
60.11 Security scanning (Trivy) — Phase 3
60.12 Summary

Generated March 16, 2026

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1 Home

2 Open-FDD

Open-FDD is an open-source knowledge graph fault-detection platform for HVAC systems that helps facilities optimize their energy usage and cost-savings. Because it runs on-prem, facilities never have to worry about a vendor hiking prices, going dark, or walking away with their data. The platform is an AFDD stack designed to run inside the building, behind the firewall, under the owner's control. It transforms operational data into actionable, cost-saving insights and provides a secure integration layer that any cloud platform can use without

vendor lock-in. U.S. Department of Energy research reports median energy savings of roughly 8–9% from FDD programs—meaningful annual savings depending on facility size and energy spend.

The building is modeled in a **unified graph**: Brick (sites, equipment, points), BACnet discovery RDF, platform config, and—as the project evolves—other ontologies such as ASHRAE 223P, in one semantic model queried via SPARQL and serialized to `config/data_model.ttl`.

2.1 What it does

Open-FDD is an **edge analytics and rules engine** for building automation. It:

- **Ingests** BACnet points via diy-bacnet-server (JSON-RPC) and weather via Open-Meteo
- **Stores** telemetry in TimescaleDB and models the building in a **unified RDF graph** (Brick + BACnet + config)
- **Runs** YAML-defined FDD rules (bounds, flatline, hunting, expression) on a configurable schedule
- **Exposes** REST APIs for sites, points, equipment, data-model export/import, bulk timeseries and fault download (Excel-friendly CSV, JSON for cloud), TTL generation, SPARQL validation
- **Visualizes** timeseries and fault results in Grafana

Operators and integrators get full control, lower cost, and no vendor lock-in. Already powering HVAC optimization and commissioning workflows.

2.2 Quick start

```
git clone https://github.com/bbartling/open-fdd.git
cd open-fdd
./scripts/bootstrap.sh
```

Endpoints (direct access):

What	URL	Notes
Frontend (React)	http://localhost:5173	Main UI: dashboard, sites, points, config, faults, plots. Use this for day-to-day workflows.

What	URL	Notes
API (REST)	http://localhost:8000/docs	Swagger UI for integration and scripts. High-level reference: Appendix: API Reference.
Caddy (reverse proxy)	http://localhost:80	Default `stack/caddy/Caddyfile` proxies to frontend only. See Protecting the entire API with Caddy below.
TimescaleDB	localhost:5432	Database `openfdd` (user `postgres`); keep internal.
BACnet (diy-bacnet-server)	http://localhost:8080/docs	JSON-RPC API; UDP 47808 for BACnet/IP.
Grafana	http://localhost:3000	**Optional:** `./scripts/bootstrap.sh --with-grafana` (admin/admin). React frontend provides equivalent views.

WebSockets: The API exposes **`/ws/events`** for live updates (faults, CRUD, FDD run). The React frontend connects with `?token=<API_KEY>` when `VITE_OFDD_API_KEY` is set; Bearer auth is used for REST. See Security & Caddy.

2.2.1 Protecting the entire API with Caddy

The default **`stack/caddy/Caddyfile`** only routes **port 80 → frontend** (no auth, API not behind Caddy). To protect the **entire** stack behind one entry point (basic auth, API + WebSocket + frontend, optional Grafana):

1. **Use a Caddyfile that:**
   - Listens on one port (e.g. `:80` or `:8088`).
   - Enables **basic_auth** for all routes (one login for the browser).
   - Proxies **`/api/*`**, **`/ws/*`**, **`/docs`**, **`/redoc`**, **`/openapi.json`**, **`/health`**, and other API paths to **`api:8000`**.
   - Proxies **`/grafana`** to **`grafana:3000`** if you use `--with-grafana`.
   - Proxies `/*` to the **frontend** (e.g. `frontend:5173`).
   - When proxying to the API, adds **`header_up X-Caddy-Auth <secret>`** so the API accepts requests that passed basic auth (set

`OFDD_CADDY_INTERNAL_SECRET` in the API container to the same value).

2. **Full steps:** See Security and Caddy — Quick bootstrap with Caddy and basic auth, default password change, and troubleshooting (401s, WebSocket behind Caddy).

3. **Caddyfile location:** `stack/caddy/Caddyfile`. A full example (basic auth, API, WebSocket, frontend) is in Security — Caddyfile for protecting the entire API.

2.3 Documentation

Section	Description
System Overview	Architecture, services, data flow
Getting Started	Install, bootstrap, first run
Using the React dashboard	Overview, Config, Points, Data model, Faults, Plots, Web weather, System resources
BACnet	Discovery, scraping, RPC, RDF/BRICK (knowledge graph, bacpypes3)
Data modeling	Sites, equipment, points (CRUD), Brick TTL, SPARQL cookbook, AI-assisted tagging
Fault rules for HVAC	Rule types, expression cookbook
Concepts	Cloud export example — how vendors pull data from the API to their cloud
How-to Guides	Grafana dashboards (optional), Grafana SQL cookbook
Security & Caddy	Basic auth, bootstrap, hardening, optional TLS
Configuration	Platform config, rule YAML
Appendix	API Reference — REST at a glance; Technical reference, Developer guide
Contributing	How to contribute; alpha/beta focus; bugs, rules, docs, drivers

Use the React frontend for config, sites, points, data model, faults, and plots. **API details** (CRUD, config, data-model, download, analytics, BACnet) are summarized in Appendix: API Reference; full OpenAPI at `/docs` and `/openapi.json`. **Grafana** is optional; the React UI provides equivalent timeseries and fault views.

2.4 Stack

Service	Port	Purpose
Caddy	80	Reverse proxy (default: frontend only; see Security to protect API + WebSocket + frontend)
API	8000	REST API, Swagger `/docs`, WebSocket `/ws/events`
Frontend	5173	React dashboard (sites, points, faults, plots)
TimescaleDB	5432	PostgreSQL + TimescaleDB
Grafana	3000	**Optional** dashboards (`--with-grafana`); React frontend has equivalent views
diy-bacnet-server	8080	JSON-RPC API (HTTP); POST server_hello returns `mqtt_bridge` when BACnet2MQTT enabled
diy-bacnet-server	47808	BACnet/IP (UDP)
Mosquitto (MQTT)	1883	Optional: `./scripts/bootstrap.sh --with-mqtt-bridge` — broker for BACnet2MQTT

2.5 Behind the firewall; cloud export is vendor-led

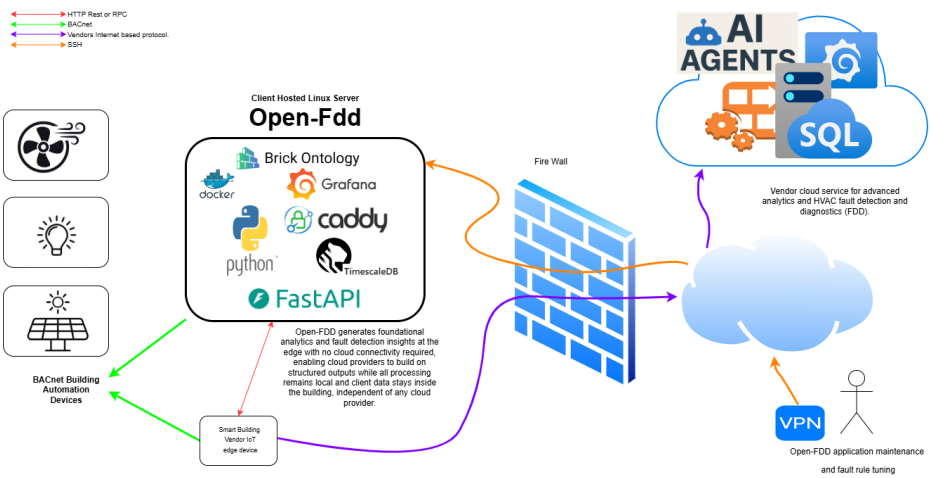
Open-FDD runs **inside the building, behind the firewall**. Analytics and rules run locally; telemetry stays on-premises unless you export it. Cloud/MSI vendors run their **own** export process (e.g. edge gateway or script) that **pulls** from the API over the LAN; Open-FDD does not push to the cloud. See the Cloud export example and `examples/cloud_export.py`. Data stays with the client; the open data model and APIs let you switch vendors without losing data or the model.

3 System Overview

4 System Overview

Open-FDD is an **open-source knowledge graph for building technology systems**, specializing in fault detection and diagnostics (FDD) for HVAC. FDD is often called AFDD when the platform runs it on a schedule (automated FDD). It runs on-premises so facilities keep control of their data and avoid vendor lock-in. This section describes the architecture, services, and data flow.

4.1 Architecture



Open-FDD Edge Platform Architecture

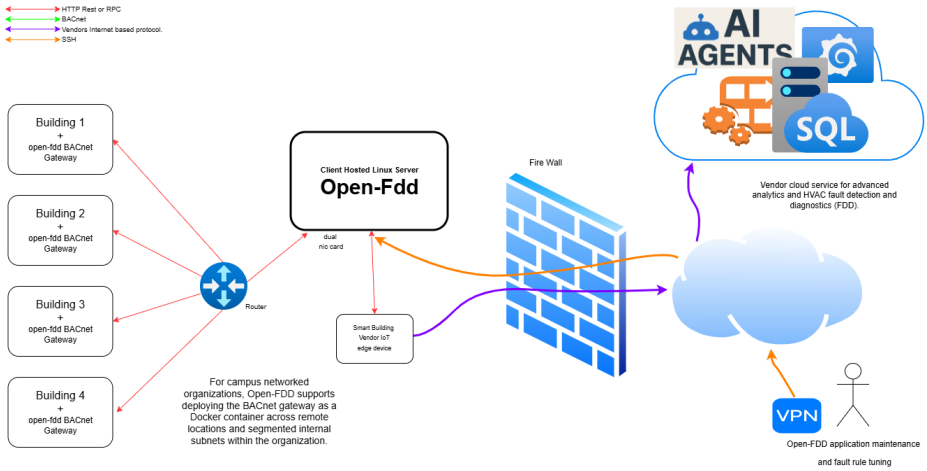
This project is an open-source stack; a cloud or MSI vendor can develop their own Docker container and deploy it on the **same client-hosted server** that runs Open-FDD, pulling from the local API over the LAN. That approach removes the need for a separate IoT or edge device dedicated to the vendor.

4.2 Services

Service	Description
API	FastAPI CRUD for sites, equipment, points. Data-model export/import, TTL generation, SPARQL validation. Swagger at `/docs`. Config UI (HA-style data model tree, BACnet test) at `/app/`.
Grafana	Pre-provisioned TimescaleDB datasource only (uid: openfdd_timescale). No dashboards; build your own with SQL from the Grafana SQL cookbook. Use `--reset-grafana` to re-apply datasource provisioning.
TimescaleDB	PostgreSQL with TimescaleDB extension. Single source of truth for metadata and time-series.
BACnet scraper	Polls diy-bacnet-server via JSON-RPC. Writes readings to `timeseries_readings`.
Weather scraper	Fetches from Open-Meteo ERA5 (temp, RH, dewpoint, wind, solar/radiation, cloud cover).
FDD loop	Runs every N hours (see `rule_interval_hours`, `lookback_days` in platform config). Pulls last N days from DB into pandas, **reloads all rules from YAML on every run** (hot reload), runs rules, writes `fault_results` back to DB. No restart needed when tuning rule params.
diy-bacnet-server	

Service	Description
	BACnet/IP JSON-RPC bridge. Discovered devices/points → CSV; scraper reads present-value via RPC.

4.3 Campus-based architecture



Open-FDD Edge Platform Architecture Campus

Remote Open-FDD BACnet gateways (e.g. diy-bacnet-server plus scraper) can be deployed **across each subnet** on the internal campus IT network. Typically each building has its own BACnet network on a unique subnet; a gateway per building or per subnet keeps BACnet traffic local while forwarding data to a **centralized** Open-FDD instance (API, Grafana, FDD loop, database). That gives the campus a single integration point for the cloud-based vendor of choice—one API and one data model for the whole portfolio, without the vendor touching each building's BACnet network directly.

How to set it up: (1) **Remote gateway per building:** On each subnet run diy-bacnet-server + scraper; set the scraper's `OFDD_DB_DSN` to the central database and `OFDD_BACNET_SITE_ID` to that building's site (create the site on the central API first). (2) **Central aggregator:** On the central host run only DB, API, Grafana, FDD loop (no local BACnet containers); set `OFDD_BACNET_GATEWAYS` to a JSON array of `{url, site_id, config_csv}` and run one scraper that polls each remote gateway. See Configuration — BACnet for keys and examples.

4.4 Data flow

1. **Ingestion:** BACnet scraper and weather scraper write to `timeseries_readings` (point_id, ts, value).
2. **Data model (unified graph):** The building is represented as a **unified graph**—one semantic model combining Brick (sites, equipment, points from the DB), BACnet RDF (from point discovery via diy-bacnet-server), platform config, and room for future ontologies (e.g. ASHRAE 223P). CRUD and **POST /bacnet/point_discovery_to_graph** update this model; SPARQL queries it. One TTL file `config/data_model.ttl` holds the graph; a background thread serializes it to disk every 5 minutes (configurable via `OFDD_GRAPH_SYNC_INTERVAL_MIN`); **POST /data-model/serialize** runs the same write on demand.
3. **FDD (Python/pandas):** The FDD loop pulls data into a pandas DataFrame, runs YAML rules, writes `fault_results` to the database. Fault logic lives in the rule runner; the database is read/write storage.
4. **Visualization:** Grafana queries TimescaleDB for timeseries and fault results.

4.5 Ways to deploy

- **Docker Compose** (recommended): `./scripts/bootstrap.sh`
- **Minimal (raw BACnet only):** DB + Grafana + BACnet server + scraper: `./scripts/bootstrap.sh --minimal` — no FDD, weather, or API.
- **Manual:** Start DB, Grafana, run scrapers and FDD loop from host

---

4.6 Key concepts

- **Sites** — Buildings or facilities.
- **Equipment** — Devices (AHUs, VAVs, heat pumps). Belong to a site.
- **Points** — Time-series references. Have `external_id` (raw name), `rule_input` (FDD column ref), optional `brick_type` (Brick class).
- **Fault rules** — YAML files (bounds, flatline, hunting, expression). Run against DataFrame; produce boolean fault flags. See Fault rules for HVAC.
- **Unified graph** — One semantic model (Brick + BACnet + platform config; future 223P or other ontologies). Stored in `config/`

`data_model.ttl`; maps Brick classes → `external_id` for rule resolution; queryable via SPARQL.

5 Getting Started

6 Getting Started

This page covers **prerequisites** and the **bootstrap script**: how to get the Open-FDD platform running. For configuration, data modeling, and rules, see the Documentation index.

6.1 Do this to bootstrap

1. **Prerequisites:** Docker, Docker Compose, and Git installed (see below).

2. **Clone and run bootstrap** (from any directory):

   ```
   git clone https://github.com/bbartling/open-fdd.git
   cd open-fdd
   ./scripts/bootstrap.sh
   ```

   That's it. The script builds and starts the full stack (DB, API, frontend, Caddy, diy-bacnet-server, BACnet scraper, weather scraper, FDD loop), waits for Postgres, runs migrations, and **seeds platform config** via the API (PUT /config) so runtime settings are in the knowledge graph. When it finishes you get:

   - **API:** http://localhost:8000/docs
   - **Frontend:** http://localhost:5173 (or via Caddy http://localhost:80). See Using the React dashboard for what each page does.
   - **BACnet Swagger:** http://localhost:8080/docs
   - **DB:** localhost:5432/openfdd (postgres/postgres)
   - **Grafana:** not started by default; use `./scripts/bootstrap.sh --with-grafana` then http://localhost:3000 (admin/admin)

3. **Optional:** Set `OFDD_*` in `stack/.env` before the first run to customize the seeded config (e.g. `OFDD_BACNET_SERVER_URL`, `OFDD_RULE_INTERVAL_HOURS`). See Configuration.

6.2 Prerequisites

- **OS:** Linux only (Ubuntu Server latest, or Linux Mint). **Tested on x86;** should work on ARM but is untested. The bootstrap script and Docker stack are not supported on Windows. Keep the system updated:

  ```
  sudo apt update && sudo apt upgrade -y
  ```

- **Docker and Docker Compose:** Required. Install Docker Engine and Docker Compose (or `docker-compose`). See Docker install for your distro.

- **Git:** To clone the project:

  ```
  git clone https://github.com/bbartling/open-fdd.git
  cd open-fdd
  ```

- **BACnet (default data driver):** The default data driver is BACnet. Bootstrap **automatically** builds and starts diy-bacnet-server as its own Docker container (plus the BACnet scraper). You must run **BACnet discovery first** and curate the resulting CSV before the platform can scrape data—the scraper uses that CSV as its config. See BACnet → Setup and BACnet overview. To run without BACnet (e.g. central-only with remote gateways), start only the services you need (e.g. `docker compose up -d db grafana api fdd-loop weather-scraper` from `stack/`).

6.3 What the bootstrap script does

`scripts/bootstrap.sh` (run from the **repo root**):

1. Ensures **diy-bacnet-server** exists as a sibling repo (clones it if missing).
2. Runs **docker compose up -d –build** from `stack/` (builds all images, starts all services).
3. Waits for **Postgres** to be ready (~15s).
4. Applies **database migrations** (idempotent; safe on existing DBs).
5. **Seeds platform config** via PUT /config (waits for API, then sends default or `stack/.env` values into the RDF graph).
6. Optionally runs **–reset-data** if you passed that flag (deletes all sites + data-model reset; for testing).

It does **not** purge or wipe the database on a normal run; only `--reset-grafana` wipes the Grafana volume.

Full stack (default): TimescaleDB, API, **diy-bacnet-server** (BACnet/IP bridge), **BACnet scraper**, weather scraper, FDD loop. Grafana is optional (`--with-grafana`). For BACnet data use the **data model** (discover via frontend or API → import points); see BACnet overview. Optional services (Caddy, host-stats) are in docker-compose; start them with `docker compose up -d` from `stack/` if needed.

Bootstrap options: Run `./scripts/bootstrap.sh --help` for the full list. Summary:

Option	Effect
(none)	Build and start full stack (DB, API, frontend, Caddy, BACnet server, scrapers, FDD loop). Prints API key if generated. Grafana **not** started by default.
`--with-grafana`	Include Grafana; then accessible at http://localhost:3000 or via Caddy at `/grafana` if using a full Caddyfile.
`--minimal`	DB + BACnet server + bacnet-scraper only. No FDD, weather, or API. Add `--with-grafana` for Grafana.
`--verify`	Health checks only: list containers, test DB; exit. Does not start or stop.
`--test`	Run tests (frontend lint + typecheck, backend pytest, Caddy validate); then exit.
`--build SERVICE ...`	Rebuild and restart only listed services, then exit. Services: `api`, `bacnet-server`, `bacnet-scraper`, `caddy`, `db`, `fdd-loop`, `frontend`, `grafana`, `host-stats`, `mosquitto` (with `--with-mqtt-bridge`), `weather-scraper`.
`--build-all`	Rebuild and restart all services; then exit.

Option	Effect
`--frontend`	Before start: stop frontend container and remove `frontend_node_modules` volume so the next `up` runs a fresh `npm install`. Use after changing `frontend/package.json`.
`--update`	Git pull open-fdd and diy-bacnet-server (sibling), then rebuild and restart (keeps DB).
`--maintenance`	Safe Docker prune only (no volume prune).
`--reset-grafana`	Wipe Grafana volume and re-apply provisioning. DB and other data retained.
`--reset-data`	Delete all sites via API and POST /data-model/reset (testing).
`--retention-days N`	TimescaleDB retention in days (default 365). Env: `OFDD_RETENTION_DAYS`.
`--log-max-size SIZE`	Docker log max size per file (default `100m`). Env: `OFDD_LOG_MAX_SIZE`.
`--log-max-files N`	Docker log max files per container (default 3). Env: `OFDD_LOG_MAX_FILES`.
`--install-docker`	Attempt Docker install (Linux) then continue.
`--skip-docker-install`	Explicitly skip Docker install (no-op; use with scripts that call bootstrap after install).
`--no-auth`	Do not generate or set `OFDD_API_KEY`; API will not require Bearer auth.

To **update** an existing clone: `git pull` then `./scripts/bootstrap.sh`, or `./scripts/bootstrap.sh --update`. Rebuild single services: `./scripts/bootstrap.sh --build api`.

6.4 After bootstrap

- **Grafana:** Open http://localhost:3000. TimescaleDB datasource and Open-FDD dashboards are provisioned from config. If dashboards or datasource are wrong, run `./scripts/bootstrap.sh --reset-grafana` (keeps DB data).
- **Minimal mode:** If you used `--minimal`, only DB, Grafana, BACnet server, and scraper run. No API; use Grafana and scraper logs to confirm data flow. To add the full stack later, run `./scripts/bootstrap.sh` without `--minimal`.

6.5 Next steps

- **How-to Guides** — Grafana dashboards (optional) and SQL cookbook.
- **Configuration** — Platform config, rule YAML, services that read config from the API.
- **Security & Caddy** — Basic auth, throttling, TLS.
- **Appendix: API Reference** — REST endpoints at a glance; Swagger at http://localhost:8000/docs.

For **BACnet** (discovery and data model): BACnet and BACnet overview. For data modeling and fault rules: Data modeling, Fault rules for HVAC.

7 Using the React dashboard

8 Using the React dashboard

The Open-FDD **React frontend** (http://localhost:5173 or via Caddy) is the main UI for sites, points, data model, faults, and timeseries. Use it after Getting started; ensure the API is running and (if auth is enabled) the frontend is built with `VITE_OFDD_API_KEY` or you access through Caddy with basic auth.

8.1 Navigation

Page	Purpose
Overview	High-level summary: site selector, FDD/weather status, links into the app. **Overview AI assistant** (read-only chat): ask questions in natural language; the assistant automatically attaches fault and sensor data (last 24h) and offers advice. It receives a **fixed excerpt of the platform docs** (capped size) plus live DB context; it does **not** use conversation history (each request is standalone). Charts and tables can be popped out and downloaded as CSV. **OpenAI is called by the backend**—you provide your API key in the UI; the frontend sends it to the Open-FDD API, which calls OpenAI server-side (key is not stored). Supported models in the UI: **GPT-5 mini** (default), **GPT-5.4 pro**. See API Reference — Overview AI context and behavior.
OpenFDD Config	View and edit platform config (GET/PUT `/config`): FDD rule interval, lookback, BACnet, Open-Meteo, graph sync. Current settings are shown in a read-only summary; use the form to change values and Save.
Points	List points (optionally by site). Shows external ID, equipment, Brick type, FDD input, unit, polling, last value and time. Use for BACnet discovery (Who-Is, point discovery, add to data model) and CRUD.
Data model	Equipment tree, points, export/import, TTL view, SPARQL. Manage sites, equipment, and points; run data-model export/import and SPARQL.
Faults	Active fault states and definitions. Filter by site/equipment.
Plots	Timeseries charts. Select site, date range, and points; view sensor and

Page	Purpose
	weather data. Download CSV for the selected range and points.
Web weather	Open-Meteo weather charts (temp, RH, wind, radiation, etc.) when weather is enabled.
System resources	Host and container metrics (when host-stats is running): memory, load, disk, **per-container CPU/memory** (table + time-series charts). Status badges (green/yellow/red) indicate resource pressure.
Stack status	Overview shows API, BACnet gateway, and MQTT bridge status with **green** / **yellow** / **red** indicators (e.g. MQTT bridge connected = green, enabled but disconnected = yellow).

8.2 Common workflows

- **Change platform config:** OpenFDD Config → edit fields → Save. Changes take effect on the next FDD run or scraper cycle.
- **Discover BACnet points:** Points → BACnet panel (Who-Is range, then point discovery for a device) → add to data model. Points appear in the data model and Points list.
- **Inspect or edit the data model:** Data model → browse equipment and points, export/import, view TTL, run SPARQL.
- **View timeseries:** Plots → pick site and date range → select points → chart. Use Download CSV for the current selection and zoom.
- **Check faults:** Faults → see active faults and definitions; combine with Plots to correlate with sensor data.

For API integration (curl, scripts), see Appendix: API Reference and Swagger at http://localhost:8000/docs.

9 BACnet Overview

10 BACnet Integration

Open-FDD uses diy-bacnet-server as a BACnet/IP-to-JSON-RPC bridge. Discovery and scrape feed the same **data model** (building as a knowledge graph). The gateway uses **bacpypes3**'s built-in RDF (BACnetGraph) for discovery-to-RDF; Open-FDD merges that TTL and queries via SPARQL.

10.1 Components

Component	Purpose
diy-bacnet-server	BACnet/IP UDP listener + HTTP JSON-RPC API. Discovers devices and objects; exposes present-value reads. Swagger: http://localhost:8080/docs
BACnet scraper	Platform service. Polls diy-bacnet-server on a schedule; **reads points from the data model** (points with `bacnet_device_id` and `object_identifier`) and writes readings to TimescaleDB.
Data model	Sites, equipment, and points with BACnet addressing (`bacnet_device_id`, `object_identifier`, `object_name`). Configured via the **React frontend** (Config, Data model, Points) or the API. Single source of truth for what to scrape.

10.2 Ports

Port	Protocol	Use
47808	UDP	BACnet/IP
8080	HTTP	JSON-RPC API, Swagger docs

Only **one** process on the host can use port 47808 (BACnet/IP). Run discovery **before** starting diy-bacnet-server so they do not conflict. diy-bacnet-server runs with `network_mode: host` so it binds to the host's network interface.

10.3 Discovery and getting points into the data model

All BACnet configuration is done via the **data model** and the **React frontend** (or the API). There is no CSV-based scrape path.

1. **Run Who-Is and point discovery** — Use the React frontend (Config or Data model → BACnet panel) or the API. From the BACnet panel you can run **Test connection**, **Who-Is range**, and **Point discovery** (these proxy to diy-bacnet-server). diy-bacnet-server must be running (e.g. `./scripts/bootstrap.sh` starts it).
2. **Graph and data model** — Use **POST /bacnet/point_discovery_to_graph** (device instance) to put BACnet devices and points into the in-memory graph and sync `config/data_model.ttl`. Create points in the DB via the frontend or CRUD (set `bacnet_device_id`, `object_identifier`, `object_name`), or use **GET /data-model/export** → LLM/human tagging → **PUT /data-model/import**.
3. **Run the scraper** — The BACnet scraper loads points that have `bacnet_device_id` and `object_identifier` from the database and polls only those via diy-bacnet-server.

See Points for the BACnet fields and Appendix: API Reference for endpoints. **Port:** Only one process can use port 47808 (BACnet/IP). For API/UI discovery, diy-bacnet-server is already running.

10.4 Data acquisition

The BACnet scraper loads points from the DB where `bacnet_device_id` and `object_identifier` are set; groups by device; calls diy-bacnet-server JSON-RPC (present-value) for each; writes `(point_id, ts, value)` to `timeseries_readings`. Throttling depends on how many points are defined and the poll interval. See Security → Throttling.

10.5 Configuration

Scraper config comes from **environment** and, when the API uses Bearer auth, from **GET /config** (the React Config page controls the interval).

- **`OFDD_BACNET_SERVER_URL`** — diy-bacnet-server base URL (e.g. http://localhost:8080). Required for RPC.
- **Scrape interval** — When **`OFDD_API_KEY`** is set (e.g. in `stack/.env`), the scraper calls GET /config and uses **`bacnet_scrape_interval_min`** from the data model (Config page). Otherwise it uses **`OFDD_BACNET_SCRAPE_INTERVAL_MIN`** env (default: 5). See Configuration → Services that read config from the API.
- **`OFDD_DB_*`** — TimescaleDB connection

11 BACnet, RDF, and BRICK

12 BACnet, RDF, and BRICK

Open-FDD treats the **building as a knowledge graph**: sites, equipment, and points (and BACnet topology) live in a single semantic model. CRUD and **POST /bacnet/point_discovery_to_graph** update this model; SPARQL queries it.

12.1 Roles

Layer	Responsibility
diy-bacnet-server	

Layer	Responsibility
	BACnet RPCs: Who-Is, point discovery (returns JSON objects per device). Open-FDD calls these.
Open-FDD	Calls point discovery, builds clean BACnet TTL from the JSON, merges into the in-memory graph (Brick from DB + BACnet). CRUD owns sites, equipment, points. SPARQL runs over the combined model; sync writes `config/data_model.ttl`.

12.2 Flow

1. **Point discovery** — Open-FDD calls **POST /bacnet/point_discovery** (gateway RPC) for a device instance; gateway returns a list of objects (object_identifier, object_name, etc.).
2. **Point discovery to graph** — **POST /bacnet/point_discovery_to_graph** (same instance, `update_graph: true`) turns that JSON into clean BACnet TTL and updates the in-memory graph; optionally writes `config/data_model.ttl`. SPARQL and **GET /data-model/ttl** see Brick + BACnet.
3. **Points in DB** — Create or edit points via CRUD (set `bacnet_device_id`, `object_identifier`, `object_name`). Optionally **GET /data-model/export** then LLM or human adds Brick types then **PUT /data-model/import**.
4. **Scrape** — Data-model driven (points with `bacnet_device_id` / `object_identifier`).

12.3 Try it (Open-FDD Swagger)

1. Who-Is
- Open **http://localhost:8000/docs**. Find **POST /bacnet/whois_range**. Body: `{"request": {"start_instance": 1, "end_instance": 3456800}}`. Execute; note a device instance (e.g. `3456789`).

2. Point discovery to graph
- Find **POST /bacnet/point_discovery_to_graph**. Body: `{"instance": {"device_instance": 3456789}, "update_graph": true, "write_file": true}`. Execute. Open-FDD calls the gateway, gets point discovery JSON, builds BACnet TTL, and updates the in-memory graph and file.

3. Check
- **GET /data-model/ttl** or **POST /data-model/sparql** (e.g. `SELECT ?dev WHERE { ?dev a bacnet:Device }`).

12.4 Summary

- **Building as knowledge graph**: one semantic model (BRICK + BACnet), updated by CRUD and by **point_discovery_to_graph**.
- **BACnet in the graph** is produced by Open-FDD from point discovery JSON (clean TTL); SPARQL validates and queries the combined model.

13 BACnet

14 BACnet

BACnet is the **default data driver** for Open-FDD. Discovery, diy-bacnet-server, and the BACnet scraper are documented here. **Swagger (diy-bacnet-server):** http://localhost:8080/docs

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14.1 Setup (do this before the platform scrapes data)

Recommended: data model

1. **Start the platform** (including diy-bacnet-server) — e.g. `./scripts/bootstrap.sh`.
2. **Discover devices and points** — In the Config UI (`/app/`) use the BACnet panel: Who-Is range, then Point discovery per device. Or call the API: `POST /bacnet/whois_range`, `POST /bacnet/point_discovery`.
3. **Graph and data model** — Call **POST /bacnet/point_discovery_to_graph** (device instance) to put BACnet devices and points into the in-memory graph and sync `config/data_model.ttl`. Create points in the DB via CRUD (set `bacnet_device_id`, `object_identifier`, `object_name`) or use **GET** /

data-model/export → LLM/human tagging → **PUT /data-model/import**.

4. **Scraper runs automatically** — The bacnet-scraper container (or `tools/run_bacnet_scrape.py`) loads points from the data model and polls only those. Configuration is via the frontend and data model only; there is no CSV path.

15 Overview

16 Data model flow

Open-FDD uses a **unified graph**: one semantic model that combines Brick (sites, equipment, points), BACnet discovery RDF (from bacpypes3 in diy-bacnet-server), platform config, and—as the project evolves—other ontologies such as ASHRAE 223P. CRUD and discovery both update this graph; **all backend queries are SPARQL-driven** (rdflib Graph parse + SPARQL; no grep or text search on the TTL). Rules resolve inputs via `ofdd:mapsToRuleInput` in the TTL.

16.1 Flow

```
Sites + Equipment + Points (DB)  ← single source of truth
         │
         ▼
  Data-model export / CRUD
         │
         ▼
  Brick TTL (config/data_model.ttl)  ← Brick section reserialized
on every create/update/delete; same file can include a BACnet
discovery section (one file for SPARQL)
         │
         ▼
  FDD column_map (external_id → rule_input)
         │
         ▼
  RuleRunner
```

CRUD and Brick TTL sync: The database is the single source of truth. Every **create**, **update**, or **delete** on **sites**, **equipment**, or **points** (via API or data-model import) triggers a reserialize: the Brick TTL file (`config/data_model.ttl`, or `OFDD_BRICK_TTL_PATH`) is regenerated from

the current DB and written to disk. So the Brick model is always in sync with CRUD. Deleting a site, device (equipment), or point also cascades to dependent data (timeseries, fault results, etc.) as in a typical CRUD app; see Danger zone — CRUD deletes.

16.2 TTL structure

```
:oat_sensor a brick:Outside_Air_Temperature_Sensor ;
    rdfs:label "OAT (°F)" ;
    brick:isPointOf :ahu_7 ;
    ofdd:mapsToRuleInput "oat" .
```

- Brick classes define sensor types
- `ofdd:mapsToRuleInput` maps to FDD DataFrame columns
- `rdfs:label` for display

16.3 Data modeling process (discover → tag → import → validate)

1. **Discover** — BACnet discovery and/or manual entry populate **sites**, **equipment**, and **points** in the DB.
2. **Export** — Use **GET /data-model/export** (or the Export card on the Data Model Setup page) to get JSON for tagging.
3. **Tag** — Either **manual**: copy JSON → use an external LLM or human → paste back; or **in-house agent**: Data Model Setup → **OpenAI API Assist** → **Tag with OpenAI** (POST /data-model/tag-with-openai), which uses retry and **prompt chaining** on validation errors (see AI-assisted tagging).
4. **Import** — **PUT /data-model/import** (or the Import card / auto-import from the agent) writes tagged points and optional equipment relationships into the DB and reserializes the Brick TTL.
5. **Validate** — Use the **Data Model Testing** page (SPARQL, “Summarize your HVAC”) to confirm the model; treat the result as pass or fail.

16.4 Data-model API

Endpoint	Description
`GET /data-model/export`	

Endpoint	Description
	Single export route: BACnet discovery + DB points (optional `?bacnet_only=true`, `?site_id=...`). Use for AI-assisted tagging.
`PUT /data-model/import`	Import JSON: **points** (required) and optional **equipment** (feeds/fed_by). Creates/updates points; does not accept sites or equipments.
`POST /data-model/tag-with-openai`	In-house AI agent: export + canonical prompt + optional user summary → OpenAI → validate (with retry and prompt chaining on failure) → return tagged JSON and agent log. Optional auto-import. See AI-assisted tagging.
`GET /data-model/ttl`	Generate Brick TTL from DB (and in-memory BACnet graph). Optional `?save=true` to write to file.
`POST /data-model/sparql`	Run SPARQL query against the current data model.

16.5 Validation

Use SPARQL to validate:

- All rule inputs have `ofdd:mapsToRuleInput`
- Equipment types and points are consistent
- Brick schema compliance (optional)

17 Sites

18 Sites

Sites represent buildings or facilities. Equipment and points are scoped to a site.

18.1 Structure

Field	Description
`id`	UUID primary key
`name`	Display name

18.2 Scoping

All time-series and equipment queries can be filtered by `site_id`. Grafana dashboards typically scope by site.

18.3 API

- `GET /sites` — List sites
- `GET /sites/{id}` — Get one
- `POST /sites` — Create
- `PATCH /sites/{id}` — Update
- `DELETE /sites/{id}` — Delete (cascades to equipment, points, timeseries, fault results; see Danger zone)

19 Equipment

20 Equipment

Equipment represents physical devices or systems (AHUs, VAVs, heat pumps, weather stations). Points may belong to equipment.

20.1 Structure

Field	Description
`id`	UUID primary key
`site_id`	FK to sites

Field	Description
`name`	Display name (e.g. “AHU-7”)
`equipment_type`	Optional type string
`external_id`	Optional raw identifier from source

20.2 Hierarchy

```
Site
 └── Equipment (AHU, VAV, etc.)
      └── Points (sensors, setpoints)
```

20.3 API

- `GET /equipment` — List equipment (filter by site)
- `GET /equipment/{id}` — Get one
- `POST /equipment` — Create
- `PATCH /equipment/{id}` — Update
- `DELETE /equipment/{id}` — Delete (cascades to points and their timeseries; see Danger zone)

21 AI-assisted data modeling

22 AI-assisted Brick tagging

Open-FDD supports **AI-assisted data modeling**: use **GET /data-model/export** to dump BACnet discovery and existing points as JSON, then have an **LLM or human** add Brick types, rule inputs, equipment relationships, and which points to poll — and send the result to **PUT /data-model/import**. The platform stays the single source of truth; tagging is the step where raw BACnet objects become a curated, rule-ready dataset.

This workflow is intended for **mechanical engineers and building operators** who need the data model tagged so that FDD rules in `stack/rules/` have the correct inputs and only the right points are logged long-term.

22.1 Workflow (export → tag → import)

1. **Discover** — Use the API: **POST /bacnet/whois_range**, then **POST /bacnet/point_discovery_to_graph** per device. The in-memory graph and `config/data_model.ttl` now contain BACnet devices and objects.

2. **Sites and equipment** — Create the building/site and equipment (AHUs, VAVs, zones) via **POST /sites** and **POST /equipment**. Note the returned `site_id` and `equipment_id` UUIDs; the import body requires real UUIDs from the API.

3. **Export** — **GET /data-model/export** (or `?bacnet_only=true` for discovery-only). Returns a single JSON array: BACnet discovery rows plus all DB points. Unimported BACnet rows have `point_id: null`, `polling: false`, and null `site_id`/`equipment_id`/`brick_type`/`rule_input`.

4. **Tag** — Send the export JSON to an LLM or edit manually. For each point set:

   - **site_id**, **external_id** (time-series key)
   - **brick_type** (e.g. `Supply_Air_Temperature_Sensor`, `Zone_Air_Temperature_Sensor`)
   - **rule_input** (name FDD rules use)
   - **unit** when known (e.g. `degF`, `%`, `cfm`, `0/1` for binary). Units are stored in the data model and TTL; the frontend uses them for Plots axis labels and grouping (e.g. temperatures on one axis, humidity on another). Use standard abbreviations so Plots and exports stay consistent.
   - **equipment_id** (optional)
   - **polling: true** for every point that must be **logged long-term** for this job (sensors/setpoints that FDD rules use); **polling: false** for points not needed. The LLM should tell the operator which points will be logged so they can confirm rules in `stack/rules/` have the required inputs.
   - For **equipment relationships** (Brick feeds/isFedBy), use the import **equipment** array: each item has `equipment_id`, optional

`feeds_equipment_id` and `fed_by_equipment_id` (UUIDs from GET /equipment).

5. **Import** — **PUT /data-model/import** with body: **points** (array) and optional **equipment** (array for feeds/fed_by only). The API accepts **only** these two keys — no `sites`, `equipments`, or `relationships`. The backend creates/updates points and equipment relationships, then rebuilds the RDF and TTL.

6. **Scraping** — The BACnet scraper (data-model path) loads points where `bacnet_device_id`, `object_identifier`, and **polling** = **true**; it calls diy-bacnet-server **client_read_multiple** per device and writes to `timeseries_readings`. Grafana and FDD use the same data model.

22.2 API contract (import)

- **points** (required): Each row is a point to **create** (omit `point_id`; set `site_id`, `external_id`, `bacnet_device_id`, `object_identifier`) or **update** (set `point_id` and fields to change). **site_id** and **equipment_id** must be real UUIDs from GET /sites and GET /equipment.
- **equipment** (optional): Updates existing equipment with Brick feeds/isFedBy. Each item: `{ "equipment_id": "<uuid>", "feeds_equipment_id": "<uuid> | null", "fed_by_equipment_id": "<uuid> | null" }`.

diy-bacnet ready: The list of points to poll is **GET /data-model/export** filtered to rows where **polling** === **true**; each has `bacnet_device_id` and `object_identifier`.

22.3 LLM prompt and agent guidelines

The **canonical prompt** lives in a single file: **`config/canonical_llm_prompt.txt`** (see README). The backend loads it when present (fallback: built-in prompt in code); you can edit the file at any time. The prompt is generic and works for **any site**: single building, campus, or tenant. The only input that changes is the export JSON (from GET /data-model/export, optionally with `?site_id=YourSiteName`). The LLM must preserve all fields, add brick_type, rule_input, **unit** (when known), and polling, and use equipment by name and site_id

from the export. The same file can be used as **model context** (e.g. in `pdf/open-fdd-docs.txt` or when building custom doc bundles for an LLM).

For exact schema details and import body (points + equipment only), see the Technical reference. It defines the primary task (Brick tagging), the export → tag → import flow, polling semantics, equipment feeds, and the **unit** field (e.g. degrees-fahrenheit, percent) used by the frontend and stored in the RDF/TTL.

22.4 Rules for this project (what to send the LLM)

To have the LLM align **rule_input** and Brick types with your FDD rules, you can include:

- **Fault rules overview** — FDD rule types and YAML format.
- **Expression Rule Cookbook** — AHU, chiller, weather, and advanced recipes; **rule_input** examples (e.g. sat, rat, zone_temp, sf_status).
- Your project's **YAML rule files** (e.g. from `stack/rules/`) — Paste snippets so the LLM uses the same input names.

See LLM workflow (export + rules + validate → import) for the full one-shot flow and validating LLM output so it parses on the backend.

22.5 See also

- LLM workflow — One prompt + export JSON + optional rules; validate with schema/Pydantic; then import and run FDD/tests.
- Data model overview — Flow (DB → TTL → FDD)

22.6 In-house AI agent (OpenAI API Assist)

Where it runs: In the frontend, open **Data Model Setup** → **Export (for AI / copy-paste)** → click **OpenAI API Assist**. The agent is **optional**; the manual export → external LLM → import flow is always available without it.

How it's invoked: When you click **Tag with OpenAI**, the frontend sends **POST /data-model/tag-with-openai** with your API key, optional chat prompt (HVAC description), model choice, and optional auto-import flag. The backend runs the agent (see below) and returns tagged JSON plus an **agent log**; the UI shows the log and pre-fills the Import textarea.

What the agent does: It uses the **same API as the manual process**: server-side **GET /data-model/export** (with optional site filter), then a single LLM call (or chunked calls for large payloads) with the **canonical prompt** from `config/canonical_llm_prompt.txt` (or built-in fallback). The response is validated with the same **DataModelImportBody** Pydantic model as **PUT /data-model/import**. If you enable auto-import, the backend then calls the import logic so the data model is updated in one step. The engineer can still validate on the **Data Model Testing** page (SPARQL / "Summarize your HVAC") and treat the result as pass or fail.

22.6.1 How the agent works: retry and prompt chaining

The agent is **not** a long-lived process; it runs inside a single **POST /data-model/tag-with-openai** request:

1. **Export** — The backend builds the same export as GET /data-model/export (optionally filtered by site).
2. **System prompt** — Loaded from **`config/canonical_llm_prompt.txt`** when the file exists (otherwise a built-in prompt in code). Edit the file to change behavior; no app restart needed for the next request.
3. **User message** — The export JSON is sent as the user message. If you filled the **chat prompt** in the UI (e.g. "Describe HVAC system and feeds or fed-by relationships for AI to tag"), that text is prepended so the LLM can use your description for Brick types and equipment relationships.
4. **LLM call** — One request to OpenAI (or, for very large exports, the backend may split into chunks of 120 rows and merge results) with `response_format={"type": "json_object"}`.
5. **Validation** — The response is parsed as JSON and validated with **DataModelImportBody** (same schema as import). If validation fails:
   - **Prompt chaining:** The error text (e.g. "OpenAI response failed schema validation: …") is appended to the **next** attempt's user message, so the model sees what went wrong and is asked to fix it.

    - **Retry** — The backend retries up to a configurable number of attempts (default 3; see `open_fdd/platform/llm_tagger.py` constants). Each attempt is a new LLM call with the same export and, from the second attempt onward, the previous validation error in the prompt.
6. **Agent log** — Each attempt and outcome (attempt, validation_failed, success) is recorded and returned in the response **meta.agent_log**; the UI shows this so you can see retries and that the agent produced valid import JSON.
7. **Optional import** — If you set **Auto-import tagged JSON** in the UI, the backend runs the same import logic as PUT /data-model/import and returns the import result in the response.

So the agent uses **rule-based retry** with **prompt chaining**: same export and rules each time, but after a validation failure the next prompt includes the error so the LLM can correct the output. Constants (max retries, chunk size) live in **`open_fdd/platform/llm_tagger.py`**; the API request can override `max_retries` (1–10).

Optional dependency: The `openai` package is required only for this endpoint. If it is not installed, the API returns 500 with a message to install it; the rest of the platform (including manual export/import) works without it.

22.7 Automated tagging via the API (summary)

1. Open the **Data Model Setup** page in the frontend.
2. Expand **OpenAI API Assist** (under Export). Optionally edit the **chat prompt** (default: "Describe HVAC system and feeds or fed-by relationships for AI to tag").
3. Enter your OpenAI API key and select a model. If you have multiple sites, optionally check **Tag specific site** and pick the site in the dropdown.
4. Click **Tag with OpenAI** — the platform calls **POST /data-model/tag-with-openai**, which runs the in-house agent (export → LLM with retry and prompt chaining → validate → optional import). The response includes **meta.agent_log**; the UI shows the log and pre-fills the Import textarea.
5. Review the tagged JSON (and agent log). If you did not auto-import, click **Import** when ready. Use the **Data Model Testing** page to

validate (SPARQL / Summarize your HVAC) and pass or fail the result.

Key handling: Your API key is sent in the request body over HTTPS and is used only for that request. It is never stored on the server. You can optionally enable *Remember key in this browser* in the UI (saved in `localStorage`) — only on a private, trusted device.

Users without an API key: The manual copy-paste workflow (Export → external LLM → Import) remains fully supported.

22.8 Possible extension: AI assist on Data Model Testing

A future **Data Model Testing** page could offer a second in-house AI assist, same chat style as on Data Model Setup: the engineer describes what they see (e.g. SPARQL summary, missing relationships, or test failure), and the model suggests **changes as import JSON** (points/equipment) so the engineer can apply and re-test. Under the hood this would use a **separate prompt** from the tagging prompt:

- **Input:** Current data model context (e.g. TTL snippet or SPARQL "Summarize your HVAC" result) plus the engineer's message (e.g. "Add feeds/fed-by between AHU-1 and VAV-1", "Fix the brick_type for SA-T").
- **Output:** Same schema as the tagging flow — valid **points** and **equipment** import JSON only — so the same **PUT /data-model/import** and validation path apply. The engineer reviews, applies, then re-runs SPARQL or predefined tests and passes/fails.

That way: **Setup** = tag from export (canonical prompt in `config/canonical_llm_prompt.txt`); **Testing** = revise from current model + chat (optional prompt e.g. `config/canonical_llm_prompt_testing.txt`). Both use the same API and retry/prompt-chaining behavior; only the system prompt and user message differ.

- Appendix: API Reference — Data model export/import, CRUD
- BACnet overview — Discovery and data-model scrape
- Fault rules — Brick-driven rules in `stack/rules/`

23 Points

24 Points

Points are time-series references in the Open-FDD data model. They link raw telemetry (BACnet objects, weather sensors) to equipment and Brick semantics.

24.1 Structure

Each point has:

Field	Description
`id`	UUID primary key
`site_id`	FK to sites
`equipment_id`	FK to equipment (nullable)
`external_id`	Raw identifier from source (e.g. BACnet object name); unique per site
`fdd_input`	Column name used by FDD rules (e.g. `oat`, `sat`); data-model API uses `rule_input`
`brick_type`	Optional Brick class (e.g. `Outside_Air_Temperature_Sensor`)
`unit`	Optional unit of measure
`description`	Optional
BACnet addressing	
`bacnet_device_id`	Optional. BACnet device instance (e.g. `3456789`). With `object_identifier`, used by the BACnet scraper to poll this point.
`object_identifier`	Optional. BACnet object ID (e.g. `analog-input,1`).
`object_name`	Optional. BACnet object name (often same as `external_id`).

24.2 Time-series data

Readings are stored in `timeseries_readings`:

Column	Type	Description
`point_id`	UUID	FK to points
`ts`	timestamp	Timestamp (UTC)
`value`	float	Numeric value

TimescaleDB hypertable, optimized for range scans and downsampling.

24.3 Layers and mapping

- **BACnet layer:** Points that have `bacnet_device_id` and `object_identifier` are scraped by the BACnet driver (data-model path). Add them via CRUD or after **POST /bacnet/point_discovery_to_graph** and data-model export/import. `external_id` is typically the BACnet object name.
- **Weather layer:** Points with `external_id` = `temp_f`, `rh_pct`, `dewpoint_f`, etc. come from the Open-Meteo weather fetch. They are linked to a synthetic equipment **Open-Meteo** (type **Weather_Service**) per site so they appear under "Open-Meteo" (web weather) in the **Data model** and **Points** tree. In the RDF graph that equipment is tagged with `ofdd:dataSource "open_meteo"` so you can query it via SPARQL.
- **Rule layer:** `fdd_input` / `rule_input` maps to DataFrame column names used by YAML rules.

The data-model API and Brick TTL coordinate `external_id` ↔ `rule_input` ↔ `brick_type`.

24.3.1 BACnet addressing

Points with `bacnet_device_id` and `object_identifier` set are used by the BACnet scraper (data-model path). Add them via CRUD or after point_discovery_to_graph and data-model export/import. See BACnet overview.

24.4 API

- `GET /points` — List points (filter by site or equipment)
- `GET /points/{id}` — Get one
- `POST /points` — Create
- `PATCH /points/{id}` — Update
- `DELETE /points/{id}` — Delete (cascades to timeseries_readings; see Danger zone)

24.5 Queries

```sql
SELECT ts, p.external_id, tr.value
FROM timeseries_readings tr
JOIN points p ON p.id = tr.point_id
WHERE p.external_id = 'temp_f'
  AND ts > NOW() - INTERVAL '1 day'
ORDER BY ts DESC;
```

25 LLM workflow (export + rules + validate → import)

26 LLM workflow: one prompt, export JSON, rules, and validated import

This page describes a **single upload** workflow for mechanical engineers: send the **canonical prompt**, the **data-model export JSON**, and (optionally) the **rules you want to use** to an LLM; get back import-ready JSON; **validate** it so you know it will parse on the Open-FDD backend; then **PUT /data-model/import** and run FDD (or Sparkl/tests) as needed.

> **Automated path available:** If you have an OpenAI API key, you can skip the copy-paste steps entirely. Enter your key in the **AI Tagging** card on the **Data Model** page — the platform runs the export, calls OpenAI with the canonical prompt, validates the

> response, and pre-fills the Import textarea. See AI-assisted tagging for details. The manual workflow below is always available for users without an API key.

26.1 What you upload to the LLM

1. **The canonical prompt** — From the README (or Technical reference): the full text that tells the LLM to return only `{"points": [...], "equipment": [...]}` with Brick types, rule_input slugs, equipment_name, feeds/fed_by, polling, and units.

2. **The export JSON** — From **GET /data-model/export** (optionally `?site_id=YourSiteName`). Example shape: one array of objects with `point_id`, `bacnet_device_id`, `object_identifier`, `object_name`, `external_id`, `site_id`, `site_name`, `equipment_id`, `equipment_name`, `brick_type`, `rule_input`, `unit`, `polling`. Unimported rows have `point_id: null` and null tagging fields; the LLM fills those and can set `site_id` if you pre-create the site.

3. **Rules for this project (optional)** — So the LLM knows which **rule_input** slugs and Brick types your FDD rules expect. You can:

   - Point the LLM at the **Fault rules overview** and **Expression Rule Cookbook** (AHU, chiller, weather, advanced recipes). The cookbook is the main reference for rule_input names and expression patterns.
   - Or paste **YAML** from your project's rules (e.g. from `stack/rules/` or your own rule files). The LLM can align `rule_input` with the inputs those rules use (e.g. `sat`, `rat`, `zone_temp`, `sf_status`).

26.2 Where the rules live

What	Where
Fault rules overview	docs/rules/overview — FDD rule types, YAML format, Brick-driven inputs.
Expression Rule Cookbook	docs/rules/expression_rule_cookbook — AHU, chiller, weather, and advanced recipes; **rule_input** examples and expression patterns.
Actual YAML rule files	

What	Where
	`stack/rules/` in the repo (or your `rules_dir`). The ME can upload or paste snippets so the LLM uses the same input names.

The cookbook is **not** a fault rule file itself; it's documentation. The **rules you want to use** are the YAML files in `stack/rules/` (or your project's rules). For the LLM, you can either paste that YAML or say "use rule_input slugs from the Expression Rule Cookbook (sat, rat, zone_temp, …)."

26.3 Feeds / fed_by (HVAC topology)

- **feeds** — This equipment supplies another (e.g. AHU feeds VAV).
- **fed_by** — This equipment is supplied by another (e.g. VAV fed_by AHU).

If topology is not known confidently, omit feeds/fed_by rather than guessing. You can add or refine relationships later.

26.4 Validate before import (so backend CRUD accepts it)

The Open-FDD **PUT /data-model/import** endpoint expects a body that matches the **DataModelImportBody** Pydantic model: exactly `points` (array) and optional `equipment` (array). If the LLM returns extra keys, wrong types, or invalid UUIDs, the API returns **422 Unprocessable Entity**.

To avoid that:

1. **Instruct the LLM** — In your prompt, add: "Return only valid JSON that conforms to the Open-FDD import schema: top-level keys `points` and `equipment` only; each point has the fields listed in the prompt; equipment items use `equipment_name`, `site_id`, `feeds`, `fed_by`."

2. **Use the API's JSON Schema** — The OpenAPI spec at **GET /openapi.json** (or **GET /docs** and "openapi.json") includes a **schema** for the import body. You can:
   - Export that schema (e.g. the `DataModelImportBody` and nested `PointImportRow` / `EquipmentImportRow` from the spec) and give it to the LLM: "Return JSON that validates against this schema."
   - Or run a local **validation step** before pasting into the UI: validate the LLM's JSON against the same schema (e.g. with a small script or tool that loads the schema and runs `jsonschema.validate`). If it passes, PUT /data-model/import will accept it (aside from referential issues like missing site_id).
3. **Pydantic in the repo** — The backend defines the import shape in **open_fdd/platform/api/data_model.py**: `DataModelImportBody`, `PointImportRow`, `EquipmentImportRow`. A script or pipeline can import those models and validate the LLM output (e.g. `DataModelImportBody.model_validate(json.loads(llm_output))`) before returning it to the human. That way the human only sees JSON that is known to parse on the backend.

26.5 Mechanical engineer flow (short)

1. **Create site** (and optionally equipment) via API or UI; note **site_id**.
2. **Export** — GET /data-model/export?site_id=YourSiteName (or no filter for full dump).
3. **Upload to LLM** — Paste (a) canonical prompt, (b) export JSON, (c) optional rules (cookbook link or YAML snippets). Optionally include the import JSON Schema so the LLM returns valid payload.
4. **Validate** — Run schema validation or Pydantic validation on the LLM reply so you know it will parse on the backend.
5. **Import** — PUT /data-model/import with the validated JSON.
6. **Run FDD / tests** — Trigger an FDD run or Sparkl (or other) tests as needed for the project.

26.6 See also

- AI-assisted data modeling — Export → tag → import and API contract.
- Technical reference — Full prompt text, PyPI vs repo, LLM tagging workflow.

- Fault rules overview and Expression Rule Cookbook — Rules and rule_input reference.

27 SPARQL cookbook

28 SPARQL cookbook

Open-FDD keeps one **knowledge graph** (Brick + BACnet + platform config) in `config/data_model.ttl`. All queries in this cookbook run **via the REST API only** — use **POST /data-model/sparql** with a JSON body `{"query": "..."}`. Do not query the TTL file directly; the API is the single entry point so the same flow works for UIs, scripts, and tests.

28.1 Data Model Setup vs Data Model Testing

The frontend has two tabs:

Tab	Route	Purpose
Data Model Setup	`/data-model`	Sites, equipment, import/export JSON, view TTL. Build or edit the data model (sites, equipment, points) and paste AI-tagged JSON to import.
Data Model Testing	`/data-model-testing`	**Summarize your HVAC** (one-click SPARQL buttons: Sites, AHUs, Zones, Building, VAV boxes, etc.) and **Custom SPARQL** (textarea, Run, upload .sparql file). Same queries as this cookbook.

Use **Data Model Testing** to run SPARQL in the browser. Use **Data Model Setup** to manage sites, equipment, and import.

28.2 How to run SPARQL

Method	Use
Data Model Testing (UI)	Open **Data Model Testing** → click a predefined button (Sites, AHUs, Zones, ...) or paste SPARQL in Custom SPARQL → Run SPARQL. Optional: check "Include BACnet device and point IDs" for `bacnet_device_id` / `object_identifier` in results.
Swagger	http://localhost:8000/docs → **POST /data-model/sparql** → body `{"query": "SELECT ..."}` → Execute.
curl	`curl -X POST http://localhost:8000/data-model/sparql -H "Content-Type: application/json" -d '{"query":"SELECT ?s ?p ?o WHERE { ?s ?p ?o } LIMIT 10"}'`
Upload .sparql file	**Data Model Testing** → Custom SPARQL → "Upload .sparql file", then Run; or **POST /data-model/sparql/upload** with a `.sparql` file.

Response: JSON with a `bindings` array; each element is a map of variable names to values (e.g. `{"site_label": {"value": "DemoSite"}}`). Use the `value` field for literals and the full object for URIs.

28.3 Prefixes and graph contents

The graph uses these namespaces. Include them in your queries:

Prefix	Namespace
`brick:`	https://brickschema.org/schema/Brick#
`rdfs:`	http://www.w3.org/2000/01/rdf-schema#
`ofdd:`	http://openfdd.local/ontology#
`bacnet:`	http://data.ashrae.org/bacnet/2020#
`:` (default)	http://openfdd.local/site#

Graph contents: Brick triples (sites, equipment, points from the DB), BACnet triples (devices and objects from discovery), and **ofdd:PlatformConfig** (platform config from GET/PUT /config). All are queryable together.

28.4 Recipe 1: Platform config

Platform config is stored as RDF (same graph). Use it to verify or audit settings after PUT /config or bootstrap.

All config triples:

```
PREFIX ofdd: <http://openfdd.local/ontology#>
SELECT ?p ?v WHERE {
  ?c a ofdd:PlatformConfig .
  ?c ?p ?v .
}
```

Example bindings: `ofdd:ruleIntervalHours`, `ofdd:bacnetServerUrl`, `ofdd:bacnetEnabled`, etc. Predicates are camelCase (e.g. `ruleIntervalHours`). Use this to confirm config is in the graph after seeding.

28.5 Recipe 2: Sites and counts

All sites with labels:

```
PREFIX brick: <https://brickschema.org/schema/Brick#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT ?site ?site_label WHERE {
```

```
  ?site a brick:Site .
  ?site rdfs:label ?site_label .
}
```

Site and aggregate counts (equipment count, point count per site):

```
PREFIX brick: <https://brickschema.org/schema/Brick#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT ?site_label (COUNT(DISTINCT ?equipment) AS ?
equipment_count) (COUNT(?point) AS ?point_count) WHERE {
  ?site a brick:Site .
  ?site rdfs:label ?site_label .
  ?equipment brick:isPartOf ?site .
  ?point brick:isPointOf ?equipment .
}
GROUP BY ?site ?site_label
```

Use this to validate the data model after CRUD or import (see Getting started).

28.6 Recipe 3: Equipment and points by site

Equipment labels for a given site (replace `"YourSiteLabel"` with your site name):

```
PREFIX brick: <https://brickschema.org/schema/Brick#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT ?eq_label WHERE {
  ?eq brick:isPartOf ?site .
  ?site rdfs:label "YourSiteLabel" .
  ?eq rdfs:label ?eq_label .
}
```

Point labels for a site (points under equipment under site):

```
PREFIX brick: <https://brickschema.org/schema/Brick#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT ?pt_label WHERE {
  ?pt brick:isPointOf ?eq .
  ?eq brick:isPartOf ?site .
  ?site rdfs:label "YourSiteLabel" .
  ?pt rdfs:label ?pt_label .
}
```

Equipment feeds (Brick `feeds` relationships for a site):

```
PREFIX brick: <https://brickschema.org/schema/Brick#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT ?eq_label ?feeds_label WHERE {
```

```
  ?eq brick:isPartOf ?site .
  ?site rdfs:label "YourSiteLabel" .
  ?eq rdfs:label ?eq_label .
  ?eq brick:feeds ?other .
  ?other rdfs:label ?feeds_label .
}
```

28.7 Recipe 4: BACnet devices and objects

After **POST /bacnet/point_discovery_to_graph**, the graph contains BACnet devices and their objects.

All BACnet devices:

```
PREFIX bacnet: <http://data.ashrae.org/bacnet/2020#>
SELECT ?dev WHERE { ?dev a bacnet:Device }
```

Object count per device:

```
PREFIX bacnet: <http://data.ashrae.org/bacnet/2020#>
SELECT ?dev (COUNT(?obj) AS ?n) WHERE {
  ?dev a bacnet:Device .
  ?dev bacnet:contains ?obj .
}
GROUP BY ?dev
```

Object names for a device (e.g. device instance 3456789):

```
PREFIX bacnet: <http://data.ashrae.org/bacnet/2020#>
SELECT ?name WHERE {
  ?dev a bacnet:Device ;
       bacnet:device-instance 3456789 ;
       bacnet:contains ?obj .
  ?obj bacnet:object-name ?name .
}
```

Use these to confirm discovery results are in the graph and to cross-check with GET /data-model/export.

BACnet device and object addresses for points with polling = true (join Brick points to BACnet via matching label and object-name):

```
PREFIX bacnet: <http://data.ashrae.org/bacnet/2020#>
PREFIX ofdd: <http://openfdd.local/ontology#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

SELECT ?device_instance ?device_address ?object_identifier ?
point_label
WHERE {
```

```
  ?point ofdd:polling true ;
         rdfs:label ?point_label .
  ?bacnet_obj bacnet:object-name ?point_label ;
              bacnet:object-identifier ?object_identifier .
  ?device bacnet:contains ?bacnet_obj ;
          bacnet:device-instance ?device_instance ;
          bacnet:device-address ?device_address .
}
ORDER BY ?device_instance ?object_identifier
```

Returns one row per polling point that has a matching BACnet object: `device_instance`, `device_address`, `object_identifier` (e.g. `analog-input, 1`), and `point_label`. Use this for scraper config or to list "which BACnet addresses are in the data model and polled."

Why "No bindings (empty result)"? The query above joins **Brick** points to **BACnet** objects. If your graph has no BACnet triples (no discovery has been run, or you only have Brick from a TTL file), there are no `bacnet:Device`, `bacnet:contains`, or `bacnet:object-name` triples, so the join returns nothing. Use the query below when you only have Brick data.

Polling points with a Brick type (no BACnet) — use this when the graph has no BACnet data; it returns points that have `ofdd:polling true` and a Brick class (e.g. `Return_Air_Temperature_Sensor`):

```
PREFIX brick: <https://brickschema.org/schema/Brick#>
PREFIX ofdd: <http://openfdd.local/ontology#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

SELECT ?brick_class ?point_label ?unit
WHERE {
  ?point ofdd:polling true ;
         rdfs:label ?point_label ;
         a ?brick_type .
  FILTER(STRSTARTS(STR(?brick_type), STR(brick:)))
  BIND(REPLACE(STR(?brick_type), "https://brickschema.org/schema/
Brick#", "") AS ?brick_class)
  OPTIONAL { ?point ofdd:unit ?unit . }
}
ORDER BY ?point_label
```

Data Model Setup: Equipment table vs "View full data model (TTL)" — The **Equipment** table on **Data Model Setup** comes from the **database** (REST API: sites, equipment, points). **Data Model Testing** (SPARQL) and **"View full data model (TTL)"** on Setup use the **in-memory graph**, which is synced from the DB (Brick) plus BACnet discovery. If you see a full TTL with equipment but "No equipment configured" in the table, the DB may be empty or out of sync: the graph

can be loaded from `config/data_model.ttl` on startup, but the UI list is always from the API/DB. Use the site selector in the top bar ("All sites" vs a specific site); ensure sites and equipment exist in the DB (e.g. create via the UI or import).

28.8 Recipe 5: FDD — points and rule mapping

FDD rules resolve Brick classes to DataFrame columns via the TTL. These queries help you validate that points have the right Brick types and rule inputs for your rules (e.g. sensor_bounds or expression rules).

Brick class → label → rule input (timeseries reference and FDD input name):

```
PREFIX brick: <https://brickschema.org/schema/Brick#>
PREFIX ofdd: <http://openfdd.local/ontology#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

SELECT ?brick_class ?label ?rule_input WHERE {
  ?point a ?brick_type .
  FILTER(STRSTARTS(STR(?brick_type), STR(brick:)))
  BIND(REPLACE(STR(?brick_type), "https://brickschema.org/schema/
Brick#", "") AS ?brick_class)
  ?point rdfs:label ?label .
  OPTIONAL { ?point ofdd:mapsToRuleInput ?rule_input . }
}
ORDER BY ?brick_class ?rule_input
LIMIT 50
```

- **`rdfs:label`** = `external_id` in the DB = timeseries key (column name in the FDD DataFrame).
- **`ofdd:mapsToRuleInput`** = `fdd_input` in the DB = name the rule uses when multiple points share a Brick class.

Brick classes used in the model (count per class):

```
PREFIX brick: <https://brickschema.org/schema/Brick#>
SELECT ?brick_class (COUNT(?point) AS ?count) WHERE {
  ?point a ?brick_type .
  FILTER(STRSTARTS(STR(?brick_type), STR(brick:)))
  BIND(REPLACE(STR(?brick_type), "https://brickschema.org/schema/
Brick#", "") AS ?brick_class)
}
GROUP BY ?brick_class
ORDER BY DESC(?count)
```

Use this to ensure every Brick class your rules reference (e.g. `Supply_Air_Temperature_Sensor`, `Zone_Temperature_Sensor`) has at least one point in the graph.

28.9 Recipe 6: FDD — time-series references for rules

Rules in `stack/rules/` (e.g. `sensor_bounds.yaml`, expression rules) declare **inputs** by Brick class only. The runner uses the Brick TTL (this graph) to resolve:

- **Brick class** → points of that type → **rdfs:label** (external_id) → column in the timeseries DataFrame.

So the **timeseries reference** for FDD is the point's **rdfs:label** (which equals `external_id` in the DB and is the key in `timeseries_readings` when joined via `point_id`). To list "which columns will this rule see for site X?":

Points that map to rule inputs for a site (equipment hierarchy + label = timeseries key):

```
PREFIX brick: <https://brickschema.org/schema/Brick#>
PREFIX ofdd: <http://openfdd.local/ontology#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

SELECT ?equipment_label ?point_class ?label ?rule_input WHERE {
  ?site a brick:Site .
  ?site rdfs:label "YourSiteLabel" .
  ?equipment brick:isPartOf ?site .
  ?equipment rdfs:label ?equipment_label .
  ?point brick:isPointOf ?equipment .
  ?point a ?brick_type .
  FILTER(STRSTARTS(STR(?brick_type), STR(brick:)))
  BIND(REPLACE(STR(?brick_type), "https://brickschema.org/schema/
Brick#", "") AS ?point_class)
  ?point rdfs:label ?label .
  OPTIONAL { ?point ofdd:mapsToRuleInput ?rule_input . }
}
ORDER BY ?equipment_label ?point_class
```

Interpretation: For each row, **`label`** is the timeseries column name (external_id). The rule runner loads timeseries by (site, external_id) and builds the DataFrame; then Brick class (and optional rule_input)

select which column is used for each rule input. So this query answers "what timeseries refs (columns) does the FDD pipeline have for this site?"

Polling points only (points that the BACnet scraper writes to; exclude non-polling):

```
PREFIX brick: <https://brickschema.org/schema/Brick#>
PREFIX ofdd: <http://openfdd.local/ontology#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT ?label ?brick_class WHERE {
  ?pt a ?brick_type .
  FILTER(STRSTARTS(STR(?brick_type), STR(brick:)))
  BIND(REPLACE(STR(?brick_type), "https://brickschema.org/schema/
Brick#", "") AS ?brick_class)
  ?pt rdfs:label ?label .
  ?pt ofdd:polling true .
}
ORDER BY ?brick_class ?label
```

These queries mirror the logic in `scripts/automated_testing/sparql/` (e.g. 05_brick_rule_mapping.sparql); run them via the API for validation or for building a small "FDD readiness" report.

28.10 Recipe 7: Integrity and completeness

Graph check — Use **GET /data-model/check** for triple counts, sites count, BACnet device count, and orphan warnings. For custom checks, use SPARQL:

Count triples:

```
SELECT (COUNT(*) AS ?n) WHERE { ?s ?p ?o }
```

Equipment hierarchy snapshot (site → equipment type → point class → rule input):

```
PREFIX brick: <https://brickschema.org/schema/Brick#>
PREFIX ofdd: <http://openfdd.local/ontology#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT ?site_label ?equipment_label ?equipment_type ?point_class ?
rule_input WHERE {
  ?site a brick:Site .
  ?site rdfs:label ?site_label .
  ?equipment brick:isPartOf ?site .
  ?equipment rdfs:label ?equipment_label .
  ?equipment ofdd:equipmentType ?equipment_type .
  ?point brick:isPointOf ?equipment .
```

```
  ?point a ?brick_type .
  FILTER(STRSTARTS(STR(?brick_type), STR(brick:)))
  BIND(REPLACE(STR(?brick_type), "https://brickschema.org/schema/
Brick#", "") AS ?point_class)
  OPTIONAL { ?point ofdd:mapsToRuleInput ?rule_input . }
}
ORDER BY ?site_label ?equipment_label ?point_class
LIMIT 100
```

28.11 Predefined buttons (Data Model Testing)

The **Data Model Testing** tab has one-click buttons under **Summarize your HVAC** that run the same SPARQL as this cookbook.

Button	Recipe
Sites	Recipe 2 (sites with labels)
AHUs, Zones, Building, VAV boxes, VAVs per AHU	Count or list by Brick type
Chillers, Cooling towers, Boilers, Central plant	Count or list equipment types
HVAC equipment, Meters, Points, Class summary	Counts and Brick class summary

Checking **Include BACnet device and point IDs** on Data Model Testing runs a variant of each query that adds `bacnet_device_id` and `object_identifier` (for telemetry and algorithms).

28.12 Saved queries

The recipes in this cookbook can be saved as `.sparql` files and run via **POST /data-model/sparql**, **POST /data-model/sparql/upload**, or **Data Model Testing** (Custom SPARQL → upload or paste). Recipe 4 covers BACnet devices, polling points with Brick type, and polling points with optional BACnet join.

scripts/automated_testing/sparql/ — `.sparql` files for FDD-oriented validation (e.g. brick rule mapping, site counts). Run via **POST /data-model/sparql/upload** or **Data Model Testing** (Custom SPARQL).

28.13 Related docs

Topic	Page
Data model flow	Overview
AI-assisted tagging	AI-assisted tagging
Rule inputs and Brick	Expression rule cookbook · Fault rules overview
Grafana (SQL, not SPARQL)	Grafana SQL cookbook
Dev reference	Technical reference

29 Data modeling

30 Data modeling

The Open-FDD data model is built around **sites**, **equipment**, and **points**: the same entities you create, read, update, and delete via the API (**CRUD** = Create, Read, Update, Delete). The REST API (Swagger at `/docs`) and the data-model endpoints (`/data-model/export`, `/data-model/import`, `/data-model/ttl`) are the main way to manage this data. The database is the single source of truth; the Brick TTL file is regenerated on every change so FDD rules and Grafana stay in sync.

Concepts (entities):

- **Sites** — Buildings or facilities. All equipment and points are scoped to a site.
- **Equipment** — Physical devices (AHUs, VAVs, heat pumps). Belong to a site; have points.
- **Points** — Time-series references (sensors, setpoints). Link to equipment and site; store `external_id`, optional `brick_type` and `rule_input` for FDD.
- **SPARQL cookbook** — Run SPARQL via POST /data-model/sparql only: config, data model, BACnet, FDD rule mapping, time-series references. Copy-paste queries for validation and UIs.
- **AI-assisted data modeling** — Export → LLM or human tagging → import (Brick types, rule_input, polling, equipment feeds). Optional **in-house agent** (Data Model Setup → OpenAI API Assist): uses the same API with retry and **prompt chaining** on validation failure;

prompt from `config/canonical_llm_prompt.txt`. See Technical reference for the full prompt and schema.
- **LLM workflow (export + rules + validate → import)** — One-shot upload: canonical prompt, export JSON, optional rules (cookbook or YAML); validate with schema or Pydantic so import parses on the backend; then PUT /data-model/import and run FDD/tests.

Framework: CRUD is provided by the FastAPI REST API. The data-model API adds bulk export/import and Brick TTL generation; see Overview for the flow (DB → TTL → FDD column_map) and Appendix: API Reference for endpoints.

31 Overview

32 Fault rules

Fault rules are YAML-defined checks run against time-series DataFrames. Each rule produces a boolean fault flag. **Open-FDD is 100% Brick-model driven:** rule inputs refer only to Brick classes; column mapping comes from the Brick TTL via SPARQL and Brick external timeseries references (timeseries storage), which Open-FDD embraces at the heart of FDD.

32.1 Where rules live: config path and how to manage them

Single directory: All FDD rule YAML files live in one directory configured as **`rules_dir`** in platform config (RDF: `ofdd:rulesDir`, e.g. `"stack/rules"`). The FDD loop loads from this path every run; the rules API (list, upload, download, delete) uses the same path. **You still need `rules_dir` in config** — it defines where files are stored. The frontend does not replace it; it lets you manage the files *in* that directory.

Two ways to manage rules:

Method	Use case
React frontend (Faults page)	Upload new YAML, download existing files, delete files, and **Sync definitions** so the fault_definitions table updates without waiting for the next FDD run. Preferred when you have UI access.
Files on disk	Edit or add files directly under the configured path (e.g. `stack/rules/` on the host or in the container). Same outcome: next FDD run (or **Sync definitions** from the UI) picks them up.

Config: `rules_dir: "stack/rules"` (GET/PUT `/config` or `OFDD_RULES_DIR` at bootstrap). If that path does not exist, the loop falls back to `stack/rules`. Default rules ship in `stack/rules/` (e.g. `sensor_bounds.yaml`, `sensor_flatline.yaml`). See the Expression Rule Cookbook to add or adapt rules.

32.2 Hot reload (edit → run → view)

Open-FDD is **AFDD** (Automated Fault Detection and Diagnostics). The project **supports hot reloading of YAML rule files** so the AFDD maintainer can tune faults without restarting. Rule YAML contains fault logic and **params** (thresholds, tolerances, windows); the FDD loop loads from `rules_dir` every run — no restart.

Fault definitions: Each FDD run syncs the loaded rules into the `fault_definitions` table (fault_id, name, category, equipment_types). When you add or edit a rule (via frontend upload or by editing a file in `rules_dir`), the next run updates the DB and the Faults UI reflects the change. From the frontend you can also click **Sync definitions** to update the definitions table immediately.

1. **Add or edit** rules: use the Faults page (upload/paste YAML, or choose file) or edit files in `stack/rules/*.yaml`. Change `params` (e.g. `tolerance`, `rolling_window`) to tune sensitivity.
2. **Run** FDD: wait for the next scheduled run (per `rule_interval_hours` and `lookback_days` in platform config), or trigger with `touch config/.run_fdd_now` or `POST /run-fdd` (see Appendix: API

Reference). Or use **Sync definitions** in the UI to only refresh the definitions table.
3. **View** fault results in the React Faults/Plots views or in Grafana (see Grafana SQL cookbook). Every run reloads all rules from disk — hot reload.

32.3 Rule types

Type	Purpose
`bounds`	Value outside `[low, high]`
`flatline`	Rolling spread < tolerance (stuck sensor)
`hunting`	Excessive state changes (PID hunting)
`expression`	Custom pandas/numpy expression
`oa_fraction`	OA fraction vs design airflow error
`erv_efficiency`	ERV effectiveness out of range

32.4 YAML structure

Inputs use **Brick class names** only; the Brick TTL (SPARQL) resolves each to a DataFrame column via external timeseries references. No `column` in YAML.

```
name: sensor_bounds
type: bounds
flag: bad_sensor
inputs:
  Outside_Air_Temperature_Sensor:
    brick: Outside_Air_Temperature_Sensor
  Supply_Air_Temperature_Sensor:
    brick: Supply_Air_Temperature_Sensor
params:
  low: 40
  high: 90
  rolling_window: 6   # optional: require N consecutive True samples before flagging
```

Per-rule rolling window: Set `params.rolling_window` (e.g. `6`) in a rule to require that many consecutive True samples before the fault is flagged. Omit for "flag on any True." See `sensor_flatline.yaml`, `weather_temp_stuck.yaml`.

32.5 Input resolution (100% Brick)

- **Brick TTL:** Each rule input declares a Brick class (e.g. `brick:Supply_Air_Temperature_Sensor`). The engine resolves Brick points via SPARQL to their external timeseries reference (`ref:TimeseriesReference` / `ref:hasTimeseriesId`), yielding the DataFrame column name. See Brick timeseries storage.
- **Column map:** Built from the Brick model (e.g. `rdfs:label` or external_id) → column names; no `column` in rule YAML.

32.6 Running rules

- **Platform:** FDD loop loads rules from `rules_dir` (default `stack/rules`) each run; edit YAML and trigger a run to see changes in Grafana.
- **Standalone:** `RuleRunner(rules_path=...)` or `RuleRunner(rules=[...])`; call `run(df, ...)`.

32.7 Cookbook

See Expression Rule Cookbook for AHU, chiller, weather, and advanced recipes.

33 Expression Rule Cookbook

34 Expression Rule Cookbook

A reference for building fault detection rules in open-fdd. Rules use **YAML** with **expression** type: when the expression evaluates to **True**, a fault is flagged. open-fdd injects **NumPy** as `np` into expression evaluation, so you can use `np.maximum`, `np.abs`, `np.sqrt`, etc. for vectorized math. **Tuning:** Change `params` in YAML and trigger an FDD run (or wait for the schedule); the platform hot-reloads rules each run, so no restart is needed. See Fault rules overview and Configuration (`rules_dir` is RDF-driven via GET `/config`).

34.1 100% Brick-model driven (no column in rules)

Rule inputs use **Brick class names** only (e.g. `Supply_Air_Temperature_Sensor`). The **Brick TTL** is the source of truth: SPARQL resolves each Brick point to its external timeseries reference (`ref:TimeseriesReference` / `ref:hasTimeseriesId`), which yields the DataFrame column. Open-FDD embraces Brick timeseries storage at the heart of FDD — **rules never reference `column`**; the data model provides the mapping. See the SPARQL cookbook (Recipe 5) and run similar queries via Data Model Testing or `scripts/automated_testing/sparql/`.

- **Platform (DB):** Points have `brick_type` and `external_id`; TTL is built from DB. PATCH points or use data-model import to set `brick_type` (e.g. `Supply_Air_Temperature_Sensor` for SA-T).
- **Disambiguation:** When multiple points share a Brick class, use `ofdd:mapsToRuleInput` in TTL; the runner resolves `BrickClass|rule_input`.

34.2 How to define expressions

1. **Inputs** — Declare Brick classes only (`brick:Supply_Air_Temperature_Sensor`). The runner resolves these via the Brick TTL (SPARQL) and Brick external timeseries references to DataFrame columns. Do not add `column` in YAML.
2. **Params** — Thresholds and constants go in `params`. Reference by name (e.g. `err_thresh`, `vfd_max`).
3. **Expression** — Must evaluate to a boolean Series (True = fault). Use `&` (AND), `|` (OR), `~` (NOT). Use `.diff()`, `.rolling()`, `.notna()` for time-series logic.

Minimal example (TTL-driven; no column):

```yaml
name: high_temp_check
type: expression
flag: high_temp_flag

inputs:
  Supply_Air_Temperature_Sensor:
    brick: Supply_Air_Temperature_Sensor

params:
  max_temp: 90.0
```

```
        expression: |
          Supply_Air_Temperature_Sensor > max_temp
```

The Brick TTL (and SPARQL) maps each Brick class to the actual column via external timeseries references. Set `points.brick_type` in the DB and sync TTL so the engine can resolve columns from the Brick model.

34.3 AHU rules (reference-style)

The following rules follow common industry practice for air-handling fault detection. Rules A through M are adapted from ASHRAE Guideline 36 (GL36) AFDD guidance. Thresholds and logic are tunable; adjust params for your site.

34.3.1 Rule A — Duct static below setpoint at full fan speed

Static pressure under setpoint while supply fan runs near maximum. May indicate duct leakage, undersized fan, or terminal damper issues. *Adapted from GL36 AFDD guidance.*

```
        name: duct_static_low_at_full_speed
        description: Static pressure below setpoint when fan at full speed
(GL36-inspired)
        type: expression
        flag: rule_a_flag
        equipment_type: [VAV_AHU]

        inputs:
          Supply_Air_Static_Pressure_Sensor:
            brick: Supply_Air_Static_Pressure_Sensor
          Supply_Air_Static_Pressure_Setpoint:
            brick: Supply_Air_Static_Pressure_Setpoint
          Supply_Fan_Speed_Command:
            brick: Supply_Fan_Speed_Command

        params:
          sp_margin: 0.12
          drv_hi_frac: 0.93
          drv_near_hi: 0.06

        expression: |
          (Supply_Air_Static_Pressure_Sensor <
```

```
Supply_Air_Static_Pressure_Setpoint - sp_margin) &
(Supply_Fan_Speed_Command >= drv_hi_frac - drv_near_hi)
```

34.3.2 Rule B — Blended air temp below expected band

Blended air temp should lie between outdoor and return. If below both (minus tolerance), suspect sensor or mixing fault. *Adapted from GL36 AFDD guidance.*

```yaml
        name: blend_temp_below_band
        description: Blended air temp below expected range (OAT/RAT)
(GL36-inspired)
        type: expression
        flag: rule_b_flag
        equipment_type: [AHU, VAV_AHU]

        inputs:
          Mixed_Air_Temperature_Sensor:
            brick: Mixed_Air_Temperature_Sensor
          Return_Air_Temperature_Sensor:
            brick: Return_Air_Temperature_Sensor
          Outside_Air_Temperature_Sensor:
            brick: Outside_Air_Temperature_Sensor
          Supply_Fan_Speed_Command:
            brick: Supply_Fan_Speed_Command

        params:
          blend_tol: 1.15
          rat_tol: 1.15
          oat_tol: 1.15

        expression: |
          (Mixed_Air_Temperature_Sensor - blend_tol <
np.minimum(Return_Air_Temperature_Sensor - rat_tol,
Outside_Air_Temperature_Sensor - oat_tol)) &
(Supply_Fan_Speed_Command > 0.01)
```

34.3.3 Rule C — Blended air temp above expected band

Blended air temp above the higher of OAT and RAT (plus tolerance) indicates mixing or sensor fault. *Adapted from GL36 AFDD guidance.*

```yaml
        name: blend_temp_above_band
        description: Blended air temp above expected range (OAT/RAT)
(GL36-inspired)
        type: expression
        flag: rule_c_flag
        equipment_type: [AHU, VAV_AHU]

        inputs:
```

```
          Mixed_Air_Temperature_Sensor:
            brick: Mixed_Air_Temperature_Sensor
          Return_Air_Temperature_Sensor:
            brick: Return_Air_Temperature_Sensor
          Outside_Air_Temperature_Sensor:
            brick: Outside_Air_Temperature_Sensor
          Supply_Fan_Speed_Command:
            brick: Supply_Fan_Speed_Command

        params:
          blend_tol: 1.15
          rat_tol: 1.15
          oat_tol: 1.15

        expression: |
          (Mixed_Air_Temperature_Sensor - blend_tol >
np.maximum(Return_Air_Temperature_Sensor + rat_tol,
Outside_Air_Temperature_Sensor + oat_tol)) &
(Supply_Fan_Speed_Command > 0.01)
```

Hunting/oscillation — see [Hunting Rule]({{ "hunting_rule" | relative_url }}).

34.3.4 Rule D — Discharge air cold when heating commanded

Discharge air temp below blended air when heating valve is open. Indicates heating coil or valve failure. *Adapted from GL36 AFDD guidance.*

```
        name: discharge_cold_when_heating
        description: Discharge air below blended air when heating active
(GL36-inspired)
        type: expression
        flag: rule_d_flag
        equipment_type: [AHU, VAV_AHU]

        inputs:
          Mixed_Air_Temperature_Sensor:
            brick: Mixed_Air_Temperature_Sensor
          Supply_Air_Temperature_Sensor:
            brick: Supply_Air_Temperature_Sensor
          Valve_Command:
            brick: Valve_Command
          Supply_Fan_Speed_Command:
            brick: Supply_Fan_Speed_Command

        params:
          blend_tol: 1.15
          sat_tol: 1.15
```

```
          fan_delta_t: 0.55

        expression: |
          (Supply_Air_Temperature_Sensor + sat_tol <=
Mixed_Air_Temperature_Sensor - blend_tol + fan_delta_t) &
(Valve_Command > 0.01) & (Supply_Fan_Speed_Command > 0.01)
```

OA fraction — see [OA Fraction Rule]({{ "oa_fraction_rule" | relative_url }}).

34.3.5 Rule E — SAT too low with full heating

Heating valve fully open but SAT remains below setpoint. Indicates undersized coil or valve failure. *Adapted from GL36 AFDD guidance.*

```
        name: sat_too_low_full_heating
        description: SAT below setpoint with heating valve fully open
(GL36-inspired)
        type: expression
        flag: rule_e_flag
        equipment_type: [AHU, VAV_AHU]

        inputs:
          Supply_Air_Temperature_Sensor:
            brick: Supply_Air_Temperature_Sensor
          Supply_Air_Temperature_Setpoint:
            brick: Supply_Air_Temperature_Setpoint
          Valve_Command:
            brick: Valve_Command
          Supply_Fan_Speed_Command:
            brick: Supply_Fan_Speed_Command

        params:
          supply_err_thres: 1.0

        expression: |
          (Supply_Air_Temperature_Sensor < Supply_Air_Temperature_Setpoint
- supply_err_thres) & (Valve_Command > 0.9) &
(Supply_Fan_Speed_Command > 0)
```

34.3.6 Rule F — SAT/MAT mismatch in economizer mode

In economizer mode (min mechanical cooling), SAT should approximate MAT. Large deviation suggests coil bypass or sensor error. *Adapted from GL36 AFDD guidance.*

```
        name: discharge_blend_mismatch_econ
        description: Discharge and blended air diverge in economizer mode
(GL36-inspired)
```

```
        type: expression
        flag: rule_f_flag
        equipment_type: [AHU, VAV_AHU]

        inputs:
          Mixed_Air_Temperature_Sensor:
            brick: Mixed_Air_Temperature_Sensor
          Supply_Air_Temperature_Sensor:
            brick: Supply_Air_Temperature_Sensor
          Damper_Position_Command:
            brick: Damper_Position_Command
          Valve_Command:
            brick: Valve_Command

        params:
          fan_delta_t: 0.55
          blend_tol: 1.15
          sat_tol: 1.15
          econ_min_open: 0.12

        expression: |
          (np.abs(Supply_Air_Temperature_Sensor - fan_delta_t -
Mixed_Air_Temperature_Sensor) > np.sqrt(sat_tol**2 +
blend_tol**2)) & (Damper_Position_Command > econ_min_open) &
(Valve_Command < 0.1)
```

34.3.7 Rule G — Ambient too warm for free cooling

Outside air temperature exceeds SAT setpoint while economizer is active and mechanical cooling is off. Economizer should not be providing “free” cooling under these conditions. *Adapted from GL36 AFDD guidance.*

```
        name: ambient_warm_free_cool
        description: Outdoor air above setpoint in free cooling mode
(GL36-inspired)
        type: expression
        flag: rule_g_flag
        equipment_type: [AHU, VAV_AHU]

        inputs:
          Outside_Air_Temperature_Sensor:
            brick: Outside_Air_Temperature_Sensor
          Supply_Air_Temperature_Setpoint:
            brick: Supply_Air_Temperature_Setpoint
          Damper_Position_Command:
            brick: Damper_Position_Command
          Valve_Command:
            brick: Valve_Command
```

```
        params:
          oat_tol: 1.15
          fan_delta_t: 0.55
          sat_tol: 1.15
          econ_min_open: 0.12

        expression: |
          (Outside_Air_Temperature_Sensor - oat_tol >
Supply_Air_Temperature_Setpoint - fan_delta_t + sat_tol) &
(Damper_Position_Command > econ_min_open) & (Valve_Command < 0.1)
```

34.3.8 Rule H — Ambient vs blended mismatch (econ + mech cooling)

When both economizer and mechanical cooling are active, MAT should approach OAT. Large deviation suggests inadequate mixing or damper fault. *Adapted from GL36 AFDD guidance.*

```
        name: ambient_blend_mismatch_econ_mech
        description: Outdoor and blended air diverge in econ+mech cooling
(GL36-inspired)
        type: expression
        flag: rule_h_flag
        equipment_type: [AHU, VAV_AHU]

        inputs:
          Outside_Air_Temperature_Sensor:
            brick: Outside_Air_Temperature_Sensor
          Mixed_Air_Temperature_Sensor:
            brick: Mixed_Air_Temperature_Sensor
          Valve_Command:
            brick: Valve_Command
          Damper_Position_Command:
            brick: Damper_Position_Command

        params:
          oat_tol: 1.15
          blend_tol: 1.15

        expression: |
          (np.abs(Mixed_Air_Temperature_Sensor -
Outside_Air_Temperature_Sensor) > np.sqrt(blend_tol**2 +
oat_tol**2)) & (Valve_Command > 0.01) & (Damper_Position_Command >
0.9)
```

34.3.9 Rule I — Ambient vs blended mismatch (econ-only)

In economizer-only mode, MAT should match OAT. Deviation indicates damper or mixing fault. *Adapted from GL36 AFDD guidance.*

```
        name: ambient_blend_mismatch_econ
        description: Outdoor and blended air diverge in economizer-only
mode (GL36-inspired)
        type: expression
        flag: rule_i_flag
        equipment_type: [AHU, VAV_AHU]

        inputs:
          Outside_Air_Temperature_Sensor:
            brick: Outside_Air_Temperature_Sensor
          Mixed_Air_Temperature_Sensor:
            brick: Mixed_Air_Temperature_Sensor
          Damper_Position_Command:
            brick: Damper_Position_Command

        params:
          oat_tol: 1.15
          blend_tol: 1.15

        expression: |
          (np.abs(Mixed_Air_Temperature_Sensor -
Outside_Air_Temperature_Sensor) > np.sqrt(blend_tol**2 +
oat_tol**2)) & (Damper_Position_Command > 0.9)
```

34.3.10 Rule J — Discharge above blended in cooling

SAT exceeds MAT when cooling (econ+mech or mech-only) is active. Indicates underperforming cooling coil or valve. *Adapted from GL36 AFDD guidance.*

```
        name: discharge_above_blend_cooling
        description: Discharge air above blended air in cooling modes
(GL36-inspired)
        type: expression
        flag: rule_j_flag
        equipment_type: [AHU, VAV_AHU]

        inputs:
          Supply_Air_Temperature_Sensor:
            brick: Supply_Air_Temperature_Sensor
          Mixed_Air_Temperature_Sensor:
            brick: Mixed_Air_Temperature_Sensor
          Valve_Command:
            brick: Valve_Command
          Damper_Position_Command:
```

```
            brick: Damper_Position_Command

        params:
          fan_delta_t: 0.55
          blend_tol: 1.15
          sat_tol: 1.15
          econ_min_open: 0.12

        expression: |
          (Supply_Air_Temperature_Sensor > Mixed_Air_Temperature_Sensor +
np.sqrt(sat_tol**2 + blend_tol**2) + fan_delta_t) &
(((Damper_Position_Command > 0.9) & (Valve_Command > 0)) |
((Damper_Position_Command <= econ_min_open) & (Valve_Command >
0.9)))
```

34.3.11 Rule K — Discharge above setpoint in full cooling

SAT above setpoint with cooling at full capacity. Suggests undersized coil or plant limits. *Adapted from GL36 AFDD guidance.*

```
        name: discharge_above_sp_full_cool
        description: Discharge air above setpoint in full cooling mode
(GL36-inspired)
        type: expression
        flag: rule_k_flag
        equipment_type: [AHU, VAV_AHU]

        inputs:
          Supply_Air_Temperature_Sensor:
            brick: Supply_Air_Temperature_Sensor
          Supply_Air_Temperature_Setpoint:
            brick: Supply_Air_Temperature_Setpoint
          Valve_Command:
            brick: Valve_Command
          Damper_Position_Command:
            brick: Damper_Position_Command

        params:
          sat_tol: 1.15
          econ_min_open: 0.12

        expression: |
          (Supply_Air_Temperature_Sensor > Supply_Air_Temperature_Setpoint
+ sat_tol) & (((Damper_Position_Command > 0.9) & (Valve_Command >
0.9)) | ((Damper_Position_Command <= econ_min_open) &
(Valve_Command > 0.9)))
```

34.3.12 Rule L — Cooling coil delta-T when inactive

Temperature drop across cooling coil when it should be off. Indicates leaking CHW valve or coil bypass. *Adapted from GL36 AFDD guidance.*

```
        name: clg_coil_drop_when_off
        description: Temperature drop across cooling coil when it should
be off (GL36-inspired)
        type: expression
        flag: rule_l_flag
        equipment_type: [AHU, VAV_AHU]

        inputs:
          Cooling_Coil_Entering_Air_Temperature_Sensor:
            brick: Cooling_Coil_Entering_Air_Temperature_Sensor
          Cooling_Coil_Leaving_Air_Temperature_Sensor:
            brick: Cooling_Coil_Leaving_Air_Temperature_Sensor
          Heating_Valve_Command:
            brick: Valve_Command
          Cooling_Valve_Command:
            brick: Valve_Command
          Damper_Position_Command:
            brick: Damper_Position_Command

        params:
          enter_tol: 1.15
          leave_tol: 1.15
          econ_min_open: 0.12

        expression: |
          ((Cooling_Coil_Entering_Air_Temperature_Sensor -
Cooling_Coil_Leaving_Air_Temperature_Sensor) >
np.sqrt(enter_tol**2 + leave_tol**2)) & (((Heating_Valve_Command >
0) & (Cooling_Valve_Command == 0) & (Damper_Position_Command <=
econ_min_open)) | ((Heating_Valve_Command == 0) &
(Cooling_Valve_Command == 0) & (Damper_Position_Command >
econ_min_open)))
```

34.3.13 Rule M — Heating coil delta-T when inactive

Temperature rise across heating coil when it should be off. Indicates leaking HW valve. *Adapted from GL36 AFDD guidance.*

```
        name: htg_coil_rise_when_off
        description: Temperature rise across heating coil when it should
be off (GL36-inspired)
        type: expression
        flag: rule_m_flag
        equipment_type: [AHU, VAV_AHU]
```

```
        inputs:
          Heating_Coil_Entering_Air_Temperature_Sensor:
            brick: Heating_Coil_Entering_Air_Temperature_Sensor
          Heating_Coil_Leaving_Air_Temperature_Sensor:
            brick: Heating_Coil_Leaving_Air_Temperature_Sensor
          Heating_Valve_Command:
            brick: Valve_Command
          Cooling_Valve_Command:
            brick: Valve_Command
          Damper_Position_Command:
            brick: Damper_Position_Command

        params:
          enter_tol: 1.15
          leave_tol: 1.15
          fan_delta_t: 0.55
          econ_min_open: 0.12

        expression: |
          ((Heating_Coil_Leaving_Air_Temperature_Sensor -
Heating_Coil_Entering_Air_Temperature_Sensor) >
np.sqrt(enter_tol**2 + leave_tol**2) + fan_delta_t) &
(((Heating_Valve_Command == 0) & (Cooling_Valve_Command == 0) &
(Damper_Position_Command > econ_min_open)) |
((Heating_Valve_Command == 0) & (Cooling_Valve_Command > 0) &
(Damper_Position_Command > 0.9)) | ((Heating_Valve_Command == 0) &
(Cooling_Valve_Command > 0) & (Damper_Position_Command <=
econ_min_open)))
```

Heat exchanger effectiveness — see [ERV/Heat Exchanger Rule] ({{ “erv_efficiency_rule” | relative_url }}).

34.4 Central plant

34.4.1 Differential pressure at max pump speed

Variable-speed pump cannot meet differential pressure setpoint at full speed. Indicates piping issues, undersized pump, or blocked strainers.

```
        name: dp_below_sp_pump_max
        description: Differential pressure below setpoint with pump at
full speed
        type: expression
        flag: dp_pump_flag

        inputs:
          Differential_Pressure_Sensor:
            brick: Differential_Pressure_Sensor
```

```
          Differential_Pressure_Setpoint:
            brick: Differential_Pressure_Setpoint
          Pump_Speed_Command:
            brick: Pump_Speed_Command

        params:
          dp_margin: 2.2
          pmp_hi_frac: 0.93
          pmp_near_hi: 0.06

        expression: |
          (Differential_Pressure_Sensor < Differential_Pressure_Setpoint -
dp_margin) & (Pump_Speed_Command >= pmp_hi_frac - pmp_near_hi)
```

34.4.2 Plant flow high at max pump

Flow unusually high with pump at high speed. Suggests short circuit or flow meter error.

```
        name: flow_high_pump_max
        description: Water flow unusually high with pump at full speed
        type: expression
        flag: flow_high_flag

        inputs:
          Water_Flow_Sensor:
            brick: Water_Flow_Sensor
          Pump_Speed_Command:
            brick: Pump_Speed_Command

        params:
          flow_hi_limit: 1100.0
          pmp_hi_frac: 0.93
          pmp_near_hi: 0.06

        expression: |
          (Water_Flow_Sensor > flow_hi_limit) & (Pump_Speed_Command >=
pmp_hi_frac - pmp_near_hi)
```

34.4.3 Plant supply temp outside deadband

CHW supply temperature outside deadband while pump runs.
Requires coil entering/leaving or plant supply temp sensors.

```
        name: plant_supply_temp_deadband
        description: Supply water temp outside deadband during pump
operation
        type: expression
        flag: chw_temp_fault
```

```
        inputs:
          Chilled_Water_Supply_Temperature_Sensor:
            brick: Chilled_Water_Supply_Temperature_Sensor
          Chilled_Water_Supply_Temperature_Setpoint:
            brick: Chilled_Water_Supply_Temperature_Setpoint
          Pump_Speed_Command:
            brick: Pump_Speed_Command

        params:
          sp_band: 2.2

        expression: |
          (Pump_Speed_Command > 0.01) &
((Chilled_Water_Supply_Temperature_Sensor <
Chilled_Water_Supply_Temperature_Setpoint - sp_band) |
(Chilled_Water_Supply_Temperature_Sensor >
Chilled_Water_Supply_Temperature_Setpoint + sp_band))
```

34.4.4 Chiller runtime over daily limit

Chiller running beyond a daily threshold (e.g. 23 hours). Often indicates over-cooling or schedules that bypass lockout. Use a rolling window sized for your data interval (e.g. 5-min data → 276 samples ≈ 23 h).

```
        name: chiller_excessive_runtime
        description: Chiller runtime exceeds daily threshold (rolling
window)
        type: expression
        flag: chiller_runtime_fault

        inputs:
          Chiller_Status:
            brick: Chiller_Status

        params:
          # For 5-min data: 23 hours ≈ 276 samples; max_runtime = count of
"on" samples in window
          rolling_samples: 276
          max_runtime_samples: 264

        expression: |
          Chiller_Status.rolling(window=rolling_samples).sum() >
max_runtime_samples
```

Note: Adjust `rolling_samples` and `max_runtime_samples` for your data interval. For 5-min data, 264 samples ≈ 22 hours.

34.5 Heat pumps

34.5.1 Discharge cold when heating

Flags when discharge air temperature is below a minimum (e.g. 80°F) while the supply fan is running. Indicates the heat pump is not heating effectively—possible issues with compressor, refrigerant, or reversing valve. Tunable via `min_discharge_temp`. Logic: if zone temp < 69°F (heating mode), the discharge should be warm; a cold discharge with the fan on and a cold zone indicates the heat pump is failing to heat.

```
        name: hp_discharge_cold_when_heating
        description: Discharge air below minimum when fan on and zone is
cold (heating mode)
        type: expression
        flag: hp_discharge_cold_flag
        equipment_type: [Heat_Pump]

        inputs:
          Supply_Air_Temperature_Sensor:
            brick: Supply_Air_Temperature_Sensor
          Zone_Temperature_Sensor:
            brick: Zone_Temperature_Sensor
          Supply_Fan_Status:
            brick: Supply_Fan_Status

        params:
          min_discharge_temp: 85
          zone_cold_threshold: 69.0
          fan_on_threshold: 0.01

        expression: |
          (Supply_Fan_Status > fan_on_threshold) &
(Zone_Temperature_Sensor < zone_cold_threshold) &
(Supply_Air_Temperature_Sensor < min_discharge_temp)
```

34.6 VAV zones

34.6.1 Excessive heating during warm weather

Reheat valve open when outdoor air is warm. Suggests over-cooling or setpoint issues.

```
        name: zone_reheat_warm_ambient
        description: Heating valve open when OAT is high
```

```yaml
        type: expression
        flag: excessive_heating_flag
        equipment_type: [VAV]

        inputs:
          Outside_Air_Temperature_Sensor:
            brick: Outside_Air_Temperature_Sensor
          Reheat_Valve_Command:
            brick: Valve_Command

        params:
          t_amb_cutoff: 78.0
          reheat_open_min: 0.52

        expression: |
          (Outside_Air_Temperature_Sensor > t_amb_cutoff) &
(Reheat_Valve_Command > reheat_open_min)
```

34.6.2 Damper or valve at full open

Damper or reheat valve consistently at full open. Indicates override, undersized equipment, or control fault.

```yaml
        name: zone_damper_valve_full_open
        description: Damper or valve at full open for extended period
        type: expression
        flag: damper_100_flag
        equipment_type: [VAV]

        inputs:
          Damper_Position_Command:
            brick: Damper_Position_Command

        params:
          full_open_pct: 97.5
          roll_samples: 105

        expression: |
          (Damper_Position_Command > full_open_pct) &
(Damper_Position_Command.rolling(roll_samples).min() >
full_open_pct)
```

34.6.3 Zone and IAQ bounds

For CO2 and zone temperature out-of-range checks, use the [Bounds Rule]({{ “bounds_rule” | relative_url }}) — `co2_bounds` and `zone_temp_bounds` examples.

34.7 Opportunistic rules (economizer & ventilation)

34.7.1 Economizing when outdoor conditions are unfavorable

Economizer active when outdoor air is too warm or humid. Use OAT (or enthalpy if available) vs. threshold.

```
        name: econ_active_warm_ambient
        description: OA damper open when outdoor conditions do not favor
economizing
        type: expression
        flag: econ_when_shouldnt_flag
        equipment_type: [AHU, VAV_AHU]

        inputs:
          Outside_Air_Temperature_Sensor:
            brick: Outside_Air_Temperature_Sensor
          Damper_Position_Command:
            brick: Damper_Position_Command

        params:
          t_amb_econ_cutoff: 63.0
          dpr_econ_min: 0.42

        expression: |
          (Outside_Air_Temperature_Sensor > t_amb_econ_cutoff) &
(Damper_Position_Command > dpr_econ_min)
```

34.7.2 Mechanical cooling when econ could suffice

Cooling valve open when outdoor air is cool enough for economizer. Opportunity to reduce mechanical cooling.

```
        name: mech_cool_when_econ_available
        description: Mechanical cooling active when economizing could
suffice
        type: expression
        flag: cooling_when_econ_flag
        equipment_type: [AHU, VAV_AHU]

        inputs:
          Outside_Air_Temperature_Sensor:
            brick: Outside_Air_Temperature_Sensor
          Damper_Position_Command:
            brick: Damper_Position_Command
          Valve_Command:
```

```
            brick: Valve_Command

        params:
          t_amb_econ_cutoff: 63.0
          dpr_not_econ_max: 0.32

        expression: |
          (Outside_Air_Temperature_Sensor < t_amb_econ_cutoff) &
(Damper_Position_Command < dpr_not_econ_max) & (Valve_Command >
0.01)
```

34.7.3 Low ventilation (estimated OA fraction)

For units without airflow meters, estimate OA fraction from OAT, MAT, RAT. Flag when below minimum design OA.

```
        name: low_oa_fraction_estimated
        description: Estimated OA fraction below minimum (OAT, MAT, RAT
method)
        type: expression
        flag: low_vent_flag
        equipment_type: [AHU, VAV_AHU]

        inputs:
          Mixed_Air_Temperature_Sensor:
            brick: Mixed_Air_Temperature_Sensor
          Return_Air_Temperature_Sensor:
            brick: Return_Air_Temperature_Sensor
          Outside_Air_Temperature_Sensor:
            brick: Outside_Air_Temperature_Sensor
          Supply_Fan_Speed_Command:
            brick: Supply_Fan_Speed_Command

        params:
          oa_min_pct: 21.0
          t_rat_oat_min_gap: 2.2

        expression: |
          (Supply_Fan_Speed_Command > 0.01) &
(np.abs(Return_Air_Temperature_Sensor -
Outside_Air_Temperature_Sensor) > t_rat_oat_min_gap) &
(((Mixed_Air_Temperature_Sensor - Return_Air_Temperature_Sensor) /
(Outside_Air_Temperature_Sensor - Return_Air_Temperature_Sensor) *
100) < oa_min_pct)
```

Note: Guard against division-by-zero when outdoor and return temps are close.

34.7.4 Preheat over-conditioning

Preheat coil leaving temp higher than needed (e.g. above OAT when OAT > SAT SP, or above SAT SP when OAT < SAT SP). Indicates wasted heating energy.

```yaml
name: preheat_excess_temp
description: Preheat coil leaving temp above required level
type: expression
flag: preheat_waste_flag
equipment_type: [AHU, VAV_AHU]

inputs:
  Preheat_Coil_Leaving_Air_Temperature_Sensor:
    brick: Preheat_Coil_Leaving_Air_Temperature_Sensor
  Supply_Air_Temperature_Setpoint:
    brick: Supply_Air_Temperature_Setpoint
  Outside_Air_Temperature_Sensor:
    brick: Outside_Air_Temperature_Sensor
  Valve_Command:
    brick: Valve_Command

params:
  excess_tol: 2.2

expression: |
  (Valve_Command > 0.01) & (((Outside_Air_Temperature_Sensor >
Supply_Air_Temperature_Setpoint) &
(Preheat_Coil_Leaving_Air_Temperature_Sensor -
Outside_Air_Temperature_Sensor > excess_tol)) |
((Outside_Air_Temperature_Sensor <
Supply_Air_Temperature_Setpoint) &
(Preheat_Coil_Leaving_Air_Temperature_Sensor -
Supply_Air_Temperature_Setpoint > excess_tol)))
```

34.7.5 Blended air damper deviation

Expected MAT (from OAT, RAT, damper positions) differs from measured MAT. Indicates damper leakage or faulty mixing.

Requires airflow or damper position data; logic is more involved. A simplified version compares MAT to a weighted blend of OAT and RAT when dampers suggest significant OA.

34.8 Weather station

weather_temp_stuck (flatline) — see [Flatline Rule]({{ “flatline_rule” | relative_url }}).

34.8.1 Unrealistic temperature spike

Temperature change between consecutive readings exceeds physical limit.

```yaml
name: weather_temp_spike
description: Unrealistic temperature change between readings
type: expression
flag: fault_temp_spike

inputs:
  Outside_Air_Temperature_Sensor:
    brick: Outside_Air_Temperature_Sensor

params:
  spike_limit: 16.0

expression: |
  Outside_Air_Temperature_Sensor.diff().abs() > spike_limit
```

34.8.2 RH bounds

For relative humidity out-of-range, use the [Bounds Rule] ({{ “bounds_rule” | relative_url }}) — `rh_bounds` example.

34.8.3 Wind gust vs sustained

```yaml
name: weather_gust_lt_wind
description: Wind gust reported lower than sustained wind
type: expression
flag: fault_gust_lt_wind

inputs:
  Wind_Gust_Speed_Sensor:
    brick: Wind_Gust_Speed_Sensor
  Wind_Speed_Sensor:
    brick: Wind_Speed_Sensor

expression: |
  Wind_Gust_Speed_Sensor.notna() & Wind_Speed_Sensor.notna() &
(Wind_Gust_Speed_Sensor < Wind_Speed_Sensor)
```

34.9 Sensor validation (bounds & flatline)

Use the [Bounds Rule]({{ “bounds_rule” | relative_url }}) and [Flatline Rule]({{ “flatline_rule” | relative_url }}) for generic sensor checks. Typical bounds:

Sensor type	Min	Max
Zone temp	40	100
Supply air temp	40	150
Air pressure (inH2O)	-5	10
RH	0	100
Chilled water temp	35	100
Hot water temp	50	212
Condenser water	50	110
CO2 (ppm)	400	2000

Next: [Flat Line Sensor Tutorial]({{ “flat_line_sensor_tutorial” | relative_url }})

35 Fault rules for HVAC

36 Fault rules for HVAC

FDD fault rule types, YAML format, and expression cookbook for HVAC and building systems. Rules are **100% Brick-model driven**: inputs reference Brick classes only; column mapping comes from the Brick TTL and Brick external timeseries references (timeseries storage), which Open-FDD embraces at the heart of FDD.

37 Cloud export example

38 Cloud export example

Open-FDD runs **inside the building, behind the firewall**. It does not push data to the cloud itself. Cloud-based FDD providers, MSI (Monitoring, Savings, and Intelligence) vendors, and commissioning or IoT contractors run their **own** export process: a small script or edge gateway on the building or OT network that **pulls** from the Open-FDD API over the LAN and then sends that data to their cloud or analytics platform. See Behind the firewall; cloud export is vendor-led on the docs home.

This page describes the **cloud export example** in the repo: a minimal Python script that demonstrates how vendor X, Y, or Z could use the Open-FDD API to get fault and timeseries data and use it as a starting point for their cloud pipeline.

38.1 What the example does

The script `examples/cloud_export.py` calls standard Open-FDD download and analytics endpoints and prints the results. In a real integration you would replace the print steps with your own logic: POST to your REST API, write to S3, push to Azure IoT Hub or AWS IoT, send to SkySpark or another analytics platform, etc.

Step	Endpoint	Purpose
1	`GET /download/faults?format=json`	Fault results for programmatic ingestion (JSON)
2	`GET /download/faults?format=csv`	Fault CSV (Excel-friendly)
3	`GET /analytics/motor-runtime`	Motor runtime (data-model driven)
4	`GET /download/csv`	Timeseries wide-format CSV
5		Fault counts by fault_id

Step	Endpoint	Purpose
	`GET /analytics/fault-summary`	

All requests use date range and optional `site_id`; the API docs at `/docs` list full parameters.

38.2 Run the example

Prerequisites: Open-FDD API running (e.g. `http://localhost:8000`), `pip install httpx`.

```
# From repo root
python examples/cloud_export.py
python examples/cloud_export.py --site default --days 14
API_BASE=http://your-openfdd-host:8000 python examples/cloud_export.py
```

Docker: See `examples/cloud_export/README.md` for building and running the example in a container (e.g. on a vendor edge device that reaches the Open-FDD API over the LAN).

38.3 Use it as your starting point

1. **Clone or copy** `examples/cloud_export.py` into your integration codebase.
2. **Keep** the HTTP calls to `/download/faults`, `/download/csv`, and the analytics endpoints; adjust `API_BASE` (or env) to point at the customer's Open-FDD instance.
3. **Replace** the `print()` / local handling with your cloud send: REST POST, message queue, blob storage, or IoT hub. Add auth (e.g. API key or TLS client cert) if you expose the puller to a wider network.
4. **Schedule** the script (cron, systemd timer, or container restart policy) so you poll the Open-FDD API at the interval that fits your product (e.g. hourly or daily fault sync).

Open-FDD does not initiate outbound cloud connections or manage your data transmission; the example shows how your process can pull data from Open-FDD and then send it to your cloud for deeper insights.

39 Concepts

40 Concepts

Conceptual guides and worked examples for how Open-FDD fits into larger workflows.

Page	Description
Cloud export example	How a cloud vendor or MSI can pull fault and timeseries data from the Open-FDD API and send it to their cloud for deeper insights. Uses the `examples/cloud_export` script as a starting point.

See also Behind the firewall; cloud export is vendor-led on the docs home.

41 How-to Guides

42 How-to Guides

- Grafana dashboards (optional) — Optional Grafana; React frontend provides equivalent timeseries and fault views. Datasource and dashboard JSON.
- Grafana SQL cookbook — SQL recipes for BACnet, faults, weather, system resources (variables, panels, sparklines).

43 Grafana SQL cookbook

44 Grafana SQL cookbook —

44.1 BACnet

- Dashboard Variables (Required for Dropdowns)

Before creating the panels below, you must set up the dashboard variables so the drop-down menus work.

In Grafana, go to **Dashboard Settings (gear icon)** > **Variables** > **Add variable**. Set the **Type** to `Query`, select your TimescaleDB datasource, and use the following queries:

44.1.1 1. Site Variable

- **Name:** `site`
- **Query:**

```
SELECT s.name AS __text, s.id::text AS __value
FROM sites s
ORDER BY s.name;
```

44.1.2 2. Device Variable

- **Name:** `device`
- **Query:**

```
SELECT DISTINCT p.bacnet_device_id::text AS __text,
p.bacnet_device_id::text AS __value
FROM points p
WHERE p.site_id::text = '$site'
  AND p.bacnet_device_id IS NOT NULL
ORDER BY 1;
```

44.1.3 3. Point Variable

- **Name:** `point`
- **Query:**

```sql
SELECT p.external_id AS __text, p.external_id AS __value
FROM points p
WHERE p.site_id::text = '$site'
  AND p.bacnet_device_id::text = '$device'
ORDER BY 1;
```

44.1.4 BACnet only data

Insert Snip here

- This dashboard uses `point`, `device`, and `site` variables.

```json
{
  "id": 1,
  "type": "timeseries",
  "title": "BACnet",
  "gridPos": {
    "x": 0,
    "y": 0,
    "h": 16,
    "w": 24
  },
  "fieldConfig": {
    "defaults": {
      "custom": {
        "drawStyle": "line",
        "lineInterpolation": "linear",
        "barAlignment": 0,
        "barWidthFactor": 0.6,
        "lineWidth": 1,
        "fillOpacity": 0,
        "gradientMode": "none",
        "spanNulls": false,
        "insertNulls": false,
        "showPoints": "auto",
        "showValues": false,
        "pointSize": 5,
        "stacking": {
          "mode": "none",
          "group": "A"
        },
        "axisPlacement": "auto",
        "axisLabel": "",
        "axisColorMode": "text",
        "axisBorderShow": false,
        "scaleDistribution": {
          "type": "linear"
        },
        "axisCenteredZero": false,
        "hideFrom": {
```

```
                "tooltip": false,
                "viz": false,
                "legend": false
              },
              "thresholdsStyle": {
                "mode": "off"
              }
            },
            "color": {
              "mode": "palette-classic"
            },
            "mappings": [],
            "thresholds": {
              "mode": "absolute",
              "steps": [
                {
                  "color": "green",
                  "value": null
                },
                {
                  "color": "red",
                  "value": 80
                }
              ]
            }
          },
          "overrides": []
        },
        "pluginVersion": "12.4.0",
        "targets": [
          {
            "editorMode": "code",
            "format": "time_series",
            "rawQuery": true,
            "rawSql": "SELECT\n
time_bucket(\n    make_interval(secs => ($__interval_ms /
1000)::int),\n    tr.ts\n  ) AS \"time\",\n  avg(tr.value) AS
\"value\",\n  p.external_id AS \"metric\"\nFROM
timeseries_readings tr\nJOIN points p ON p.id =
tr.point_id\nWHERE $__timeFilter(tr.ts)\n  AND p.site_id::text IN
(${site:sqlstring})\n  AND p.bacnet_device_id::text IN ($
{device:sqlstring})\n  AND p.external_id IN (${point:sqlstring})
\nGROUP BY 1, 3\nORDER BY 1, 3;",
            "refId": "A",
            "sql": {
              "columns": [
                {
                  "parameters": [],
                  "type": "function"
                }
              ],
              "groupBy": [
```

```
            {
              "property": {
                "type": "string"
              },
              "type": "groupBy"
            }
          ],
          "limit": 50
        }
      }
    ],
    "datasource": {
      "uid": "openfdd_timescale"
    },
    "options": {
      "tooltip": {
        "mode": "single",
        "sort": "none",
        "hideZeros": false
      },
      "legend": {
        "showLegend": true,
        "displayMode": "list",
        "placement": "bottom",
        "calcs": []
      }
    }
  }
```

44.2 BACnet Plus Fault Data

```
  {
    "id": null,
    "type": "timeseries",
    "title": "BACnet Telemetry w/ Fault Overlays",
    "gridPos": {
      "x": 0,
      "y": 16,
      "h": 12,
      "w": 24
    },
    "fieldConfig": {
      "defaults": {
        "custom": {
          "drawStyle": "line",
          "lineInterpolation": "linear",
          "barAlignment": 0,
          "barWidthFactor": 0.6,
          "lineWidth": 2,
          "fillOpacity": 10,
```

```
            "gradientMode": "none",
            "spanNulls": false,
            "insertNulls": false,
            "showPoints": "never",
            "showValues": false,
            "pointSize": 5,
            "stacking": {
              "mode": "none",
              "group": "A"
            },
            "axisPlacement": "left",
            "axisLabel": "",
            "axisColorMode": "text",
            "axisBorderShow": false,
            "scaleDistribution": {
              "type": "linear"
            },
            "axisCenteredZero": false,
            "hideFrom": {
              "tooltip": false,
              "viz": false,
              "legend": false
            },
            "thresholdsStyle": {
              "mode": "off"
            }
          },
          "color": {
            "mode": "palette-classic"
          },
          "mappings": [],
          "thresholds": {
            "mode": "absolute",
            "steps": [
              {
                "value": null,
                "color": "green"
              },
              {
                "value": 80,
                "color": "red"
              }
            ]
          }
        },
        "overrides": [
          {
            "matcher": {
              "id": "byFrameRefID",
              "options": "B"
            },
            "properties": [
```

```
                    {
                      "id": "custom.axisPlacement",
                      "value": "right"
                    },
                    {
                      "id": "custom.drawStyle",
                      "value": "bars"
                    },
                    {
                      "id": "custom.fillOpacity",
                      "value": 30
                    },
                    {
                      "id": "min",
                      "value": 0
                    },
                    {
                      "id": "max",
                      "value": 1.2
                    },
                    {
                      "id": "custom.axisLabel",
                      "value": "Fault Active (1 = Yes)"
                    }
                  ]
                }
              ]
            },
            "pluginVersion": "12.4.0",
            "targets": [
              {
                "refId": "A",
                "format": "time_series",
                "rawQuery": true,
                "editorMode": "code",
                "rawSql": "SELECT\n  time_bucket(make_interval(secs =>
($__interval_ms / 1000)::int), tr.ts) AS \"time\",\n
avg(tr.value) AS \"value\",\n  p.external_id AS \"metric\"\nFROM
timeseries_readings tr\nJOIN points p ON p.id =
tr.point_id\nWHERE $__timeFilter(tr.ts)\n  AND p.site_id::text IN
(${site:sqlstring})\n  AND p.bacnet_device_id::text IN ($
{device:sqlstring})\n  AND p.external_id IN (${point:sqlstring})
\nGROUP BY 1, 3\nORDER BY 1, 3;"
              },
              {
                "refId": "B",
                "format": "time_series",
                "rawQuery": true,
                "editorMode": "code",
                "rawSql": "SELECT\n  time_bucket(make_interval(secs =>
($__interval_ms / 1000)::int), fr.ts) AS \"time\",\n
max(fr.flag_value) AS \"value\",\n  fd.name AS \"metric\"\nFROM
```

```
fault_results fr\nJOIN fault_definitions fd ON fd.fault_id =
fr.fault_id\nWHERE $__timeFilter(fr.ts)\n  AND (fr.site_id IN ($
{site:sqlstring}) OR fr.site_id IN (SELECT name FROM sites WHERE
id::text IN (${site:sqlstring})))\nGROUP BY 1, 3\nORDER BY 1, 3;"
            }
          ],
          "datasource": {
            "uid": "openfdd_timescale"
          },
          "options": {
            "tooltip": {
              "mode": "multi",
              "sort": "none",
              "hideZeros": false
            },
            "legend": {
              "showLegend": true,
              "displayMode": "list",
              "placement": "bottom",
              "calcs": []
            }
          }
        }
```

44.3 Host stats (stack-host-stats)

The **stack-host-stats** container scrapes host memory, load, swap, disk usage, and Docker container stats into `host_metrics`, `disk_metrics`, and `container_metrics`. Use the same TimescaleDB datasource (`openfdd_timescale`).

44.3.1 Dashboard variable (Host)

- **Name:** `hostname`
- **Type:** Query
- **Datasource:** openfdd_timescale
- **Query:**

```sql
SELECT DISTINCT hostname AS __text, hostname AS __value
FROM host_metrics
WHERE $__timeFilter(ts)
ORDER BY 1;
```

44.3.2 Host — Memory (used / available / total)

```
        {
          "id": null,
```

```
          "type": "timeseries",
          "title": "Host — Memory (used / available / total)",
          "gridPos": { "x": 0, "y": 0, "h": 10, "w": 24 },
          "fieldConfig": {
            "defaults": {
              "custom": {
                "drawStyle": "line",
                "lineInterpolation": "linear",
                "lineWidth": 2,
                "fillOpacity": 12,
                "axisPlacement": "left",
                "scaleDistribution": { "type": "linear" }
              },
              "unit": "bytes",
              "color": { "mode": "palette-classic" },
              "thresholds": { "mode": "absolute", "steps": [{"color":
"green", "value": null}] }
            },
            "overrides": [
              {
                "matcher": { "id": "byName", "options":
"mem_used_bytes" },
                "properties": [{ "id": "displayName", "value": "Used" }]
              },
              {
                "matcher": { "id": "byName", "options":
"mem_available_bytes" },
                "properties": [{ "id": "displayName", "value":
"Available" }]
              },
              {
                "matcher": { "id": "byName", "options":
"mem_total_bytes" },
                "properties": [{ "id": "displayName", "value": "Total" }]
              }
            ]
          },
          "targets": [
            {
              "refId": "A",
              "format": "time_series",
              "rawQuery": true,
              "editorMode": "code",
              "rawSql": "SELECT ts AS \"time\", mem_used_bytes AS
\"mem_used_bytes\", mem_available_bytes AS
\"mem_available_bytes\", mem_total_bytes AS \"mem_total_bytes\"
FROM host_metrics WHERE $__timeFilter(ts) AND hostname IN ($
{hostname:sqlstring}) ORDER BY ts;"
            }
          ],
          "datasource": { "uid": "openfdd_timescale" },
          "options": { "legend": { "showLegend": true, "placement":
```

```
"bottom" } }
        }
```

44.3.3 Host — Load (1 / 5 / 15 min)

```
        {
          "id": null,
          "type": "timeseries",
          "title": "Host — Load (1 / 5 / 15 min)",
          "gridPos": { "x": 0, "y": 10, "h": 8, "w": 24 },
          "fieldConfig": {
            "defaults": {
              "custom": { "drawStyle": "line", "lineWidth": 2,
"fillOpacity": 8, "axisPlacement": "left" },
              "color": { "mode": "palette-classic" },
              "thresholds": { "mode": "absolute", "steps": [{"color":
"green", "value": null}] }
            },
            "overrides": [
              { "matcher": { "id": "byName", "options": "load_1" },
"properties": [{ "id": "displayName", "value": "Load 1m" }] },
              { "matcher": { "id": "byName", "options": "load_5" },
"properties": [{ "id": "displayName", "value": "Load 5m" }] },
              { "matcher": { "id": "byName", "options": "load_15" },
"properties": [{ "id": "displayName", "value": "Load 15m" }] }
            ]
          },
          "targets": [
            {
              "refId": "A",
              "format": "time_series",
              "rawQuery": true,
              "editorMode": "code",
              "rawSql": "SELECT ts AS \"time\", load_1 AS \"load_1\",
load_5 AS \"load_5\", load_15 AS \"load_15\" FROM host_metrics
WHERE $__timeFilter(ts) AND hostname IN (${hostname:sqlstring})
ORDER BY ts;"
            }
          ],
          "datasource": { "uid": "openfdd_timescale" },
          "options": { "legend": { "showLegend": true, "placement":
"bottom" } }
        }
```

44.3.4 Host — Swap (used / total)

```
        {
          "id": null,
          "type": "timeseries",
          "title": "Host — Swap (used / total)",
```

```
          "gridPos": { "x": 0, "y": 18, "h": 6, "w": 24 },
          "fieldConfig": {
            "defaults": {
              "custom": { "drawStyle": "line", "lineWidth": 2,
"fillOpacity": 10, "axisPlacement": "left" },
              "unit": "bytes",
              "color": { "mode": "palette-classic" },
              "thresholds": { "mode": "absolute", "steps": [{"color":
"green", "value": null}] }
            },
            "overrides": [
              { "matcher": { "id": "byName", "options":
"swap_used_bytes" }, "properties": [{ "id": "displayName",
"value": "Swap used" }] },
              { "matcher": { "id": "byName", "options":
"swap_total_bytes" }, "properties": [{ "id": "displayName",
"value": "Swap total" }] }
            ]
          },
          "targets": [
            {
              "refId": "A",
              "format": "time_series",
              "rawQuery": true,
              "editorMode": "code",
              "rawSql": "SELECT ts AS \"time\", swap_used_bytes AS
\"swap_used_bytes\", swap_total_bytes AS
\"swap_total_bytes\" FROM host_metrics WHERE $__timeFilter(ts) AND
hostname IN (${hostname:sqlstring}) ORDER BY ts;"
            }
          ],
          "datasource": { "uid": "openfdd_timescale" },
          "options": { "legend": { "showLegend": true, "placement":
"bottom" } }
        }
```

44.3.5 Host — Disk space (free / used / total)

Disk metrics come from `disk_metrics` (mount path from env `OFDD_DISK_MOUNT_PATHS`, default `/`). The table is created by the same migration as `host_metrics` (`stack/sql/006_host_metrics.sql`); re-run migrations if you added host-stats after an older deploy.

```
        {
          "id": null,
          "type": "timeseries",
          "title": "Host — Disk space (free / used / total)",
          "gridPos": { "x": 0, "y": 24, "h": 10, "w": 24 },
          "fieldConfig": {
            "defaults": {
```

```
              "custom": { "drawStyle": "line", "lineWidth": 2,
"fillOpacity": 12, "axisPlacement": "left" },
              "unit": "bytes",
              "color": { "mode": "palette-classic" },
              "thresholds": { "mode": "absolute", "steps": [{"color":
"green", "value": null}] }
            },
            "overrides": [
              { "matcher": { "id": "byName", "options": "free_bytes" },
"properties": [{ "id": "displayName", "value": "Free" }] },
              { "matcher": { "id": "byName", "options": "used_bytes" },
"properties": [{ "id": "displayName", "value": "Used" }] },
              { "matcher": { "id": "byName", "options": "total_bytes" },
"properties": [{ "id": "displayName", "value": "Total" }] }
            ]
          },
          "targets": [
            {
              "refId": "A",
              "format": "time_series",
              "rawQuery": true,
              "editorMode": "code",
              "rawSql": "SELECT ts AS \"time\", free_bytes AS
\"free_bytes\", used_bytes AS \"used_bytes\", total_bytes AS
\"total_bytes\" FROM disk_metrics WHERE $__timeFilter(ts) AND
hostname IN (${hostname:sqlstring}) AND mount_path = '/' ORDER BY
ts;"
            }
          ],
          "datasource": { "uid": "openfdd_timescale" },
          "options": { "legend": { "showLegend": true, "placement":
"bottom" } }
        }
```

44.3.6 Host — Disk available (gauge, latest)

```
        {
          "id": null,
          "type": "gauge",
          "title": "Host — Disk space available (latest)",
          "gridPos": { "x": 0, "y": 34, "h": 6, "w": 8 },
          "fieldConfig": {
            "defaults": {
              "unit": "bytes",
              "min": 0,
              "color": { "mode": "thresholds" },
              "thresholds": { "mode": "absolute", "steps": [{"color":
"red", "value": null}, {"color": "yellow", "value": 5368709120},
{"color": "green", "value": 10737418240}] }
            }
          },
```

```
          "targets": [
            {
              "refId": "A",
              "format": "time_series",
              "rawQuery": true,
              "editorMode": "code",
              "rawSql": "SELECT ts AS \"time\", free_bytes AS \"value\"
FROM disk_metrics WHERE hostname IN (${hostname:sqlstring}) AND
mount_path = '/' ORDER BY ts DESC LIMIT 1;"
            }
          ],
          "datasource": { "uid": "openfdd_timescale" },
          "options": { "showThresholdLabels": false }
        }
```

44.3.7 Containers — CPU %

```
        {
          "id": null,
          "type": "timeseries",
          "title": "Containers — CPU %",
          "gridPos": { "x": 0, "y": 40, "h": 8, "w": 24 },
          "fieldConfig": {
            "defaults": {
              "custom": { "drawStyle": "line", "lineWidth": 1,
"fillOpacity": 5, "axisPlacement": "left" },
              "unit": "percent",
              "min": 0,
              "color": { "mode": "palette-classic" },
              "thresholds": { "mode": "absolute", "steps": [{"color":
"green", "value": null}] }
            }
          },
          "targets": [
            {
              "refId": "A",
              "format": "time_series",
              "rawQuery": true,
              "editorMode": "code",
              "rawSql": "SELECT ts AS \"time\", cpu_pct AS \"value\",
container_name AS \"metric\" FROM container_metrics WHERE
$__timeFilter(ts) ORDER BY ts, container_name;"
            }
          ],
          "datasource": { "uid": "openfdd_timescale" },
          "options": { "legend": { "showLegend": true, "placement":
"bottom" } }
        }
```

44.3.8 Containers — Memory %

```
        {
          "id": null,
          "type": "timeseries",
          "title": "Containers — Memory %",
          "gridPos": { "x": 0, "y": 48, "h": 8, "w": 24 },
          "fieldConfig": {
            "defaults": {
              "custom": { "drawStyle": "line", "lineWidth": 1,
"fillOpacity": 5, "axisPlacement": "left" },
              "unit": "percent",
              "min": 0,
              "max": 100,
              "color": { "mode": "palette-classic" },
              "thresholds": { "mode": "absolute", "steps": [{"color":
"green", "value": null}] }
            }
          },
          "targets": [
            {
              "refId": "A",
              "format": "time_series",
              "rawQuery": true,
              "editorMode": "code",
              "rawSql": "SELECT ts AS \"time\", COALESCE(mem_pct, 0) AS
\"value\", container_name AS \"metric\" FROM container_metrics
WHERE $__timeFilter(ts) ORDER BY ts, container_name;"
            }
          ],
          "datasource": { "uid": "openfdd_timescale" },
          "options": { "legend": { "showLegend": true, "placement":
"bottom" } }
        }
```

44.3.9 Containers — Table (latest)

```
        {
          "id": null,
          "type": "table",
          "title": "Containers — Latest stats",
          "gridPos": { "x": 0, "y": 56, "h": 12, "w": 24 },
          "fieldConfig": {
            "defaults": { "custom": { "align": "auto", "displayMode":
"auto" }, "mappings": [], "thresholds": { "mode": "absolute",
"steps": [{"color": "green", "value": null}] } },
            "overrides": [
              { "matcher": { "id": "byName", "options": "cpu_pct" },
"properties": [{ "id": "unit", "value": "percent" }] },
              { "matcher": { "id": "byName", "options": "mem_pct" },
"properties": [{ "id": "unit", "value": "percent" }] },
```

```
              { "matcher": { "id": "byName", "options":
"mem_usage_bytes" }, "properties": [{ "id": "unit", "value":
"bytes" }] }
            ]
          },
          "targets": [
            {
              "refId": "A",
              "format": "table",
              "rawQuery": true,
              "editorMode": "code",
              "rawSql": "SELECT container_name, ts, cpu_pct, mem_pct,
mem_usage_bytes, pids FROM (SELECT *, row_number() OVER (PARTITION
BY container_name ORDER BY ts DESC) AS rn FROM container_metrics
WHERE $__timeFilter(ts)) sub WHERE rn = 1 ORDER BY
container_name;"
            }
          ],
          "datasource": { "uid": "openfdd_timescale" },
          "options": { "showHeader": true }
        }
```

44.4 Weather

44.4.1 Temp RH Dewpoint

```
        {
          "id": 21,
          "type": "timeseries",
          "title": "Weather — Temp / RH / Dewpoint",
          "gridPos": {
            "x": 0,
            "y": 0,
            "h": 10,
            "w": 24
          },
          "fieldConfig": {
            "defaults": {
              "custom": {
                "drawStyle": "line",
                "lineInterpolation": "linear",
                "barAlignment": 0,
                "barWidthFactor": 0.6,
                "lineWidth": 2,
                "fillOpacity": 12,
                "gradientMode": "none",
                "spanNulls": false,
                "insertNulls": false,
                "showPoints": "never",
```

```
            "showValues": false,
            "pointSize": 4,
            "stacking": {
              "mode": "none",
              "group": "A"
            },
            "axisPlacement": "left",
            "axisLabel": "°F",
            "axisColorMode": "text",
            "axisBorderShow": false,
            "scaleDistribution": {
              "type": "linear"
            },
            "axisCenteredZero": false,
            "hideFrom": {
              "tooltip": false,
              "viz": false,
              "legend": false
            },
            "thresholdsStyle": {
              "mode": "off"
            }
          },
          "color": {
            "mode": "palette-classic"
          },
          "mappings": [],
          "thresholds": {
            "mode": "absolute",
            "steps": [
              {
                "color": "green",
                "value": null
              },
              {
                "color": "red",
                "value": 80
              }
            ]
          }
        },
        "overrides": [
          {
            "matcher": {
              "id": "byName",
              "options": "temp_f"
            },
            "properties": [
              {
                "id": "unit",
                "value": "fahrenheit"
              },
```

```
            {
              "id": "custom.axisPlacement",
              "value": "left"
            },
            {
              "id": "custom.axisLabel",
              "value": "°F"
            }
          ]
        },
        {
          "matcher": {
            "id": "byName",
            "options": "dewpoint_f"
          },
          "properties": [
            {
              "id": "unit",
              "value": "fahrenheit"
            },
            {
              "id": "custom.axisPlacement",
              "value": "left"
            }
          ]
        },
        {
          "matcher": {
            "id": "byName",
            "options": "rh_pct"
          },
          "properties": [
            {
              "id": "unit",
              "value": "percent"
            },
            {
              "id": "custom.axisPlacement",
              "value": "right"
            },
            {
              "id": "custom.axisLabel",
              "value": "RH (%)"
            },
            {
              "id": "min",
              "value": 0
            },
            {
              "id": "max",
              "value": 100
            }
```

```
                ]
              }
            ]
          },
          "pluginVersion": "12.4.0",
          "targets": [
            {
              "refId": "A",
              "editorMode": "code",
              "rawQuery": true,
              "format": "time_series",
              "rawSql": "SELECT\n  time_bucket(make_interval(secs =>
($__interval_ms / 1000)::int), tr.ts) AS \"time\",\n
avg(tr.value) AS \"value\",\n  p.external_id AS \"metric\"\nFROM
timeseries_readings tr\nJOIN points p ON p.id =
tr.point_id\nWHERE $__timeFilter(tr.ts)\n  AND p.site_id::text IN
(${site:sqlstring})\n  AND p.external_id IN
('temp_f','rh_pct','dewpoint_f')\nGROUP BY 1, 3\nORDER BY 1, 3;"
            }
          ],
          "datasource": {
            "uid": "openfdd_timescale"
          },
          "options": {
            "tooltip": {
              "mode": "single",
              "sort": "none",
              "hideZeros": false
            },
            "legend": {
              "showLegend": true,
              "displayMode": "list",
              "placement": "bottom",
              "calcs": []
            }
          }
        }
```


44.4.2 Wind Speed Gusts Direction


```
        {
          "id": 22,
          "type": "timeseries",
          "title": "Weather — Wind (Speed / Gust / Direction)",
          "gridPos": {
            "x": 0,
            "y": 10,
            "h": 9,
            "w": 24
          },
```

```
"fieldConfig": {
  "defaults": {
    "custom": {
      "drawStyle": "line",
      "lineInterpolation": "linear",
      "barAlignment": 0,
      "barWidthFactor": 0.6,
      "lineWidth": 2,
      "fillOpacity": 10,
      "gradientMode": "none",
      "spanNulls": false,
      "insertNulls": false,
      "showPoints": "never",
      "showValues": false,
      "pointSize": 4,
      "stacking": {
        "mode": "none",
        "group": "A"
      },
      "axisPlacement": "left",
      "axisLabel": "mph",
      "axisColorMode": "text",
      "axisBorderShow": false,
      "scaleDistribution": {
        "type": "linear"
      },
      "axisCenteredZero": false,
      "hideFrom": {
        "tooltip": false,
        "viz": false,
        "legend": false
      },
      "thresholdsStyle": {
        "mode": "off"
      }
    },
    "color": {
      "mode": "palette-classic"
    },
    "mappings": [],
    "thresholds": {
      "mode": "absolute",
      "steps": [
        {
          "color": "green",
          "value": null
        },
        {
          "color": "red",
          "value": 80
        }
      ]
```

```
      }
    },
    "overrides": [
      {
        "matcher": {
          "id": "byName",
          "options": "wind_mph"
        },
        "properties": [
          {
            "id": "unit",
            "value": "mph"
          },
          {
            "id": "custom.axisPlacement",
            "value": "left"
          },
          {
            "id": "custom.axisLabel",
            "value": "mph"
          },
          {
            "id": "min",
            "value": 0
          }
        ]
      },
      {
        "matcher": {
          "id": "byName",
          "options": "gust_mph"
        },
        "properties": [
          {
            "id": "unit",
            "value": "mph"
          },
          {
            "id": "custom.axisPlacement",
            "value": "left"
          },
          {
            "id": "min",
            "value": 0
          }
        ]
      },
      {
        "matcher": {
          "id": "byName",
          "options": "wind_dir_deg"
        },
```

```
                "properties": [
                  {
                    "id": "unit",
                    "value": "degrees"
                  },
                  {
                    "id": "custom.axisPlacement",
                    "value": "right"
                  },
                  {
                    "id": "custom.axisLabel",
                    "value": "°"
                  },
                  {
                    "id": "min",
                    "value": 0
                  },
                  {
                    "id": "max",
                    "value": 360
                  }
                ]
              }
            ]
          },
          "pluginVersion": "12.4.0",
          "targets": [
            {
              "refId": "A",
              "editorMode": "code",
              "rawQuery": true,
              "format": "time_series",
              "rawSql": "SELECT\n  time_bucket(make_interval(secs =>
($__interval_ms / 1000)::int), tr.ts) AS \"time\",\n
avg(tr.value) AS \"value\",\n  p.external_id AS \"metric\"\nFROM
timeseries_readings tr\nJOIN points p ON p.id =
tr.point_id\nWHERE $__timeFilter(tr.ts)\n  AND p.site_id::text IN
(${site:sqlstring})\n  AND p.external_id IN
('wind_mph','gust_mph','wind_dir_deg')\nGROUP BY 1,
3\nORDER BY 1, 3;"
            }
          ],
          "datasource": {
            "uid": "openfdd_timescale"
          },
          "options": {
            "tooltip": {
              "mode": "single",
              "sort": "none",
              "hideZeros": false
            },
            "legend": {
```

```
        "showLegend": true,
        "displayMode": "list",
        "placement": "bottom",
        "calcs": []
      }
    }
  }
```

44.4.3 Wind Speed Gust Direction

```
  {
    "id": 22,
    "type": "timeseries",
    "title": "Weather — Wind (Speed / Gust / Direction)",
    "gridPos": {
      "x": 0,
      "y": 10,
      "h": 9,
      "w": 24
    },
    "fieldConfig": {
      "defaults": {
        "custom": {
          "drawStyle": "line",
          "lineInterpolation": "linear",
          "barAlignment": 0,
          "barWidthFactor": 0.6,
          "lineWidth": 2,
          "fillOpacity": 10,
          "gradientMode": "none",
          "spanNulls": false,
          "insertNulls": false,
          "showPoints": "never",
          "showValues": false,
          "pointSize": 4,
          "stacking": {
            "mode": "none",
            "group": "A"
          },
          "axisPlacement": "left",
          "axisLabel": "mph",
          "axisColorMode": "text",
          "axisBorderShow": false,
          "scaleDistribution": {
            "type": "linear"
          },
          "axisCenteredZero": false,
          "hideFrom": {
            "tooltip": false,
            "viz": false,
            "legend": false
```

```
            },
            "thresholdsStyle": {
              "mode": "off"
            }
          },
          "color": {
            "mode": "palette-classic"
          },
          "mappings": [],
          "thresholds": {
            "mode": "absolute",
            "steps": [
              {
                "color": "green",
                "value": null
              },
              {
                "color": "red",
                "value": 80
              }
            ]
          }
        },
        "overrides": [
          {
            "matcher": {
              "id": "byName",
              "options": "wind_mph"
            },
            "properties": [
              {
                "id": "unit",
                "value": "mph"
              },
              {
                "id": "custom.axisPlacement",
                "value": "left"
              },
              {
                "id": "custom.axisLabel",
                "value": "mph"
              },
              {
                "id": "min",
                "value": 0
              }
            ]
          },
          {
            "matcher": {
              "id": "byName",
              "options": "gust_mph"
```

```
        },
        "properties": [
          {
            "id": "unit",
            "value": "mph"
          },
          {
            "id": "custom.axisPlacement",
            "value": "left"
          },
          {
            "id": "min",
            "value": 0
          }
        ]
      },
      {
        "matcher": {
          "id": "byName",
          "options": "wind_dir_deg"
        },
        "properties": [
          {
            "id": "unit",
            "value": "degrees"
          },
          {
            "id": "custom.axisPlacement",
            "value": "right"
          },
          {
            "id": "custom.axisLabel",
            "value": "°"
          },
          {
            "id": "min",
            "value": 0
          },
          {
            "id": "max",
            "value": 360
          }
        ]
      }
    ]
  },
  "pluginVersion": "12.4.0",
  "targets": [
    {
      "refId": "A",
      "editorMode": "code",
      "rawQuery": true,
```

```
              "format": "time_series",
              "rawSql": "SELECT\n  time_bucket(make_interval(secs =>
($__interval_ms / 1000)::int), tr.ts) AS \"time\",\n
avg(tr.value) AS \"value\",\n  p.external_id AS \"metric\"\nFROM
timeseries_readings tr\nJOIN points p ON p.id =
tr.point_id\nWHERE $__timeFilter(tr.ts)\n  AND p.site_id::text IN
(${site:sqlstring})\n  AND p.external_id IN
('wind_mph','gust_mph','wind_dir_deg')\nGROUP BY 1,
3\nORDER BY 1, 3;"
            }
          ],
          "datasource": {
            "uid": "openfdd_timescale"
          },
          "options": {
            "tooltip": {
              "mode": "single",
              "sort": "none",
              "hideZeros": false
            },
            "legend": {
              "showLegend": true,
              "displayMode": "list",
              "placement": "bottom",
              "calcs": []
            }
          }
        }
```

44.4.4 Solar Radiation

```
        {
          "id": 23,
          "type": "timeseries",
          "title": "Weather — Solar / Radiation (W/m²)",
          "gridPos": {
            "x": 0,
            "y": 19,
            "h": 9,
            "w": 24
          },
          "fieldConfig": {
            "defaults": {
              "custom": {
                "drawStyle": "line",
                "lineInterpolation": "linear",
                "barAlignment": 0,
                "barWidthFactor": 0.6,
                "lineWidth": 2,
                "fillOpacity": 10,
                "gradientMode": "none",
```

```
            "spanNulls": false,
            "insertNulls": false,
            "showPoints": "never",
            "showValues": false,
            "pointSize": 4,
            "stacking": {
              "mode": "none",
              "group": "A"
            },
            "axisPlacement": "left",
            "axisLabel": "W/m²",
            "axisColorMode": "text",
            "axisBorderShow": false,
            "scaleDistribution": {
              "type": "linear"
            },
            "axisCenteredZero": true,
            "hideFrom": {
              "tooltip": false,
              "viz": false,
              "legend": false
            },
            "thresholdsStyle": {
              "mode": "off"
            }
          },
          "color": {
            "mode": "palette-classic"
          },
          "mappings": [],
          "thresholds": {
            "mode": "absolute",
            "steps": [
              {
                "color": "green",
                "value": null
              },
              {
                "color": "red",
                "value": 80
              }
            ]
          },
          "unit": "wattperm2",
          "min": 0
        },
        "overrides": [
          {
            "matcher": {
              "id": "byName",
              "options": "shortwave_wm2"
            },
```

```
          "properties": [
            {
              "id": "min",
              "value": 0
            }
          ]
        },
        {
          "matcher": {
            "id": "byName",
            "options": "direct_wm2"
          },
          "properties": [
            {
              "id": "min",
              "value": 0
            }
          ]
        },
        {
          "matcher": {
            "id": "byName",
            "options": "diffuse_wm2"
          },
          "properties": [
            {
              "id": "min",
              "value": 0
            }
          ]
        },
        {
          "matcher": {
            "id": "byName",
            "options": "gti_wm2"
          },
          "properties": [
            {
              "id": "min",
              "value": 0
            }
          ]
        }
      ]
    },
    "pluginVersion": "12.4.0",
    "targets": [
      {
        "refId": "A",
        "editorMode": "code",
        "rawQuery": true,
        "format": "time_series",
```

```
              "rawSql": "SELECT\n  time_bucket(make_interval(secs =>
($__interval_ms / 1000)::int), tr.ts) AS \"time\",\n
avg(tr.value) AS \"value\",\n  p.external_id AS \"metric\"\nFROM
timeseries_readings tr\nJOIN points p ON p.id =
tr.point_id\nWHERE $__timeFilter(tr.ts)\n  AND p.site_id::text IN
(${site:sqlstring})\n  AND p.external_id IN
('shortwave_wm2','direct_wm2','diffuse_wm2','gti_wm2')\nGROUP BY
1, 3\nORDER BY 1, 3;"
            }
          ],
          "datasource": {
            "uid": "openfdd_timescale"
          },
          "options": {
            "tooltip": {
              "mode": "single",
              "sort": "none",
              "hideZeros": false
            },
            "legend": {
              "showLegend": true,
              "displayMode": "list",
              "placement": "bottom",
              "calcs": []
            }
          }
        }
```

44.4.5 Cloud Cover

```
        {
          "id": 24,
          "type": "timeseries",
          "title": "Weather — Cloud Cover (%)",
          "gridPos": {
            "x": 0,
            "y": 28,
            "h": 7,
            "w": 24
          },
          "fieldConfig": {
            "defaults": {
              "custom": {
                "drawStyle": "line",
                "lineInterpolation": "linear",
                "barAlignment": 0,
                "barWidthFactor": 0.6,
                "lineWidth": 2,
                "fillOpacity": 35,
                "gradientMode": "opacity",
```

```
          "spanNulls": false,
          "insertNulls": false,
          "showPoints": "never",
          "showValues": false,
          "pointSize": 4,
          "stacking": {
            "mode": "none",
            "group": "A"
          },
          "axisPlacement": "left",
          "axisLabel": "Cloud (%)",
          "axisColorMode": "text",
          "axisBorderShow": false,
          "scaleDistribution": {
            "type": "linear"
          },
          "axisCenteredZero": false,
          "hideFrom": {
            "tooltip": false,
            "viz": false,
            "legend": false
          },
          "thresholdsStyle": {
            "mode": "off"
          }
        },
        "color": {
          "mode": "palette-classic"
        },
        "mappings": [],
        "thresholds": {
          "mode": "absolute",
          "steps": [
            {
              "color": "green",
              "value": null
            },
            {
              "color": "red",
              "value": 80
            }
          ]
        },
        "unit": "percent",
        "min": 0,
        "max": 100
      },
      "overrides": []
    },
    "pluginVersion": "12.4.0",
    "targets": [
      {
```

```
              "refId": "A",
              "editorMode": "code",
              "rawQuery": true,
              "format": "time_series",
              "rawSql": "SELECT\n  time_bucket(make_interval(secs =>
($__interval_ms / 1000)::int), tr.ts) AS \"time\",\n
avg(tr.value) AS \"value\",\n  p.external_id AS \"metric\"\nFROM
timeseries_readings tr\nJOIN points p ON p.id =
tr.point_id\nWHERE $__timeFilter(tr.ts)\n  AND p.site_id::text IN
(${site:sqlstring})\n  AND p.external_id IN ('cloud_pct')\nGROUP
BY 1, 3\nORDER BY 1, 3;"
            }
          ],
          "datasource": {
            "uid": "openfdd_timescale"
          },
          "options": {
            "tooltip": {
              "mode": "single",
              "sort": "none",
              "hideZeros": false
            },
            "legend": {
              "showLegend": true,
              "displayMode": "list",
              "placement": "bottom",
              "calcs": []
            }
          }
        }
```

44.5 Faults

44.5.1 Fault Definitions

```
        {
          "id": 1,
          "type": "table",
          "title": "Fault Definitions",
          "gridPos": {
            "x": 0,
            "y": 0,
            "h": 10,
            "w": 14
          },
          "fieldConfig": {
            "defaults": {
              "custom": {
                "align": "auto",
```

```
                "footer": {
                  "reducers": []
                },
                "cellOptions": {
                  "type": "auto"
                },
                "inspect": false,
                "hideFrom": {
                  "viz": false
                }
              },
              "mappings": [],
              "thresholds": {
                "mode": "absolute",
                "steps": [
                  {
                    "color": "green",
                    "value": null
                  },
                  {
                    "color": "red",
                    "value": 80
                  }
                ]
              },
              "color": {
                "mode": "thresholds"
              }
            },
            "overrides": []
          },
          "pluginVersion": "12.4.0",
          "targets": [
            {
              "dataset": "openfdd",
              "datasource": {
                "type": "grafana-postgresql-datasource",
                "uid": "openfdd_timescale"
              },
              "editorMode": "code",
              "format": "table",
              "rawQuery": true,
              "rawSql": "SELECT \r\n  fault_id, \r\n  name, \r\n
category, \r\n  severity,
\r\n  -- Shows what Brick equipment this rule binds to (e.g., AHU,
VAV)\r\n  array_to_string(equipment_types, ', ') AS
target_equipment,
\r\n  -- Shows the raw JSON of the specific Brick points it needs
(e.g., sat, mat)\r\n  inputs AS required_points,\r\n
updated_at\r\nFROM fault_definitions\r\nWHERE
$__timeFilter(updated_at)\r\nORDER BY updated_at DESC;",
              "refId": "A",
```

```
        "sql": {
          "columns": [
            {
              "parameters": [],
              "type": "function"
            }
          ],
          "groupBy": [
            {
              "property": {
                "type": "string"
              },
              "type": "groupBy"
            }
          ],
          "limit": 50
        }
      }
    ],
    "datasource": {
      "type": "grafana-postgresql-datasource",
      "uid": "openfdd_timescale"
    },
    "options": {
      "showHeader": true,
      "cellHeight": "sm"
    }
  }
```

44.5.2 Fault Definition Count

```
  {
    "id": 2,
    "type": "stat",
    "title": "Fault Definition Count",
    "gridPos": {
      "x": 0,
      "y": 15,
      "h": 15,
      "w": 24
    },
    "fieldConfig": {
      "defaults": {
        "mappings": [],
        "thresholds": {
          "mode": "absolute",
          "steps": [
            {
              "color": "green",
              "value": null
```

```
                    },
                    {
                      "color": "red",
                      "value": 80
                    }
                  ]
                },
                "color": {
                  "mode": "thresholds"
                }
              },
              "overrides": []
            },
            "pluginVersion": "12.4.0",
            "targets": [
              {
                "dataset": "openfdd",
                "datasource": {
                  "type": "grafana-postgresql-datasource",
                  "uid": "openfdd_timescale"
                },
                "editorMode": "code",
                "format": "table",
                "rawQuery": true,
                "rawSql": "SELECT count(*)::int AS \"Fault
definitions\"\r\nFROM fault_definitions;",
                "refId": "A",
                "sql": {
                  "columns": [
                    {
                      "parameters": [],
                      "type": "function"
                    }
                  ],
                  "groupBy": [
                    {
                      "property": {
                        "type": "string"
                      },
                      "type": "groupBy"
                    }
                  ],
                  "limit": 50
                }
              }
            ],
            "datasource": {
              "type": "grafana-postgresql-datasource",
              "uid": "openfdd_timescale"
            },
            "options": {
              "reduceOptions": {
```

```
          "values": false,
          "calcs": [
            "lastNotNull"
          ],
          "fields": ""
        },
        "orientation": "auto",
        "textMode": "auto",
        "wideLayout": true,
        "colorMode": "value",
        "graphMode": "area",
        "justifyMode": "auto",
        "showPercentChange": false,
        "percentChangeColorMode": "standard"
      }
    }
```

45 Grafana dashboards

46 Grafana dashboards (optional)

Grafana is optional. Install with `./scripts/bootstrap.sh --with-grafana`. The React frontend provides equivalent timeseries, fault, and system-resource views. Use Grafana when you want custom SQL dashboards or existing Grafana workflows.

No dashboards are provisioned by default. Only the **datasource** (TimescaleDB, uid: `openfdd_timescale`) is prebuilt.

To build your own dashboards using SQL and the Open-FDD database, see the **Grafana SQL cookbook**.

You can also place dashboard JSON files in `stack/grafana/dashboards/` (or the directory mounted at `/var/lib/grafana/dashboards` in the Grafana container) and restart Grafana to load them.

46.1 React frontend parity

The React UI provides equivalent time-series and stat views for FDD workflows:

- **Trending** — Multi-point time-series (site/equipment/point picker, date range, dual Y-axis). Uses the same `timeseries_readings` data as the Grafana BACnet timeseries dashboard.
- **Faults** — Active fault table (with **Device** and **Sensor/point** columns so you can see which bad sensor was found on which device), fault definitions, fault count (period) stat, and fault flags over time chart (from `fault_results` via `GET /analytics/fault-timeseries` and `GET /analytics/fault-summary`).
- **Plots (trend + fault overlay)** — Same idea as the Grafana "BACnet Plus Fault Data" panel: one chart with point time-series (left axis) and fault data overlaid. In the React UI, select a site, add points and **Add faults**; fault activity is drawn as **vertical colored bands** on the same plot (data from `GET /analytics/fault-timeseries`, i.e. `fault_results`). So fault data does show on the plot in parity with the Grafana version.
- **Overview** — Site cards with fault counts, equipment/points stats, and FDD status.
- **System resources** — A dedicated sidebar page that mirrors the Grafana "System Resources (Host & Containers)" dashboard: host memory used/available, load average (1m/5m/15m), swap used (stat cards with threshold colors), host memory and load time-series charts, container memory (MB) and CPU % charts, **Containers (latest)** table (CPU %, Mem MB, Mem %, PIDs), and **disk usage** per mount (used/free/total GB, % used, progress bar). Data comes from `host_metrics`, `container_metrics`, and `disk_metrics` via `GET /analytics/system/*`. Requires the **host-stats** service to be running so those tables are populated.

If you prefer building custom SQL dashboards in Grafana, use the cookbook and the same tables (`host_metrics`, `container_metrics`, `disk_metrics`, `timeseries_readings`, `fault_results`, etc.).

47 Configuration

48 Configuration

48.1 Platform config (RDF + CRUD)

Platform config lives in the **same RDF graph** as Brick and BACnet (`config/data_model.ttl`). No YAML file. **Where:** Bootstrap seeds via PUT /config; API GET/PUT /config and POST /data-model/sparql use the graph. The **entire app is bootstrapped from this data model** (sites, equipment, points, and platform config such as `ofdd:rulesDir`, `ofdd:bacnetScrapeIntervalMin`, etc.). **`rules_dir` (ofdd:rulesDir) remains required**: it is the path where FDD rule YAML files are stored; the frontend upload/download UI manages files *in* that path and does not replace the need for the path itself. **Individual rule YAML files are not stored in the data model**—only the single directory path (e.g. `ofdd:rulesDir "stack/rules"`) is; the files themselves live on disk under that path.

- **Bootstrap:** `./scripts/bootstrap.sh` seeds config via PUT /config (defaults or `OFDD_*` from `stack/.env`).
- **API:** GET /config, PUT /config; query via POST /data-model/sparql.
- **Env at seed:** Set `OFDD_*` in `stack/.env` before first bootstrap to customize the seed. **Docker:** Set env in `stack/docker-compose.yml` or `stack/.env`; bootstrap uses these when calling PUT /config.

Rename / multiple configs: Use different env files (e.g. `platform-prod.env`, `platform-building-a.env`) and start with `docker compose --env-file platform-building-a.env up -d`, or set `OFDD_*` in that file. No built-in “config name” selector; use env files or env vars per deployment.

48.2 Platform keys (RDF / GET-PUT /config)

Example keys (GET/PUT /config or OFDD_* at bootstrap seed):

Key	Default	Description
`rule_interval_hours`	3	FDD loop run interval (when no trigger file: run every N hours)
`lookback_days`	3	Days of data loaded per run (each run pulls last N days into the rule engine)
`rules_dir`	stack/rules	**Single directory for your rules** (hot reload)
`brick_ttl_dir`	—	Optional. Directory containing Brick model TTL (e.g. `config/`); platform uses first `.ttl` or brick_ttl path for FDD column mapping. Optional if using points `brick_type`/fdd_input. See Data modeling.
`bacnet_enabled`	true	Enable BACnet scraper
`graph_sync_interval_min`	5	Minutes between serializing the in-memory graph to `data_model.ttl` (API background thread). Env: `OFDD_GRAPH_SYNC_INTERVAL_MIN`.
`bacnet_scrape_interval_min`	5	Poll interval (minutes)
`bacnet_site_id`	default	Site to tag when scraping (use on **remote gateways** so data is attributed to the right building on the central DB)
`bacnet_gateways`	—	Optional. **Central aggregator:** JSON array of `{url, site_id}`; scraper polls each remote diy-bacnet-server in turn. Env: `OFDD_BACNET_GATEWAYS`.
`open_meteo_enabled`	true	Enable weather; when true, **FDD loop runs a weather fetch at the start of each run** (same cadence as rules, every `rule_interval_hours`).
`open_meteo_interval_hours`	24	**Standalone weather-scraper only.** Poll interval (hours) when using the optional weather-scraper container.

Key	Default	Description
		Ignored when weather is run from the FDD loop.
`open_meteo_latitude`	41.88	Site latitude
`open_meteo_longitude`	-87.63	Site longitude
`open_meteo_timezone`	America/Chicago	Timezone
`open_meteo_days_back`	3	**Standalone weather-scraper only.** Days of archive to fetch per run. When weather runs with FDD, a 1-day lookback is used (no separate config).
`open_meteo_site_id`	default	Site ID for weather points

Weather fetch: Weather is normally fetched **with each FDD run** (same interval as fault rules, e.g. every 3 h), using a **1-day lookback** so the API is not over-used. A standalone weather scraper is only for setups that do not run the FDD loop; do not run both to avoid redundant fetches. The standalone scraper reads config from **GET /config** (like the BACnet scraper) when available.

Where rules live: Platform config includes **`rules_dir`** (e.g. `stack/rules`). This path is **required**: the FDD loop and the rules API both use it. Rule YAML files are stored there; you can manage them in two ways. (1) **React frontend (Faults page):** upload, download, delete YAML and **Sync definitions** so the fault_definitions table updates without waiting for the next FDD run. (2) **Files on disk:** edit or add files under the configured path (e.g. `stack/rules/`). Both target the same `rules_dir`; the frontend is the preferred path when you have UI access. See Fault rules overview and the Expression Rule Cookbook.

Hot reload (AFDD tuning): The FDD loop loads rules from `rules_dir` on every run (no cache). Edit YAML on disk or upload via the frontend → trigger a run (or wait for the schedule) or click **Sync definitions** in the UI → results and the definitions table reflect the new rules. The `rules_dir` path is **RDF-driven** (GET/PUT `/config`, same graph as the rest of platform config). Unit tests: `open_fdd/tests/platform/test_rules_loader.py`, `open_fdd/tests/platform_api/test_rules.py`.

Rolling window (per rule): Set `params.rolling_window` in each rule YAML; see Fault rules for HVAC.

48.3 Services that read config from the API (BACnet scraper)

Platform config (e.g. **Scrape interval (min)**) is stored in the data model and served by **GET /config**. Some services need to call the API to get that config so changes in the Config UI take effect.

BACnet scraper and **weather scraper (standalone):** Both can read config from **GET /config** (with a short cache) so the Config UI and data model control their behaviour. The BACnet scraper uses `bacnet_scrape_interval_min`; the standalone weather scraper uses `open_meteo_enabled`, `open_meteo_interval_hours`, `open_meteo_days_back`, and the geo/site fields. If GET /config fails (e.g. no `OFDD_API_KEY`), they fall back to env vars.

BACnet scraper: The scraper runs on a fixed interval (e.g. every 1, 5, or 10 minutes). It can get that interval in two ways:

1. **From the API (dynamic)** — The scraper calls **GET /config** (with a short cache). It then uses `bacnet_scrape_interval_min` from the response, so whatever you set in the Config UI is what the scraper uses. **This only works when the scraper can authenticate:** the API requires Bearer auth when `OFDD_API_KEY` is set, so the scraper must have **`OFDD_API_KEY`** in its environment (same value as in `stack/.env`). The stack's `docker-compose.yml` passes `OFDD_API_KEY: ${OFDD_API_KEY:-}` to the bacnet-scraper service for this reason.
2. **Fallback (env)** — If GET /config fails (e.g. 401 because no API key, or API unreachable), the scraper falls back to **`OFDD_BACNET_SCRAPE_INTERVAL_MIN`** from its environment (e.g. 5 in compose). **So if you set an interval in the Config UI but the scraper does not have `OFDD_API_KEY`, it will ignore the UI and keep using the env default.** Rebuild/restart the bacnet-scraper after ensuring `OFDD_API_KEY` is in `stack/.env` and in the scraper's env: `./scripts/bootstrap.sh --build bacnet-scraper`.

Summary: For the Config UI (and data model) to control the BACnet scrape interval, the API must have `OFDD_API_KEY` set and the bacnet-scraper container must receive the same key so it can call GET /config successfully.

48.4 BACnet: single gateway, remote gateways, central aggregator

- **Single building (or one remote gateway):** Set `OFDD_BACNET_SERVER_URL` and optionally `OFDD_BACNET_SITE_ID` (e.g. `building-a`). The scraper tags all readings with that site. On a **remote** gateway (diy-bacnet-server + scraper on another subnet), point `OFDD_DB_DSN` at the central Open-FDD DB and set `OFDD_BACNET_SITE_ID` to the site name/UUID used on the central API so data is attributed to that building.
- **Central aggregator (multiple remote gateways):** On the central host, do **not** run local bacnet-server/bacnet-scraper; run only DB, API, Grafana, FDD loop, and (optional) weather. Set `OFDD_BACNET_GATEWAYS` to a JSON array, e.g. `[{"url":"http://10.1.1.1:8080","site_id":"building-a"},{"url":"http://10.1.2.1:8080","site_id":"building-b"}]`, and run one scraper container (or cron) that polls each URL and writes to the central DB with the given `site_id`. Alternatively, deploy one scraper per building elsewhere, each with `OFDD_DB_DSN=central` and `OFDD_BACNET_SITE_ID=<that building>`.

Open-Meteo weather points (stored in `timeseries_readings`, `external_id`):

Point	Description
`temp_f`	Temperature (°F)
`rh_pct`	Relative humidity (%)
`dewpoint_f`	Dew point (°F)
`wind_mph`	Wind speed (mph)
`gust_mph`	Wind gusts (mph)
`wind_dir_deg`	Wind direction (degrees)
`shortwave_wm2`	Shortwave radiation (W/m²)
`direct_wm2`	Direct radiation (W/m²)
`diffuse_wm2`	Diffuse radiation (W/m²)
`gti_wm2`	Global tilted irradiance (W/m²)
`cloud_pct`	Cloud cover (%)

In the **data model** and **Points** UI these appear under equipment **Open-Meteo** (type **Weather_Service**); the RDF graph marks that equipment with `ofdd:dataSource "open_meteo"`.

48.5 Edge / resource limits

For edge deployments with limited disk, set these at bootstrap (or in `stack/.env`). See Getting Started — Bootstrap options.

Setting	Default	Bootstrap arg	Env (stack/.env)
Data retention	365 days	`--retention-days N`	`OFDD_RETENTION_DAYS`
Docker log size	100m per file	`--log-max-size SIZE`	`OFDD_LOG_MAX_SIZE`
Docker log files	3	`--log-max-files N`	`OFDD_LOG_MAX_FILES`

Data retention: TimescaleDB drops chunks older than the configured interval from `timeseries_readings`, `fault_results`, `host_metrics`, `container_metrics`. Set at first run, e.g. `./scripts/bootstrap.sh --retention-days 180`, or set `OFDD_RETENTION_DAYS` in `stack/.env` before running bootstrap.

Log rotation: Containers use `json-file` driver; size and file count come from the table above. Restart the stack after changing so new values apply.

Accessing logs: `docker logs <container> --tail 50` (e.g. `openfdd_weather_scraper`, `openfdd_bacnet_scraper`).

48.6 Environment

`OFDD_`-prefixed vars override YAML:

Variable	Description
`OFDD_DB_HOST`	TimescaleDB host
`OFDD_DB_PORT`	TimescaleDB port
`OFDD_DB_NAME`	Database name
`OFDD_DB_USER`	Database user
`OFDD_DB_PASSWORD`	Database password
`OFDD_BACNET_SERVER_URL`	diy-bacnet-server base URL (e.g. http://localhost:8080)
`OFDD_BACNET_SITE_ID`	Site to tag when scraping (default: default; use on remote gateways)

Variable	Description
`OFDD_BACNET_GATEWAYS`	JSON array of {url, site_id} for central aggregator
`OFDD_RULES_DIR`	Rules directory (default: stack/rules)

49 Contributing

50 Contributing to Open-FDD

First off, thanks for taking the time to contribute!

Open-FDD is in **Alpha**. The most valuable contributions right now are **bug reports** and **FDD rules**—especially from mechanical engineers and building professionals who can add and refine fault-detection rules using the Expression Rule Cookbook. All types of contributions are encouraged.

Phase focus: Alpha emphasizes platform stability, driver implementation beyond BACnet, and API changes for specific integration needs. Beta (planned) will focus more on Brick data modeling (e.g. ASHRAE 223P), mechanical engineering and consulting input into the expression rule cookbook, and better default Grafana dashboards for HVAC analytics. See the Table of Contents for different ways to help and how this project handles them. Please read the relevant section before contributing; it helps maintainers and keeps things smooth for everyone.

> If you like the project but don't have time to contribute, that's fine. Other ways to support it: - Star the repository - Share it with colleagues or at meetups - Refer to Open-FDD in your project's readme or documentation

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50.1 Table of Contents

- Code of Conduct
- I Have a Question
- I Want To Contribute
- Reporting Bugs

- Suggesting Enhancements
- Contributing Rules (Expression Rule Cookbook)
- Your First Code Contribution
- Improving the Documentation
- Styleguides
- Git workflow (branch → PR → sync)
- Commit Messages
- Join the Project Team

50.2 Code of Conduct

This project expects everyone to be respectful and constructive. By participating, you agree to uphold a positive environment. Please report unacceptable behavior by opening an issue or contacting the repository maintainers.

50.3 I Have a Question

Before asking, please check the documentation and search existing issues to see if your question is already answered.

If you still need help:

- Open an issue.
- Provide as much context as you can (what you're trying to do, what you ran, what happened).
- Include relevant versions: Python, Docker, OS (e.g. Ubuntu, Linux Mint), and Open-FDD commit or release if known.

We'll respond as soon as we can.

50.4 I Want To Contribute

50.4.1 Legal notice

When contributing, you agree that you have authored 100% of the content, have the necessary rights to it, and that your contribution may be provided under the project license (MIT).

50.5 Reporting Bugs

Bug testing is a top priority in Alpha. Good bug reports help us fix issues quickly.

50.5.1 Before submitting a bug report

- Use the latest version (main branch or latest release).
- Confirm the bug is in Open-FDD and not in your environment (e.g. wrong Python version, missing config). Check the documentation and I Have a Question first.
- Search issues to see if the bug is already reported.
- Collect:
  - **Stack trace** (Traceback) if applicable
  - **OS and platform** (e.g. Linux, macOS, Windows; x86, ARM)
  - **Versions**: Python, Docker, Docker Compose, and how you ran Open-FDD (pip, Docker Compose, etc.)
  - **Steps to reproduce** and, if possible, sample input/config
  - Whether you can reproduce it every time and on an older version

50.5.2 How to submit a good bug report

- **Security issues:** Do **not** report security vulnerabilities in public issues. Email the repository owner or open a private security advisory on GitHub.
- Open a new issue. Don't assume it's a bug yet—avoid using the word "bug" in the title until it's confirmed.
- Describe **expected behavior** vs **actual behavior**.
- Provide **reproduction steps** so someone else can recreate the issue. Isolate the problem when possible (e.g. minimal rule YAML, minimal config).
- Paste the information you collected above.

After you file:

- Maintainers will label the issue. If we can't reproduce it, we may ask for more steps and mark it `needs-repro`. Reproducible issues may be marked `needs-fix` and opened for implementation.

50.6 Suggesting Enhancements

Enhancements are tracked as GitHub issues.

50.6.1 Before submitting

- Use the latest version and read the documentation to see if the behavior already exists or can be configured.
- Search issues to see if the enhancement was already suggested; if so, add to that discussion.
- Consider whether the idea fits Open-FDD's scope (edge AFDD, rules, API, BACnet, Grafana). Make a clear case for why it would help most users.

50.6.2 How to submit a good enhancement suggestion

- Use a **clear, descriptive title**.
- Describe the **current behavior** and what you **want instead**, and why.
- Provide **step-by-step detail** where useful; screenshots or example YAML/config can help.
- Explain why this would be useful to the community.

50.7 Contributing Rules (Expression Rule Cookbook)

We especially welcome contributions from mechanical engineers and building professionals who can add or improve FDD rules.

- **Where rules live:** Fault rules overview — put project rules in `stack/rules/` (YAML). The FDD loop reloads them every run; no restart needed.
- **How to write rules:** Expression Rule Cookbook — expression-type rules use YAML with BRICK-style inputs, params, and pandas/NumPy expressions. The cookbook includes AHU-style rules (e.g. GL36-inspired) and patterns you can adapt.
- **What to contribute:**
  - **New rule YAMLs** for common faults (AHU, VAV, plant, sensors) that others can reuse.
  - **Improvements to existing cookbook rules** (thresholds, logic, descriptions).

- **New cookbook sections or examples** (e.g. for a specific equipment type or standard).

To contribute a rule or cookbook change:

1. Add or edit files under `stack/rules/` for new/updated rules, or under `docs/expression_rule_cookbook.md` (and related docs) for cookbook content.
2. Follow the existing YAML style and BRICK naming used in the cookbook.
3. In a PR, describe the fault the rule targets and any assumptions (e.g. Brick types, units). If it's from a standard (e.g. ASHRAE), note the reference.

50.8 Your First Code Contribution

- Look for issues labeled `good first issue` or `help wanted` (if we've added them).
- The codebase is Python 3.11+, with FastAPI (API), pandas/NumPy (rules), and Docker for the platform. See Getting Started and the README for setup.
- For rule changes, run the test suite and, if possible, the FDD loop locally with sample data before submitting a PR.

50.9 Improving the Documentation

Documentation lives in the `docs/` directory and is published at bbartling.github.io/open-fdd (Just the Docs theme). Improvements are welcome:

- Fix typos, clarify wording, or update steps.
- Add examples or how-tos (e.g. for the Expression Rule Cookbook, How-to Guides, or BACnet).
- Keep code blocks and links in sync with the codebase.

Building a PDF: To generate a single PDF of the docs (e.g. for offline use or to push to the repo), run `python3 scripts/build_docs_pdf.py`. Requires pandoc and either **weasyprint** (`pip install weasyprint`) or LaTeX. Output is `pdf/open-fdd-docs.pdf` by default; use `--output path` to change it. You can commit and push the PDF if you want it in the repo.

Open a PR with your changes; for large edits, an issue first can help align with maintainers.

50.10 Styleguides

- **Python:** We use Black for formatting. Run `black` on changed files before committing.
- **YAML:** Match the style in `stack/rules/` and the expression rule cookbook (indentation, key order).
- **Markdown:** Use clear headings and lists; link to existing docs where relevant.

50.11 Git workflow (branch → PR → sync)

Create the branch **before** committing so that after the PR is merged, `git pull` on `main`/`master` fast-forwards and you avoid divergent-branch errors (protected `main`/`master`).

1. **Create a branch**

   ```
   git checkout -b feature/short-name
   ```

2. **Stage, commit, and push**

   ```
   git add .
   git commit -m 'short note describing the change'
   git push -u origin feature/short-name
   ```

3. **Open a PR manually on GitHub** from `feature/short-name` into `main` (or `master`). Create the PR in the GitHub UI; we do not use a CLI for this step.

4. **After the PR is merged** — fetch prune, switch to default branch, and pull to sync:

   ```
   git fetch --prune
   git checkout main
   git pull
   ```

   Use `master` instead of `main` if that is your default branch. If you ever committed on `main`/`master` before creating the branch, local and remote can diverge after the merge; run `git reset --hard origin/main` (or `origin/master`) to match the remote.

50.12 Commit Messages

- Use a clear, short summary line. Optionally add a body with details.
- Reference issues/PRs when relevant (e.g. `Fix bounds rule when input is all NaN (#123)`).

50.13 Join the Project Team

If you're interested in ongoing maintenance or a larger role, say so in an issue or reach out to the maintainers. We're especially interested in folks who can triage bugs, review rule contributions, or improve docs and the cookbook.

50.14 Attribution

This contributing guide was inspired by the contributing.md project. You don't need to pay to create or use a CONTRIBUTING file—it's just a markdown file in your repo that helps contributors and maintainers.

51 API Reference

52 API Reference

The Open-FDD platform exposes a **REST API** (port 8000) for CRUD, config, data model, bulk download, analytics, BACnet proxy, faults, and jobs. **Use the React frontend** (http://localhost:5173) for normal workflows—sites, points, config, faults, plots. Use the API when integrating scripts or cloud export.

Interactive docs: When the API is running, open **Swagger UI** at http://localhost:8000/docs or **ReDoc** at http://localhost:8000/redoc. Full **OpenAPI 3.1** spec: http://localhost:8000/openapi.json. When `OFDD_API_KEY` is set (e.g. in `stack/.env`), click **Authorize** in Swagger and paste the key so **Try it out** works.

52.1 High-level overview

Area	Purpose
CRUD	**Sites** — list, create, get, update, delete (cascade to equipment, points, timeseries, fault results). **Equipment** — list, create, get, update, delete (cascade to points). **Points** — list, create, get, update, delete (cascade to timeseries). Points can include `bacnet_device_id` and `object_identifier` so the BACnet scraper polls them.
Config	**GET/PUT /config** — platform config (rule interval, BACnet scrape interval, Open-Meteo, etc.) stored in the RDF graph. The React Config page uses these.
Data model	**GET /data-model/export** — BACnet discovery + DB points (for LLM tagging). **PUT /data-model/import** — bulk create/update points and equipment. **GET /data-model/ttl** — Brick + BACnet TTL. **POST /data-model/sparql** — run SPARQL. **POST /data-model/reset** — reset graph to DB-only.
Download	**GET/POST /download/csv** — timeseries CSV (wide or long). **GET /download/faults** — fault results (CSV or JSON). Excel-friendly; timestamps in UTC (Z).
Analytics	**GET /analytics/fault-summary**, **GET /analytics/fault-timeseries** — fault counts and time-series for charts. **GET /analytics/motor-runtime** — fan/VFD runtime (data-model driven). **GET /analytics/system/*** — host/container/disk metrics.
BACnet	**GET /bacnet/gateways** — list gateways. **POST /bacnet/server_hello**, **POST /bacnet/whois_range**, **POST /bacnet/point_discovery**, **POST /bacnet/point_discovery_to_graph** — proxy to diy-bacnet-server; discovery-to-graph feeds the data model. **POST /bacnet/**

Area	Purpose
	write_point — write value (audited; third-party integrations use this for audited writes).
Faults	**GET /faults/active**, **GET /faults/state**, **GET /faults/definitions** — active state, full state, definitions.
Jobs	**POST /jobs/bacnet/discovery**, **POST /jobs/fdd/run** — async BACnet discovery and FDD run. **GET /jobs/{job_id}** — status.
Run FDD	**POST /run-fdd** — trigger FDD run now. **GET /run-fdd/status** — last run.
AI	**POST /ai/agent** — Overview chat (mode `overview_chat`). See Overview AI context and behavior below.

Base URL: `http://localhost:8000` (use your host or IP when remote). **Auth:** When `OFDD_API_KEY` is set, send `Authorization: Bearer <key>`; the React frontend does this when built with `VITE_OFDD_API_KEY`.

52.2 Overview AI context and behavior

POST /ai/agent (mode `overview_chat`) powers the Overview AI assistant in the React UI. The backend calls OpenAI with the following context; the browser never talks to OpenAI directly.

52.2.1 What the agent receives on each request

1. **Live context (JSON)** — Built from the database and graph on every request: data model summary (sites, equipment, points, rule definitions), active fault count, last FDD run status (and error message if the run failed), BACnet summary, graph serialization status.
2. **Platform documentation excerpt** — The first **~28,000 characters** of `pdf/open-fdd-docs.txt` (or the file at `OFDD_DOCS_PATH` if set). This gives the agent enough to explain what Open-FDD is, quick start, stack, and key concepts. The **full doc is not sent** (the file is ~213k characters / ~56k tokens); a fixed cap keeps token usage and cost reasonable. The excerpt is loaded once per request; there is no “fetch more doc as needed” (no RAG or chunk retrieval).

3. **Attached data** — The backend always attaches fault and sensor charts and tabular data for the last 24 hours; the model is told these are attached and should describe what they show.

52.2.2 What the agent does *not* use

- **Conversation history** — Each request is **stateless**. The API does not send prior messages or replies to OpenAI. The UI may show a scroll of past Q&A, but the backend has no access to that; every call is a single user question and a single assistant reply. Multi-turn or "get more context as needed" in a thread is not implemented.

52.2.3 Models and keys

- **Models** — The API accepts any `model` string; the frontend offers **GPT-5 mini** (default, cost-efficient) and **GPT-5.4 pro** (for more complex tasks). No 4o or other model names are assumed in the docs.
- **OpenAI API key** — Sent in the request body; used by the backend for that request only and never stored.

53 Appendix

54 Appendix

Technical deep dives and maintainer reference material.

Page	Description
API Reference	REST API at a glance: CRUD, config, data model, download, analytics, BACnet, faults, jobs. Full spec at Swagger /docs and /openapi.json.
Technical reference	Directory structure, environment variables, unit tests, BACnet scrape, bootstrap, database schema, LLM tagging workflow.
Developer guide	Front-end development, database schema (migrations, tables, cascade deletes), where to go next.

55 Frontend Evaluation

56 React frontend evaluation

This document summarizes the quality and feature coverage of the Open-FDD React frontend (CRUD UI), its alignment with the API and the Grafana cookbook, and recommendations for WebSockets and the legacy static UI.

56.1 Quality and stack

- **Stack:** React 19, TypeScript, Vite, TanStack Query, React Router, Recharts, Tailwind CSS, shadcn-style UI (Card, Table, Badge, Skeleton). Fits a modern SPA setup.
- **Structure:** Clear separation of pages, hooks (data + WebSocket), contexts (site, theme), and shared UI. API types in `src/types/api.ts` match the OpenAPI schemas (Site, Equipment, Point, FaultState, FaultDefinition, etc.).
- **Auth:** Bearer token from `VITE_OFDD_API_KEY` is sent on all API requests and as `?token=` on the WebSocket; no auth UI (key is build-time env).
- **Missing pieces (added in this pass):** `src/lib/api.ts` (apiFetch with Bearer), `src/lib/csv.ts` (fetchCsv, parseLongCsv, pivotForChart for trending), and `src/lib/utils.ts` (cn, timeAgo, severityVariant) were not in the PR and are required for the app to build and run.

56.2 Feature coverage vs OpenAPI CRUD

API area	Frontend coverage
Sites	✅ List, select site; overview cards (equipment/points/faults per site). Create/update/delete not in UI (API-only or future).
Equipment	✅ List by site or all; table with point count and fault count.
Points	✅ List by site or all; table with external_id, equipment, brick_type, fdd_input, unit.

API area	Frontend coverage
Faults	✅ Active faults list; fault definitions table + count (Grafana cookbook parity).
Trending	✅ Site-scoped point picker, date presets (24h, 7d, 30d, custom), line chart via POST /download/csv (long format) and pivot.
FDD status	✅ Last run in banner; run FDD is API-only (could add button).
Config, data-model, BACnet, jobs	❌ Not in UI; use Swagger or API.

So the frontend covers the main "operator" workflows: sites, equipment, points, active faults, fault definitions, and trending. Advanced flows (config, data-model export/import, BACnet discovery, jobs) stay in Swagger or the API.

56.3 Grafana cookbook parity

The Grafana SQL cookbook describes:

- **BACnet:** Variables (site, device, point), timeseries panel, BACnet + fault overlays.
  Frontend: Site selector + point picker + trending chart (same data via `/download/csv`). No separate "device" (bacnet_device_id) filter in the UI; points are chosen by site/equipment. Fault overlays on the chart could be a future enhancement.
- **Host stats:** Memory, load, swap, disk, containers.
  Frontend: No host-stats views; those remain in Grafana (or a future "System" page).
- **Weather:** Temp/RH/Dewpoint, wind, solar, cloud.
  Frontend: Covered indirectly: weather points (e.g. temp_f, rh_pct) appear in the point list and can be trended like any other point.
- **Faults:** Fault definitions table, fault definition count.
  Frontend: Implemented: "Fault definitions (N)" table (fault_id, name, category, severity, target equipment) and active faults table.

So the frontend is capable of the main cookbook workflows that are data-model/CRUD driven (sites, points, faults, trending). Grafana remains the place for raw SQL, host metrics, and pre-built dashboard JSON.

56.4 WebSockets: keep or remove?

- **Current use:** The React app uses `/ws/events` with token auth; on `fault.*`, `fdd.run`, `crud.*` it invalidates TanStack Query so lists and banners refresh. The HA integration optionally uses the same WebSocket for coordinator refresh.
- **Recommendation: keep WebSockets.** They give the React UI live updates without polling. If you later add MQTT for some integrations, you can keep WebSockets for the browser and use MQTT elsewhere; they are complementary (WS for real-time UI, MQTT for pub/sub or constrained clients). No need to remove the WebSocket feature.

56.5 Legacy static UI and BACnet tree

- The old config UI in `open_fdd/platform/static` (data-model tree Site → Equipment → Points, BACnet test/whois/discovery) was removed so the stack uses a single frontend: the React app, served from its own container and via Caddy.
- **BACnet tree / BAS-style tree:** The static UI had a tree that mirrored the data model and building automation. To get that in the React app, add a "Data model" or "Tree" page that:
  - Fetches `/sites`, then for each site `/equipment?site_id=`, then for each equipment `/points?equipment_id=`,
  - Renders a collapsible tree (sites → equipment → points) with the same behavior (expand/collapse, optional right-click delete).
  - Optionally add a "BACnet" section: server URL, Test (server_hello), Who-Is range, Point discovery, using the same API endpoints the static UI used (`/bacnet/server_hello`, `/bacnet/whois_range`, `/bacnet/point_discovery`). That would bring the legacy config + BACnet flows into the React app without serving static files from FastAPI.

56.6 Caddy and security

- With the React frontend in its own container, **Caddy** is the single entry point (e.g. :8088): basic auth, then proxy to API (and `/ws/*`) and to the frontend (`/*`). Optional Grafana at `/grafana` when started with `--with-grafana`. See Security and Caddy.

56.7 Summary

- **Quality:** Solid React/TS/Vite setup; missing `lib` (api, csv, utils) was added so the app builds and runs.
- **Features:** Good coverage of sites, equipment, points, faults, definitions, and trending; config/data-model/BACnet/jobs stay in API/Swagger.
- **Grafana cookbook:** Parity on fault definitions and trending; host stats and raw SQL stay in Grafana; optional to add fault overlays and a data-model/BACnet tree page later.
- **WebSockets:** Keep; they support real-time UI and HA. MQTT can be added alongside.
- **Static UI:** Removed; React is the primary UI, served by Caddy. BACnet tree and discovery can be reimplemented as a React page if desired.

57 Plan Rules Crud And Hot Reload

58 Plan: Default Rules, Fix Phantom Definitions, Rules CRUD + Frontend (Hot Reload)

58.1 Goals

1. **Fix phantom "AHU Short Cycling"** — Definitions table and Faults matrix show only what exists in `rules_dir`; remove stale rows when a rule file is gone.
2. **Default rules** — Ship with only `sensor_bounds.yaml` and `sensor_flatline.yaml` in `stack/rules`; no extra files that create phantom definitions.
3. **Rules CRUD + frontend** — Users can upload, download, and delete rule YAML files via the UI; changes hot-reload (next FDD run picks them up).

58.2 1. Fix phantom definitions (backend)

Problem: `fault_definitions` is upserted from YAML on each FDD run but never pruned. If a rule file is removed (or never existed on this server), its row remains and the UI shows e.g. “AHU Short Cycling”.

Change: In `open_fdd/platform/loop.py`, extend `_sync_fault_definitions_from_rules(rules)` so that after upserting from the current rule list it **prunes** definitions that no longer have a corresponding rule:

- Collect `fault_id` for every rule in `rules` (same logic: `r.get("flag") or f"{r.get('name', 'rule')}_flag"`).
- After the upsert loop, run: `DELETE FROM fault_definitions WHERE fault_id != ALL(%s)` (or equivalent) with the list of current fault_ids.
- No FK from `fault_results` / `fault_state` to `fault_definitions` (they store `fault_id` as text), so deleting definitions is safe; historical results keep their fault_id for display/analytics.

Result: Definitions list and Faults matrix only show faults that have a YAML file in `rules_dir`. One-time fix for existing DBs: next FDD run will prune “AHU Short Cycling” (and any other orphan).

58.3 2. Default rules (no code change)

- **Repo:** `stack/rules` already contains only `sensor_bounds.yaml` and `sensor_flatline.yaml`. Keep it that way; do not add an AHU Short Cycling YAML unless you want that rule.
- **Server:** If the Linux server has extra files in `stack/rules` that you don’t want, remove them; after the prune logic above and one FDD run, definitions will match the remaining files.
- **New deployments:** Same two files as defaults; anything else is added by users via the new CRUD (or manually).

58.4 3. Backend: Rules CRUD (upload / delete)

Existing:

- `GET /rules` — list `.yaml` filenames and `rules_dir` (from config).

- `GET /rules/{filename}` — return file content (plain text). Use for **download**.
- `POST /rules/test-inject`, `DELETE /rules/test-inject/{filename}` — test-only (gated by `OFDD_ALLOW_TEST_RULES`); keep for automation.

Add (production, no env gate):

Method	Endpoint	Purpose
POST	`/rules` or `/rules/upload`	Upload a new or replace an existing rule file.
DELETE	`/rules/{filename}`	Delete a rule file from `rules_dir`.

Upload (POST):

- **Body:** Either `multipart/form-data` (file) or JSON `{ "filename": "my_rule.yaml", "content": "..." }`. JSON is easier for "copy/paste YAML" in the UI and matches test-inject.
- **Validation:**
  - `filename`: must end in `.yaml`, no `..`, `/`, `\`.
  - Resolve path under `_rules_dir_resolved()`; ensure path stays inside `rules_dir` (path traversal check).
  - Optional but recommended: parse YAML and check for minimal keys (`name` or `flag`) so invalid rules are rejected.
- **Write:** `path.write_text(content)` (same as test-inject). Create `rules_dir` if missing (optional; currently API returns 503 if not a dir).
- **Response:** 200 + `{ "filename": "...", "path": "..." }` or 201 for create.
- **Auth:** Same as rest of API (Bearer when `OFDD_API_KEY` set).

Delete (DELETE):

- **Path:** `DELETE /rules/{filename}` (filename in path; same safety as GET: `.yaml` only, no path traversal).
- **Behavior:** `path.unlink()`. Return 404 if file doesn't exist.
- **Optional:** After delete, call a small "sync definitions from rules_dir" helper that runs `load_rules_from_dir` + `_sync_fault_definitions_from_rules` so the definitions table (and matrix) update immediately instead of waiting for the next FDD run. If you skip this, the next FDD run will prune the definition anyway.

Optional helper: Expose `POST /rules/sync-definitions` that loads all YAML from `rules_dir` and runs `_sync_fault_definitions_from_rules` (no FDD run). Then upload/delete can call it so the UI updates without waiting for the next run. Low priority if FDD runs every few hours.

58.5 4. Frontend: Upload, Download, Delete

Current (Faults page – FDD rule files section):

- Lists filenames from `GET /rules`.
- Click filename → fetch `GET /rules/{filename}` and show content in a `<pre>`.
- Shows `rules_dir` from the list response.

Add:

1. **Download**
   - Per file: "Download" link that opens `GET /rules/{filename}` (e.g. in a new tab or via `fetch` + blob download). Already supported by existing endpoint.
2. **Upload**
   - "Upload rule" (or "Add rule file"):
     - Option A: File input → read file as text → POST JSON `{ filename: file.name, content }` (normalize filename to `.yaml` if needed).
     - Option B: Textarea "Paste YAML" + filename input → POST same JSON.
   - On success: invalidate `GET /rules` (and definitions query if you add sync), show success toast; optionally refetch rules list and select the new file.
3. **Delete**
   - Per file: "Delete" button (with confirmation: "Remove rule file X? Definition will be removed after next FDD run.").
   - Call `DELETE /rules/{filename}`; on success invalidate list and definitions, clear selection if the deleted file was selected.
4. **Copy from existing**
   - Optional: "Duplicate" that copies current file content into the upload form with a new name (e.g. `sensor_bounds_copy.yaml`) so users can duplicate and edit.

UX: Keep the existing "view content" flow; add Upload above the list and Download/Delete next to each filename. Use existing auth (e.g. `apiFetch` with API key).

58.6 5. Hot reload (no change)

- FDD loop already loads rules from `rules_dir` on **every** run (`load_rules_from_dir(rules_path)` in `run_fdd_loop`). No caching.
- So: user uploads a new YAML → file is on disk in `rules_dir` → next FDD run loads it and runs it; new rule appears in definitions (after sync) and in results. Same for delete: file gone → next run doesn't load it → prune step removes the definition.
- Optional `POST /rules/sync-definitions` only makes the definitions table (and matrix) update immediately; it doesn't change hot-reload behavior of the FDD engine.

58.7 6. Implementation order

Step	Task	Effort
1	Backend: Prune stale definitions in `_sync_fault_definitions_from_rules`	Small
2	Backend: `POST /rules` (upload) with filename + content, validation	Small
3	Backend: `DELETE /rules/{filename}`	Small
4	Optional: `POST /rules/sync-definitions` and call after upload/delete	Small
5	Frontend: Download link per file	Small
6	Frontend: Upload form (file or paste + filename), call POST /rules	Medium
7	Frontend: Delete button + confirm, call DELETE /rules/{filename}	Small
8	Tests: API tests for upload/delete; e2e or manual for UI	Small

Total: Backend ~half day; frontend ~half day; testing ~half day. Optional sync-definitions adds a bit more.

58.8 7. Security / validation

- **Path traversal:** Reject `..`, `/`, `\` in filename; resolve path and ensure `path.relative_to(base)` (already done in test-inject).

- **Overwrite:** Upload can overwrite existing files (same as test-inject). Optional: return 409 if file exists and require explicit overwrite flag.
- **YAML validation:** At least parse YAML; optionally require `name` or `flag` and reject if missing so broken rules don't get into the matrix until the next run fails.

58.9 8. Summary

- **Fix:** Prune `fault_definitions` in sync so only current rule files have rows → phantom "AHU Short Cycling" goes away after one FDD run.
- **Defaults:** Keep `stack/rules` with only `sensor_bounds.yaml` and `sensor_flatline.yaml`.
- **CRUD:** Add POST (upload) and DELETE for rule files; reuse GET for download; list already exists.
- **Frontend:** Add upload (file or paste), download link, delete with confirm; reuse existing list + view.
- **Hot reload:** Already works; no change. Optional sync-definitions endpoint improves UX by updating definitions immediately after upload/delete.

59 Security

60 Security and Caddy Bootstrap

This document describes how to protect Open-FDD endpoints with **Caddy** (basic auth), which services are **unencrypted by default**, and **hardening** best practices. The project defaults to **non-TLS**; TLS (including self-signed certificates) is optional.

What is Caddy? A small web server in front of the API and the **React frontend**—one entry point (e.g. port 8088) with optional **basic auth** and **HTTPS**. Optional **Grafana** (when started with `--with-grafana`) is available at `/grafana`. By default Open-FDD does not run Caddy; this doc explains how to enable it for a protected entry point.

60.1 Architecture: frontend, API, and Caddy

- **React frontend** runs in its own container (port 5173 in dev; built and served via Caddy in production-style setups). It uses **Bearer token** auth against the Open-FDD API when `OFDD_API_KEY` is set: all requests (and the WebSocket at `/ws/events`) send `Authorization: Bearer <key>`. See Frontend API key (Bearer) below for where the key is sent.
- **API** (FastAPI) serves REST and WebSocket; when `OFDD_API_KEY` is set, it requires Bearer auth on all endpoints except `/health`, `/`, `/docs`, `/redoc`, `/openapi.json`, and `/app` (legacy static config UI).
- **Caddy** sits in front of both: it enforces **basic auth** (one login for the browser), then proxies to the API and frontend. When proxying to the API, Caddy adds the **X-Caddy-Auth** header (`header_up` in the Caddyfile) so the API accepts those requests without a Bearer token. So you sign in once with openfdd/xyz and the dashboard and API calls work without a second login. The web app can still send Bearer tokens when built with `VITE_OFDD_API_KEY`; when behind Caddy, the API trusts either Bearer or X-Caddy-Auth. **Best practice:** keep the frontend in its own container; do not serve the compiled React app from FastAPI as static files. Caddy (or another reverse proxy) serves the frontend and proxies API/WebSocket.

60.2 Frontend API key (Bearer)

When `VITE_OFDD_API_KEY` is set (at frontend build time), the frontend sends it everywhere it talks to the backend: `apiFetch()` in `frontend/src/lib/api.ts` adds `Authorization: Bearer <key>` to all REST calls (sites, equipment, points, faults, config, FDD status, data model, etc.), `fetchCsv()` in `frontend/src/lib/csv.ts` adds the same header for CSV downloads, and the WebSocket in `frontend/src/hooks/use-websocket.ts` connects to `/ws/events?token=<key>`, which the backend accepts for WebSocket auth when an API key is configured.

60.3 Quick bootstrap with Caddy and basic auth

1. **Start the stack** (from repo root):

```
./scripts/bootstrap.sh
```

This starts DB, API (8000), frontend (5173), BACnet server (8080), scrapers, FDD loop, and **Caddy on port 8088**. **Grafana is not started by default**; use `./scripts/bootstrap.sh --with-grafana` to include it (then access at `http://localhost:8088/grafana` or http://localhost:3000).

2. **Access through Caddy** (single entry point with basic auth):

   - **URL:** `http://localhost:8088`
   - **Default username:** `openfdd`
   - **Default password:** `xyz` (for local/testing only; change before any production or exposed use)

   The bundled Caddyfile routes API paths to the Open-FDD API, **WebSocket** `/ws/*` to the API, and `/*` to the **React frontend**. If Grafana was started with `--with-grafana`, it is available at `http://localhost:8088/grafana`. All routes require this one basic-auth credential. The React app then uses the **API key** (from `VITE_OFDD_API_KEY` at build time) for Bearer auth to the API.

3. **Without Caddy:** You can still use the services directly (no auth if `OFDD_API_KEY` is unset):

   - Frontend: http://localhost:5173
   - API: http://localhost:8000/docs
   - Grafana (if started): http://localhost:3000
     Do not expose these ports to untrusted networks without a reverse proxy and auth.

4. **Browser "Not secure" or "Your connection is not private":** Over plain HTTP (e.g. http://192.168.204.16:8088/), browsers show a "Not secure" or "connection not private" warning. That is expected without TLS. Use **http://** (not https://) so you are not blocked by a bad certificate. For production, add HTTPS (e.g. Caddy with TLS).

60.4 Caddyfile for protecting the entire API

The default **`stack/caddy/Caddyfile`** in the repo only proxies **port 80 → frontend** (no basic auth, API not behind Caddy). To protect the **entire** API (REST, WebSocket, docs) and frontend behind one entry point with basic auth, replace the Caddyfile with a block like the following. Use

the same **secret** in `header_up X-Caddy-Auth` and in the API container env **`OFDD_CADDY_INTERNAL_SECRET`** so the API trusts requests that passed Caddy's basic auth.

```
# Listen on port 80 (or use :8088 and map 8088:8088 in docker-
compose).
:80 {
    # Optional: basic auth (generate hash: docker run --rm caddy:2
caddy hash-password --plaintext 'your_password')
    # basic_auth /* {
    #   openfdd <bcrypt_hash>
    # }

    # API: REST, Swagger, WebSocket. Set
OFDD_CADDY_INTERNAL_SECRET in the API container to match the value
below.
    handle /ws/* {
        header_up X-Caddy-Auth "your_internal_secret"
        reverse_proxy api:8000
    }
    handle /docs {
        header_up X-Caddy-Auth "your_internal_secret"
        reverse_proxy api:8000
    }
    handle /redoc {
        header_up X-Caddy-Auth "your_internal_secret"
        reverse_proxy api:8000
    }
    handle /openapi.json {
        header_up X-Caddy-Auth "your_internal_secret"
        reverse_proxy api:8000
    }
    handle /health {
        header_up X-Caddy-Auth "your_internal_secret"
        reverse_proxy api:8000
    }
    handle /config* {
        header_up X-Caddy-Auth "your_internal_secret"
        reverse_proxy api:8000
    }
    handle /sites* {
        header_up X-Caddy-Auth "your_internal_secret"
        reverse_proxy api:8000
    }
    handle /equipment* {
        header_up X-Caddy-Auth "your_internal_secret"
        reverse_proxy api:8000
    }
    handle /points* {
        header_up X-Caddy-Auth "your_internal_secret"
        reverse_proxy api:8000
```

```
    }
    handle /faults* {
        header_up X-Caddy-Auth "your_internal_secret"
        reverse_proxy api:8000
    }
    handle /data-model* {
        header_up X-Caddy-Auth "your_internal_secret"
        reverse_proxy api:8000
    }
    handle /download* {
        header_up X-Caddy-Auth "your_internal_secret"
        reverse_proxy api:8000
    }
    handle /analytics* {
        header_up X-Caddy-Auth "your_internal_secret"
        reverse_proxy api:8000
    }
    handle /bacnet* {
        header_up X-Caddy-Auth "your_internal_secret"
        reverse_proxy api:8000
    }
    handle /jobs* {
        header_up X-Caddy-Auth "your_internal_secret"
        reverse_proxy api:8000
    }
    handle /capabilities* {
        header_up X-Caddy-Auth "your_internal_secret"
        reverse_proxy api:8000
    }
    handle /ai* {
        header_up X-Caddy-Auth "your_internal_secret"
        reverse_proxy api:8000
    }

    # Optional: Grafana when started with --with-grafana
    # handle /grafana/* {
    #   reverse_proxy grafana:3000
    # }

    # Frontend (React)
    handle /* {
        reverse_proxy frontend:5173
    }
}
```

Then restart Caddy and recreate the API so it has `OFDD_CADDY_INTERNAL_SECRET` set: `./scripts/bootstrap.sh --build caddy api` (from repo root). See Troubleshooting if the API returns 401 or the WebSocket fails.

60.5 Default password and how to change it

- The Caddyfile uses a **bcrypt hash** for the password. The default hash in the repo corresponds to password **`xyz`**.
- **Change the password** (recommended before production):

  1. Generate a new hash:

     ```
     docker run --rm caddy:2 caddy hash-password --plaintext 'your_secure_password'
     ```

  2. Open `stack/caddy/Caddyfile` and replace the hash on the `openfdd` line with the new output.

  3. Restart Caddy: `docker compose -f stack/docker-compose.yml restart caddy` (from repo root).

- **Advanced / multiple users:** Edit the `basic_auth` block in the Caddyfile and add more lines (one per `username hash`). Use `caddy hash-password` for each password. For SSO or advanced auth, consider Caddy's JWT or forward-auth and an IdP instead of basic auth.

60.6 Services: encryption and exposure

Service	Port	Encrypted by default?	Notes
API	8000	No (HTTP)	Put behind Caddy; do not expose directly. Bearer auth when `OFDD_API_KEY` is set.
Frontend	5173	No (HTTP)	React app; use Caddy to serve and protect. Sends Bearer token to API.
Grafana	3000	No (HTTP)	

Service	Port	Encrypted by default?	Notes
			Optional (–with-grafana); use Caddy at /grafana or Grafana's own auth.
TimescaleDB	5432	No (plain PostgreSQL)	Not TLS by default. Keep on internal network only.
BACnet server	8080	No (HTTP API)	Host network; protect with firewall or Caddy on host.

- **Recommendation:** Expose only Caddy (e.g. 8088 or 443) to the building/remote network. Do not expose 5432, 8000, 5173, 3000, or 8080 to the internet. Use Caddy basic auth (and optionally TLS) for the frontend and API; keep the database and BACnet server behind the firewall. The React frontend runs in its own container and is served by Caddy (or another reverse proxy), not as static files from the FastAPI process.

60.7 Hardening best practices

1. **Passwords**
   - Do not use default `openfdd` / `xyz` in production. Change via `caddy hash-password` and update the Caddyfile.
   - Use strong, unique passwords (or a secrets manager) for Grafana (`GF_SECURITY_ADMIN_PASSWORD`), Postgres (`POSTGRES_PASSWORD`), and Caddy basic auth.
2. **Network**
   - Run TimescaleDB and internal services on a private Docker network; bind DB to localhost or internal IP only if needed.

    - Restrict firewall so only Caddy's port (and optionally SSH) is reachable from outside.
3. **Reverse proxy**
    - Use Caddy in front of API and Grafana so one place enforces auth and (optionally) TLS.
    - Basic auth over plain HTTP is weak; prefer HTTPS when possible (see below).
4. **Principle of least privilege**
    - Run containers as non-root where possible; avoid sharing host network unless required (e.g. BACnet).
    - Limit Grafana sign-up and use strong admin credentials.
5. **Updates and hygiene**
    - Keep Caddy, Grafana, TimescaleDB, and the OS updated. Rotate credentials after compromise or periodically.

60.8 Throttling and rate limiting

60.8.1 1. No API rate limiting by default

Open-FDD does **not** rate-limit incoming HTTP requests to the API. There is no built-in cap on how often clients can call the API. If you need to limit how often external clients (e.g. a busy integration or cloud poller) can call the API, add rate limiting at the reverse proxy (e.g. Caddy with a rate-limit module) or with rate-limit middleware in front of the app.

60.8.2 2. Outbound: OT/building network is paced

The application **does** throttle its own outbound traffic to the building and OT network. We do not continuously hammer BACnet or other building systems. Load is paced by configuration **and by which points you scrape**:

Component	Config	Effect
BACnet scraper	`bacnet_scrape_interval_min` (e.g. 5)	Polls points on a fixed interval (e.g. every 5 minutes), not in a burst.
BACnet scraper	**Data model or CSV**	The scraper polls only the points it is configured with: **by default, points in the data model** that have

Component	Config	Effect
		`bacnet_device_id` and `object_identifier` (e.g. added via CRUD or after **POST /bacnet/point_discovery_to_graph** and data-model export/import). If none exist, it can fall back to a **curated CSV**. Throttling depends on **how many points** are defined (in the DB or in the CSV). Best practice: scrape only the points needed for FDD and HVAC health. See BACnet overview.
FDD rule loop	`rule_interval_hours` (e.g. 3)	Runs fault detection on a schedule (e.g. every 3 hours); each run pulls data from the DB, not from BACnet.
Weather scraper	`open_meteo_interval_hours` (e.g. 24)	Fetches weather once per interval (e.g. daily).

So outbound load on the OT network is predictable and tunable. **Define only the points you need** (in the data model via CRUD or point_discovery_to_graph + export/import, or in a curated CSV), then adjust intervals via `OFDD_*` environment variables. See Configuration and BACnet overview.

60.8.3 3. Inbound: rate limiting at the reverse proxy (e.g. Caddy)

If you need to **throttle incoming traffic** to the API—for example to protect the API and OT network from aggressive polling, misconfigured integrators, or abuse—enforce rate limiting at the reverse proxy or with middleware. Open-FDD does not implement this itself; use Caddy (with a rate-limit module), nginx, or application middleware.

Using Caddy for rate limiting

Caddy can throttle requests per client (e.g. per IP or per authenticated user) when configured with a rate-limiting capability. The standard Caddy build does not include an HTTP rate-limit directive; you add it by building Caddy with a module (e.g. `xcaddy build --with github.com/mholt/caddy-ratelimit`) or by using a distribution that includes one. Once available, you configure a rate-limit zone in the Caddyfile (e.g. limit by `{remote_host}` or by a header), then apply it to the blocks that proxy to the API (and optionally Grafana). Typical choices are a cap per IP (e.g. 60 requests per minute) or per client identity when using auth. Exceeding the limit returns `429 Too Many Requests`; the client can retry after the window resets.

Example (conceptual; syntax depends on the module you use):

```
:8088 {
  rate_limit {
    zone api { key {remote_host} events 60 duration 1m }
  }
  handle /api/* {
    rate_limit api
    reverse_proxy api:8000
  }
  # ... basic_auth and other handle blocks
}
```

Adjust the zone (events per duration) to match your OT network and integration requirements. For official options and current syntax, see Caddy's module documentation and the documentation for the rate-limit module you build or install.

60.9 TLS (optional): self-signed certificates

The project **default is non-TLS**. If you want HTTPS in front of Caddy:

1. **Option A – Caddy automatic HTTPS (public hostname)**
   If you have a public hostname and port 80/443, change the Caddyfile address to your domain; Caddy will obtain and renew Let's Encrypt certs automatically.

2. **Option B – Self-signed certificate for Caddy**
   For internal or lab use:

   - Generate a self-signed cert (e.g. with `openssl`).
   - In the Caddyfile, use `tls /path/to/cert.pem /path/to/key.pem` and listen on `:443` (or another port).

- Mount the cert and key into the Caddy container and point Caddyfile to those paths.
- Browsers will show a certificate warning; accept or install the CA for your environment.

Example Caddyfile snippet for self-signed (conceptual):

```
https://0.0.0.0:8443 {
  tls /etc/caddy/cert.pem /etc/caddy/key.pem
  basic_auth /* { ... }
  # same handle/reverse_proxy blocks as :8088
}
```

Generate self-signed (host):

```
        openssl req -x509 -newkey rsa:4096 -keyout key.pem -out cert.pem -
days 365 -nodes -subj "/CN=openfdd.local"
```

Mount `cert.pem` and `key.pem` into the Caddy container and reference them in `tls`.

60.10 Troubleshooting

API returns 401 (Unauthorized) after you pass Basic auth in the browser

The API trusts requests that carry the internal header `X-Caddy-Auth` (set by Caddy after Basic auth). The API only checks this when the env var `OFDD_CADDY_INTERNAL_SECRET` is set in the API container and matches the value Caddy sends. If you added or changed this after the stack was already running, the API container may not have the variable. Recreate the API so it picks up the env:

```
        cd stack && docker compose up -d --force-recreate api
```

Or from the repo root: `./scripts/bootstrap.sh --build api`. Then hard-refresh or use an incognito window and sign in again with Basic auth.

WebSocket to `/ws/events` fails or closes immediately

When access is through Caddy with Basic auth, the browser may not send credentials on the WebSocket upgrade. The API also accepts the same `X-Caddy-Auth` header on the WebSocket endpoint, so if Caddy is forwarding the request (after Basic auth), the WebSocket should work. Ensure the API has been recreated with `OFDD_CADDY_INTERNAL_SECRET` as

above. If the WebSocket still fails (e.g. due to browser or proxy behavior), the UI will keep working; live updates will resume when the connection succeeds.

60.11 Security scanning (Trivy) — Phase 3

During development, use **Trivy** to scan container images and the repo for vulnerabilities and misconfigurations. Run it after building images, when changing Dockerfiles or dependencies, and optionally in CI. See **How-to: Trivy security scanning** for install, when to run, image and filesystem scan commands, and keeping docs up to date. High-level roadmap: NOTES.md (Phase 2: stack/DB/Caddy; Phase 3: Trivy in the loop).

60.12 Summary

- **Bootstrap:** Start the stack; use `http://localhost:8088` (or your Caddy port, e.g. 80) with user `openfdd` and default password `xyz` (change before production).
- **Passwords:** Change default by running `caddy hash-password` and updating the Caddyfile; use strong passwords for Grafana and Postgres.
- **Unencrypted by default:** API, Grafana, TimescaleDB, and BACnet API are plain HTTP/TCP; protect them with network isolation and Caddy (and optional TLS).
- **Hardening:** Strong passwords, expose only Caddy, keep DB and internal services off the public internet, keep software updated.
- **Throttling:** (1) No API rate limiting by default. (2) Outbound traffic to the OT/building network is paced (BACnet scrape, FDD loop, weather intervals). (3) To throttle incoming API traffic, use the reverse proxy (e.g. Caddy with a rate-limiting module) or middleware.
- **TLS:** Optional; default is non-TLS. Add self-signed or Let's Encrypt via Caddy when you need HTTPS.